ANNEX 1

Horizon Europe (HORIZON)

Description of the action (DoA)

Part A

Part B

DESCRIPTION OF THE ACTION (PART A)

COVER PAGE

Part A of the Description of the Action (DoA) must be completed directly on the Portal Grant Preparation screens.

PROJECT *Grant Preparation (General Information screen) — Enter the info.*	
Project number:	101181988
Project name:	SORT4CIRC - Intelligent Textile SORting for enable CIRCularity
Project acronym:	SORT4CIRC
Call:	HORIZON-CL6-2024-CIRCBIO-02
Topic:	HORIZON-CL6-2024-CircBio-02-1-two-stage
Type of action:	HORIZON-RIA
Service:	REA/B/03
Project starting date:	fixed date: 1 December 2025
Project duration:	36 months

TABLE OF CONTENTS

Project summary 3
List of participants 3
List of work packages 5
Staff effort 20
List of deliverables 21
List of milestones (outputs/outcomes) 39
List of critical risks 40
Project reviews 43

PROJECT SUMMARY

Project summary

Grant Preparation (General Information screen) — Provide an overall description of your project (including context and overall objectives, planned activities and main achievements, and expected results and impacts (on target groups, change procedures, capacities, innovation etc)). This summary should give readers a clear idea of what your project is about.

Use the project summary from your proposal.

The EU generates 12.6 million tonnes of textile waste per year, with only 22% currently collected for reuse or recycling —the rest often incinerated or landfilled. With mandating separate textile collection across the EU by 2025, the collected amount is expected to increase and we must upscale sorting and recycling capacity and efficiency to ensure sustainable recovery of the material. SORT4CIRC's overall ambition is to improve value creation and cost efficiency through automated sorting and innovative recycling. This will generate new revenue streams for sorters and recyclers by turning non-re-wearable textiles into valuable feedstock for high-value recycling. The project will strive to unlock the current technological gaps for textile identification and value-chain traceability, while providing an overall sound business case driving circularity within the textile value chain (TVC). The use of modern digital technology in the sorting process will make it easier to achieve comprehensive information of the materials and fabrication process steps and integrate it to automated presorting and sorting for recycling processes. The different & complementary techonolgies are interlinked with each other with interoperable architecture to achieve complete traceability. Results generated are cross-checked to ensure reliability and consistency, to make decision over its optimal channel for reuse or recycling. SORT4CIRC gathers all relevant expertise to develop a systemic transformative solution enabling a traceable, circular textile industry: industrial collector and sorter, design for recycling by allowing disassembly, technology developers (machine learning, Artificial Intelligence (AI), imaging, hyperspectral, NIR, multisensors), recyclers (chemical, thermo-mechanical and mechanical), along with specialists in supply chain traking and tracing, digital passports, blockchain technology and IoT, textile business models, trade and LCC/LCA.

LIST OF PARTICIPANTS

PARTICIPANTS

Grant Preparation (Beneficiaries screen) — Enter the info.

Number	Role	Short name	Legal name	Country	PIC
1	COO	CU	CONSTRUCTOR UNIVERSITY BREMEN GGMBH	DE	999897147
2	BEN	IZCS	INTERZERO CIRCULAR SOLUTIONS GERMANY GMBH	DE	890257368
2.1	AE	IZCT	IZ Circular Textiles Gmbh	DE	873039092
3	BEN	ARIS	ARISTENG SARL	LU	890617044
4	BEN	RECY	RECYC ELIT SAS	FR	898416620
5	BEN	RESOR	REGENERATION	BE	893491348
6	BEN	IOM	LEIBNIZ-INSTITUT FUR OBERFLACHENMODIFIZIERUNG EV	DE	968926308
7	BEN	CEF	STICHTING CIRCLE ECONOMY	NL	939754528
7.1	AE	CGS	Circularity Gap Solutions B.V.	NL	877344631
8	BEN	CEA	COMMISSARIAT A L ENERGIE ATOMIQUE ET AUX ENERGIES ALTERNATIVES	FR	999992401

PARTICIPANTS *Grant Preparation (Beneficiaries screen) — Enter the info.*					
Number	**Role**	**Short name**	**Legal name**	**Country**	**PIC**
9	BEN	SEM	SEMSO	FR	876371333
10	BEN	PURFI	PURFI MANUFACTURING BELGIUM	BE	887835957
11	BEN	PSB	PARIS SCHOOL OF BUSINESS	FR	944056866
12	BEN	TRA	TEXTILE RECYCLING ASSOCIATION LIMITED	UK	878690991
13	BEN	IRIS	IRIS TECHNOLOGY SOLUTIONS, SOCIEDAD LIMITADA	ES	998993786
14	BEN	REFA	reverse.fashion GmbH	DE	878874321
15	BEN	TXHO	TEXTILE HOUSE FOR EURO TRADE, S.R.O.	SK	870613413

LIST OF WORK PACKAGES

Work packages *Grant Preparation (Work Packages screen) — Enter the info.*						
Work Package No	**Work Package name**	**Lead Beneficiary**	**Effort (Person-Months)**	**Start Month**	**End Month**	**Deliverables**
WP1	Management	1 - CU		1	36	D1.1 – Project Management Plan D1.2 – Data Management Plan D1.3 – Project meeting minutes and decision records M1-M18 D1.4 – Data Management Plan M18 D1.5 – Project meeting minutes and decision records M18-M24 D1.6 – Project meeting minutes and decision records M24-M36 D1.7 – Data Management Plan M36
WP2	Pre-sorting & textile classification	9 - SEM		1	24	D2.1 – Development of a complete machine for PSFS D2.2 – Sorting system for PSFS D2.3 – Automated detection for a PSFS garment sorting system D2.4 – Usage Guidelines D2.5 – Summaries of Learning and Deployment Insights
WP3	Textile recognition for automatic sorting in real-time	13 - IRIS		1	24	D3.1 – Report on characterisation of textile samples D3.2 – Report on the selection of best techniques for the recognition of complex garments D3.3 – Multirange Spectroscopic smart sensing system D3.4 – Eco-design guidelines and report on the process to recognise DfD garments

Work packages *Grant Preparation (Work Packages screen) — Enter the info.*						
Work Package No	**Work Package name**	**Lead Beneficiary**	**Effort (Person-Months)**	**Start Month**	**End Month**	**Deliverables**
						D3.5 – Tests report
WP4	Textile DPP	1 - CU		1	36	D4.1 – TTRFID assessment D4.2 – TTRFID design and selection guideline D4.3 – DPP development guidelines D4.4 – DPP smart contract guidelines for PSFS & PSSR compatibility D4.5 – Tokenised Ecosystem framework analysis and development guideline D4.6 – DfD solution development over DPP framework D4.7 – Guidelines for recyclability and reusability
WP5	Demonstrator	15 - TXHO		19	36	D5.1 – TTRFID technology deployment D5.2 – DPP Blockchain demonstration deployment with tokenised LCA assessment dashboard D5.3 – Automatised PSFS pilot D5.4 – Upgrade Automatised Sorting Line D5.5 – Validation of Sorted Feedstock D5.6 – Successful recycling of sorted feedstock
WP6	Technical and economic feasibility, environmental and social assessment, design and efficiency improvement	11 - PSB		1	36	D6.1 – Data collection for SORT4CIRC D6.2 – Report on environmental impact assessment D6.3 – Report on cost-benefit analysis D6.4 – Report on the BM settings D6.5 – Report on the efficient design and operation of the automated sorting system

Work packages *Grant Preparation (Work Packages screen) — Enter the info.*						
Work Package No	**Work Package name**	**Lead Beneficiary**	**Effort (Person-Months)**	**Start Month**	**End Month**	**Deliverables**
						D6.6 – Holistic technology benchmarking among different textile sorting processes D6.7 – Public report on environmental impact assessment
WP7	Communication, Dissemination, and Exploitation	7 - CEF		1	36	D7.1 – Communication, Dissemination, and Exploitation Plan M3 D7.2 – SORT4CIRC Brand Book D7.3 – SORT4CIRC website D7.4 – Communication, Dissemination, and Exploitation Plan M12 D7.5 – SORT4CIRC policy briefing document M18 D7.6 – Communication, Dissemination, and Exploitation Plan M36 D7.7 – SORT4CIRC circular textiles toolbox D7.8 – SORT4CIRC policy briefing document M36 D7.9 – Report on capacity building webinars D7.10 – Summary of clustering activties

Work package WP1 – Management

Work Package Number	WP1	Lead Beneficiary	1 - CU
Work Package Name	Management		
Start Month	1	End Month	36

Objectives
• To ensure smooth collaboration and integration of all partners applying methodologies and tools for efficient implementation of activities as well as ensuring decision-making and steering within governing bodies; • To anticipate, monitor and assess project progress vs. work plan and initiate, if necessary, prompt corrective actions and assure timely reporting and communication with the Project Officer and the EC; • To guarantee a maximised collaborative leverage effect and benefits for the consortium and the EC; • To implement a performing risk and quality management as well as efficient data management. CU will be the Project Coordinator (PCo) and lead global project coordination (strategic, contractual, animation, communication, reporting, etc.).

Description
T1.1: Strategic & technical project management [CU, WP leaders] (M1-M36) This task is dedicated to the strategic and technical management of the project, including the coordination of partners through governing bodies animation, WPs follow-up, progress monitoring (cost and time), assessment of the quality of the results and their impact, and, if necessary, application of contingency plans. It also covers relationships with relevant stakeholders beyond participant organisations (EC, Advisory Board - see 3.2, other projects, etc.), as needed by the project to guarantee its success, impact, and leverage effect. The results of Task 1.1 will be reported in D1.3, D1.5 and D1.6. T1.2: Administrative, financial, and contractual coordination [CU, WP leaders] (M1-M36) This task comprises all legal, financial, and administrative activities within the project, both among the partners and concerning the relationship with the EC. CU will oversee the contractual and administrative management of the project: Management of the Grant Agreement (including amendment requests); Keeping updated the Consortium Agreement, especially know-how and access right management; Managing partner's access and withdrawal together with governing bodies. CU will also oversee financial reporting: Handling grant provided by EC regarding distribution between partners and activities; A 6M-reporting (both from technical and financial aspects will be proposed) synchronised with annual and mid-year reviews; Maintaining a day-to-day relationship with partners and answering requests regarding cost eligibility, financial reporting, and official process for fund transfer; Ensuring payments to all partners in due time. In addition, the activities of the ethics mentor willbe included in the Periodic technical reports. The results of Task 1.1 will be reported in D1.3, D1.5 and D1.6. T1.3: Internal communication [CU, WP leaders] (M1-M36) In this task, CU will promote efficient communication flows within the consortium. The consortium will have access to a SharePoint platform for efficient project information flow. Official documents, templates, working documents, technical and financial progress reports will be available on that platform. They will be accessible to partners with different levels of responsibility (partner, WP leader, guests, users). The results of Task 1.1 will be reported in D1.3, D1.5 and D1.6. T1.4: Quality assurance, data and risk management [CU] (M1-M36) CU will set a quality management policy covering all aspects related to practical management (document exchanges & internal workflows) of the project. It will rule the risk management plan and associated contingency plans, the critical path of the project monitoring, communication and quality control activities, gender indicators according to the Gender Equality Plan, internal processes related to standards of documents, proof-reading, validation workflow for the approval of deliverables, reports, dissemination, exploitation and IPR items, etc. Key indicators will also be proposed to evaluate the project performance and impact. In parallel, CU will establish data management procedures describing how project data will be handled, what data will be collected, processed, or generated, and methodologies and standards followed

regarding sharing, duration, and preservation, ensuring full compliance with open science policy. The procedures will be gathered in a Data Management Plan to be submitted to the EC in M6 with regular updates.

The results of Task 1.4 will be reported in D1.1, D1.2, D1.4 and D1.7.

Work package WP2 – Pre-sorting & textile classification

Work Package Number	WP2	**Lead Beneficiary**	9 - SEM
Work Package Name	Pre-sorting & textile classification		
Start Month	1	**End Month**	24

Objectives

• Develop the automatic recognition of garment types further and make detection quality more robust.
• Develop models to evaluate the degree of soiling and wear condition with a perspective of interoperability with the second-hand market further.
• Develop a SKU retrieval system to access product data without requiring a visible or accessible identifier.
• Identifying and addressing data gaps in the textile industry to enhance transparency and traceability throughout the TVC.

Description

T2.1: Maturing AI-based vision detection system for Garment PSFS and developing textile classifications that are interoperable with the second-hand market [REFA, TXHO] (M1-M24)
T2.1.1: Automated AI-Based PSFS System for Garment Attributes
REFA's sorting technology accurately scans and identifies garment types, wear conditions, style, brands, colours, fabric construction and sizes through AI-based image analysis. Feeding the conveyor belt with the garments by placing them unfolded on it. This technology will be taken as a basis for this WP. REFA will improve and adapt the AI technology to increase its accuracy and efficiency. This will involve refining the AI algorithms, improving image processing capabilities, and enchaining overall system integration. By the end of the project, the AI-based PSFS system will be significantly optimised. This will result in a more efficient, accurate, and scalable solution for garment PSFS, benefiting the sorting and textile industries.
T2.1.2: Development of a SKU retrieval system
REFA builds on existing AI algorithms and further develops them to recognise individual products and infer their SKU based on image similarity. REFA is creating a database of image feature vectors that links image embeddings to product data. Through the additional product data that can be processed in the sorting decision, the sorting quality can be significantly increased. This is particularly useful for sorting unique and high-value garments that would otherwise require economically unfeasible sorting steps.

The results of T2.1.1 and T2.1.2 will be reported in D2.3, D2.4 and D2.5.

T2.2: Development of an integrated semi-automated garment PSFS [SEM, TXHO] (M1-M20)
SEM will develop an integrated automated system design for the PSFS, encompassing both the automatic recognition and separation of garment types. TXHO will contribute to this process by providing access to its logistics and sorting lines for practical testing and supplying a collection of pre-sorted garments across 15 article classes in various quality levels to support the development and validation of the designed system.

The results of Task 2.2 will be reported in D2.2.

T2.3: Development of a complete machine consisting of PSFS system [SEM, REFA, TXHO] (M1-M12)
SEM will combine the components developed for the new type of PSFS and build a compact all-in-one solution from them. The requirements, such as available space and the existing manual sorting line, will be considered. REFA and TXHO will support SEM in integrating their process technology into the overall system with the IT interfaces, documentation of the system control, and technical assistance with the installation of the hardware.

The results of Task 2.3 will be reported in D2.1.

Work package WP3 – Textile recognition for automatic sorting in real-time

Work Package Number	WP3	**Lead Beneficiary**	13 - IRIS
Work Package Name	Textile recognition for automatic sorting in real-time		
Start Month	1	**End Month**	24

Objectives
• Studies and selection of the most adequate optical techniques for recognising complex textiles. • Leveraging a novel Multirange Spectroscopic Platform to design and develop the most suitable spectroscopic configuration to be engineered into a compact and affordable inline device for the real-time monitoring of the textiles' critical parameters, overcoming the limitations of current NIR technology. • Develop a process to recognise DfD garments and sort disassembled pieces.

Description
Description of work T3.1: Definition of textile typologies to be studied, collection of textile waste samples and characterisation and creation of complex textiles databases [TXHO, RECY, PURFI, CEA] (M1-M06) TXHO will define an action plan to collect the different textile residues, including collection, transportation, storage and logistics. Dirty samples from PCT complex waste (from all types of garments with interest for recyclers) will be collected by TXHO. Clean textile samples will be collected, characterised and used for the tests. CEA will use a database of clean samples from previous projects to develop models capable of estimating the composition of garments and to validate the results obtained through the techniques tested in the following tasks. PURFI and RECY will provide specifications to drive the development of the technology. The results of Task 3.1 will be reported in D3.1. T3.2: Selection of best techniques for the recognition of complex textiles and development of chemometric, AI and ML models [CEA, IOM, IRIS] (M1-M18) T3.2.1: Selection of optical techniques for identification of complex textiles: CEA, IOM and IRIS will study a comprehensive series of spectroscopic techniques for detecting complex parameters in garments. CEA and IOM will focus on certain aspects of the spectroscopic problems that must be solved. CEA will extensively test the performance of the MWIR camera for carbon black textiles, which is assumed to have potential due to its wavelength range being far away from the VIS range. CEA will optimise the SWIR HSI technique to detect low-content materials (<3%). IOM will test the performance of NIR imaging and spectroscopy and the potential of other spectroscopic techniques, such as Raman. IRIS will study the identification of complex compositions of multilayer garments through NIR hyperspectral imaging and DfD solutions (Smart Stitch™), i.e. NIR tackled by the project, as well as foreign materials with metallic parts (zippers, buttons, etc.) through LIBS. T3.2.2: Development of chemometric, AI and ML models: CEA, IOM and IRIS will develop chemometric models based on AI and statistical aspects for extracting actionable information from the spectroscopic data capable of accurately identifying complex textile waste through the selected optical techniques selected in subtask T3.2.1. The results of T3.2.1 and T3.2.2 will be reported in D3.2. T3.3: Development of a cost-effective and reliable inline multirange spectroscopic smart sensing system for the recognition of complex textiles in real-time [IRIS, TXHO, CEA, IOM] (M19-M24) T3.3.1: Development of an inline Multirange Spectroscopic smart sensing system: IRIS will develop a multirange spectroscopic platform incorporating the most suitable optical techniques for relevant parameters for proper sorting. CEA, IOM and IRIS will commonly evaluate the specific informative value of the results obtained by the various spectroscopic methods for final decision-making in the sorting process and derive from that evaluation the optimum configuration of methods to be implemented in the VISUM Fusion multirange spectroscopic platform. Once the chemometric models are built through T3.2.2 and lab-tested and validated through FAT, this task will culminate with the sensing system being installed in the pilot unit in WP5. In addition to multivariate data from spectroscopy analysers, univariate data from available traditional sensors installed in the pilot unit in WP5 will also be assessed to explore the data fusion opportunities. T3.3.2: Data Management platform and predictive modelling: In this task, IRIS, will develop a data management platform whereby data will be acquired from the Multirange Spectroscopic smart sensing system and traditional sensors. IRIS will adapt its current Scadalytics software for the data collection, storage, fusion and processing.

The results of T3.3.1 and T3.3.2 will be reported in D3.3.

T3.4. Recognition of DfD solutions of garments with Smart Stitch™ technology and automated assessment of disassembly efficiency while sorting by composition after smart disassembly [RESO, IRIS]. (M6-M24)
T3.4.1: Recognition of DfD eco-design prototypes: RESO and IRIS will study the best method(s) of identification techniques (RFID, NIR, RGB) that could be coupled with AI for Smart Stitch™ identification, fulfilling speed and accuracy requirements (500 kg/hour infeed, >95% accuracy) and will work on an effective pre-sorting assessment (categories, integration in the ecosystem chain, automation level). RESO will work as well on a Failure Mode and Effects Analysis (FMEA) and risk assessment, and they will test the speed, efficiency and accuracy of the system developed and validate the device in the relevant environment.
T3.4.2: Automated assessment of disassembly efficiency while sorting by composition after Smart Disassembly™: RESO will work on the integration of method(s) in machinery after the Smart Disassembly™ process to assess effective disassembly while sorting in mono-material streams. With the support of IRIS, the best method(s) of identification (NIR, RGB with AI, electromagnetic sensors…) that fulfil speed and accuracy requirements (500 kg/h infeed, >95% accuracy) will be studied. The acceptance criteria (sorting categories, purity level, type of flow, speed, accuracy…) will be defined. FMEA and risk assessment will be performed, and the speed, efficiency and accuracy of the system will be tested and validated.

The results of T3.4.1 and T3.4.2 will be reported in D3.4.

T3.5. Validation by recyclers [RECY, PURFI] (M18-M24)
RECY will validate, at a laboratory scale, if the composition of the four families (PET/PA; PET/EA: PET/cotton; multilayers) of feedstocks sorted by TXHO matches the specifications provided in T3.1 (polyester content). Co-materials will be analysed to validate the composition and absence of any contaminants. PURFI will validate if the composition of cotton feedstocks matches with the specifications provided in T3.1.

The results of Task 3.5 will be reported in D3.5.

Work package WP4 – Textile DPP

Work Package Number	WP4	**Lead Beneficiary**	1 - CU
Work Package Name	Textile DPP		
Start Month	1	**End Month**	36

Objectives

- Develop a customised TTRFID tag model specifically for eco-friendly and recyclable textiles.
- Design an adaptable/scalable blockchain architecture for the traceability of textile data over the TVC ecosystem.
- Integrate DPP data retrieval with automated PSFS & PSSR systems for textile recycling solutions.
- Establish a tokenised ecosystem over the TVC to incentivise sustainable practices.
- Implement DfD solutions to enhance textile recycling and reuse.

Description

Description of work
T4.1: Development of a TTRFID tag model [CU, PURFI, RECY] (M1-M12)
T4.1.1. TTRFID technology assessment and developing design and selection guidelines: CU will assess existing RFID tag technologies to identify the most suitable options for textile applications. This will include a detailed review of available chipless RFID tags (RFID tags without chips and textile recycling compatible material composition), focusing on their eco-friendliness, recyclability, and operational frequency ranges. Various RFID tags, such as those identified in the methodology section (e.g., biopolymer-based, textile-based), will be included for their durability through consumer usage, data storage capacity, and transmission efficiency. Moreover, the implementation and deployment requirements, besides compatibility with current technologies like QR codes, will be developed, as well as the complementary requirements for integration with blockchain technology for dynamic/static public key embedding. CU also investigates innovative solutions like ink-based RFID tags, which offer the advantage of being less obtrusive and potentially non-irritating for wearers, as survey feedback suggests.
T4.1.2. TTRFID design for recycling analysis and assessment: PURFI and RECY focus on analysing the recycling

compatibilities of 3 technologies of TTRFID tags (provided by CU). This includes assessing the inclusion of TTRFID in recycling processes at several scales (from lab to pilot) and analysing the impact of TTRFID in recycling outcomes.
T4.1.3. TTRFID economic and societal analysis and assessment: T6.4 will provide complementary information for the required readiness for SC configuration for supplying the TTRFID and related ROI analysis. Moreover, the results of the societal study for acceptance of TTRFID-equipped products will also be conducted in T6.4, contributing to the completion of the requirements for the guidelines. They will integrate the technological assessment in T4.1.1, besides the outcomes of T4.1.2 and T4.1.3, as guidelines for designing and procuring TTRFID models. The guidelines will explain the required technological characteristics, the economic and feasibility analysis, and the criteria for fulfilling consumer data privacy. The guidelines will explain the processes and methodologies for integrating TTRFID tags into various textiles, such as different textile processing scenarios, including washing and heat exposure. This ensures that the TTRFID tags can be seamlessly integrated into manufacturing processes without compromising the quality or functionality of the textile products.

The results of the whole Task 4.1 will be reported in D4.1 and D4.2.

T4.2: Design of an eco-friendly blockchain architecture for the DPP with traceability of data over TVC ecosystem [CU] (M1-M12)
T4.2.1 Developing Guidelines for Assessment and Selection of Blockchain Platform: CU will develop guidelines for assessing and selecting blockchain platforms based on eco-friendliness, scalability, and support for smart contracts, as described in the methodology section. A comparative analysis of platforms like EBSI, Algorand, IOTA, and Ethereum will be conducted, considering factors such as energy consumption, transaction speed, cost and fulfilment with sustainability criteria for eco-friendly PoS consensus and support for smart contracts.
Sub-task 4.2.2 Developing guidelines for fulfilling the DPP solution with Semantic Web framework: CU will focus on the DPP's interoperability over blockchain with TVC IT solutions. The guideline will be extended to enhance data interoperability and machine readability over the Semantic Web framework. This involves guidelines for data model development using RDF & OWL. A shared vocabulary and data model will be created, ensuring all participants in the TVC ecosystem can uniformly interpret and utilise the data. Mappings to existing data formats (e.g. JSON, XML) will also be developed to facilitate smooth data integration.
T4.2.3 Developing guidelines for DPP Semantic Data Format and API Standardisation: CU will extend the DPP development guidelines to support standardised data formats and enable seamless communication between different IT systems within the TVC ecosystem. RESTful APIs requirements will be designed and documented, and data translation layer implementation will be included to ensure compatibility of ERP and MES systems. This will involve defining standardised protocols and data formats for real-time updates and data retrieval based on standards like IEC 62264: (ISA-95) and IEC 62443: (ISA -99), and ISO 27001 certification for data security. The integration will connect the DPP and PSSR technologies with existing ERP and MES systems. This connection will streamline operations by making data from the DPP accessible to relevant stakeholders within the TVC ecosystem.
T4.2.4 Developing guidelines for interoperability of DPP with IoT, TTRFID and Web3 Technologies: CU will extend the guideline to enable the communication of IoT and TTRFID technologies with the DPP blockchain system. The guidelines will detail the requirements for IoT-enabled sensors through TVC at key stages of the PSSR. The guidelines also include data container standards like XML headers to ensure data is structured in a standardised format. As detailed in the methodology section, the fulfilment of Web3 technologies will be included in guidelines to facilitate decentralised data exchange and interactions, ensuring secure and transparent operations.
T4.2.5 Developing the guidelines for DPP Scalability and Flexibility for TVC: CU will develop the required guidelines for the scalability of DPP over the TVC. This is to enable comprehensive data coverage of the whole TVC ecosystem. The guideline will define the requirements of the DPP model over blockchain technologies for supporting both the increase of volumes of data when the input of textiles clothes increases and when more stakeholders are joining the TVC. Flexibility will be incorporated into the system design for easy updates and adaptations to accommodate new sorting technologies, regulatory changes, and evolving industry practices. Scalability testing criteria will ensure the system's capability to efficiently process growing volumes of textiles, maintaining performance and reliability as demand increases.

The results of the whole Task 4.2 will be reported in D4.3.

T4.3: Integration of the DPP data retrieval with the automatised PSFS & PSSR solutions [CU, CEA, IRIS, IOM, REFA] (M13-M24)
T4.3.1 Developing APIs and Interface interoperability guidelines with PSFS & PSSR technologies: CU will extend the guideline for supporting the seamless connection and communication of technologies developed in PSFS & PSSR technologies with the DPP. This targets the DPP interoperability for data retrieval and communication with automated sorting machines. Efficient data exchange will be facilitated by utilising APIs and standardised data formats such as JSON or XML. This will allow for real-time updates and retrieval of textile data, ensuring that sorting machines have the necessary information for efficient processing.

T4.3.2 Developing Blockchain and Smart Contract Integration guidelines for bidirectional PSSR customised queries: CU will develop guidelines for enabling the design of customised queries over DPP Smart contracts. This includes the possibility of using the DPP smart contract semantic model schema for developing customised queries over different TVC data by using the holonic structure explained in the methodology section ranging from Garment, Yarn, Fabric, Filament, types like Cotton, Synthetic and also the Energy and CO2 emission data. The PSFS & PSSR technologies can develop customised queries to retrieve customised inquiries. The outcomes will be released through DPP smart contract guidelines. REFA, IOM, CEA, and IRIS will develop sample queries for sorting for reuse/recycling technologies in SORT4CIRC deployment, embed them into their data processing technologies, and test and validate the guidelines for integrity in deployment at WP5.

The results of the whole Task 4.3 will be reported in D4.4.

T4.4: Establishment of a tokenised ecosystem over TVC [CU] (M20-M34)
T4.4.1 Token Economy Scenario Design: CU designs the mechanisms for token issuance, distribution, and trading using the methodological approach explained. A comprehensive token economic scenario model will be developed to clarify how tokens are issued, distributed among participants, and traded within the TVC ecosystem. This model will also outline the incentive structures encouraging stakeholders through the TVC and consumers to adopt sustainable behaviours. The scenarios will be analysed from economic aspects, and the sustainability strategies, including energy and CO2 footprint tracking through the DPP model, will be examined to ensure the scalability and sustainability of the token circulation model.
T4.4.2 Token Management System Development: CU develops the guidelines for implementing smart contracts for token management through DPP. Smart contracts will be coded to manage token issuance, transfer, and deduction securely. Integration with the DPP blockchain will ensure that all transactions are transparent and immutable. Security measures will be put in place to protect the integrity of the token management system, as well as the scalability of the token model over TVC. The result includes the details of a token exchange over the TVC SC, covering the holistic methodological approach for including tokens including Garment, Yarn, Fabric, Filament, Cotton, Synthetic and even Energy and CO2 emission throughout the whole TVC.
T4.4.3. LCA, TEF interoperability framework with DPP token economy model: CU establishes guidelines for compatibility of ISO 14040 and ISO 14044 LCA assessment over the token economy model. This will enable the LCA assessment models to use the DPP token as a backbone for assessment. CU develops guidelines for enabling the design of customised queries over the DPP token model and semantic model schema for developing customised queries over different TVC data by using the holonic structure explained in the methodology section ranging from Garment, Yarn, Fabric, Filament, types like Cotton, Synthetic and also the Energy and CO2 emission data. T6.2 and T6.3 outputs will give basis to CU for validating customised queries and activating smart contract actuators for regulatory context mechanisms. This will enable the TVC stakeholder to conduct LCA/TEF assessment and simulate and analyse their token initiation and distribution strategies from a TEF assessment perspective. CU validates the guidelines by designing LCA/TEF assessment scenarios over the DPP token strategy for the WP5 pilot demonstration.

The results of the whole Task 4.4 will be reported in D4.5.

T4.5: Development of DfD solutions over the DPP model [RESO, CU, RECY, PURFI] (M18-M36)
T4.5.1 DfD Principles to eco-design new textile garments with Smart Stitch™ to ensure recyclability: RESO will gather requirements from RECY as well as use insights of previous projects and pilots to write preliminary eco-design guidelines for design-for-disassembly and design-for-recycling that will be used in T4.5.2. RECY will perform tests at lab and pilot scale with 3 families of feedstocks (PET/CO; PET/PA and PET/EA) provided by TXHO to identify the influence of Smart Stitch™ technology on the recycling of identified textiles. For each family, 3 experiments will be done: 1) feedstock alone without any pre-treatment (reference), 2) feedstock with a thermal treatment (done by RESO) simulating the machine required for the disassembly of garments including the RESO' technology) feedstock + Smart Stitch™ threads (provided by RESO). Analyses will be done to assess the quality of the obtained monomer and the co-materials to evaluate the impact of the thermal process. The results from the tests done by RECY will be used to define preliminary guidelines taking into account recyclers' needs.
T4.5.2 DfD Principles Extractor Guideline over DPP: CU creates comprehensive guidelines for using DPP historical data over PSSR results. As explained in T4.3.2, the PSSR technologies will have a bi-directional interoperability with DPP. This enables the capture of PSSR results in DPP. CU will develop data analytical guidelines to extract insights about clothes' sorting and recycling compatibilities. The guidelines will result in the extraction of DfD principles. RESO will participate in the drafting of the guidelines by leveraging the preliminary guidelines defined in T4.5.1 for DfD rules, which focus on active disassembly methods such as thermally trigger stitching threads and adhesives as well as other non-permanent fastening methods and how they have improved recyclability. This task aims to provide comprehensive guidelines to enhance the recyclability of textile products and explain how DPP is crucial to enabling it. The principles can be used for researching best practices and consulting with industry experts to draft guidelines that prioritise the

principles, like how the use of non-permanent fastening methods, such as screws, clips, or snaps, instead of adhesives or stitches can improve the recyclability. The guidelines will also cover layering strategies where different materials can be easily separated, ensuring that each layer can be processed individually during recycling. The guidelines will be validated with PURFI.
T4.5.3 DfD Technology Development and Integration with DPP: RESO analyses tools and technologies for assisting textile disassembly, like heat-dissolvable bio-based and synthetic stitching threads. This also includes innovating disassembly tools that are efficient and cost-effective and help optimise the time for disassembly and recycling. Research into self-disassembling materials, which change properties under specific conditions, can facilitate easier disassembly will be conducted. Guidelines for detecting such disassembling materials and related technologies will be embedded into the DPP model for effective PSSR processes by CU. This will be verified by successfully recognising DfD-enabled textile products through PSSR processes in WP5.

The results of the whole Task 4.5 will be reported in D4.6 and D4.7.

Work package WP5 – Demonstrator

Work Package Number	WP5	**Lead Beneficiary**	15 - TXHO
Work Package Name	Demonstrator		
Start Month	19	**End Month**	36

Objectives

• Develop a comprehensive layout design for the pre-sorting pilot, integrating technologies.
• Set up the pilot plan adjacent to the manual pre-sorting system and run a test using an offline bypass to ensure independent operation.
• Replace the manual pre-sorting system with the pilot plant after successful testing and validation.
• Establish a data connection to integrated operations' higher-level plant control system.
• Equip the system with conveyor belts, stable frames for the mounting measuring devices, and RFID detectors.
• Demonstrate the operational capabilities of the advanced sorting line and provide ongoing support for technology integration and measurement geometries.
• Develop a token pilot model and demonstrate how textile product separation results in taken circularity.
• Coordinate with recyclers to provide sufficient quantities of sorted feedstocks for testing.

Description

T5.1: Demonstration of an automatised PSFS pilot [SEM, TXHO, IRIS, CEA, REFA] (M19-M36)
As gravity is used to transport the material from the pre-sorting line to the sorting line, the developed machine must be supported on a suitable metal frame so that the containers can be positioned under the outlet of the discharge chutes. SEM will design and build this metal frame, considering the spatial conditions of the factory halls at the selected plant. TXHO will support the machine manufacturer in the design of the automatic pre-sorting pilot. TXHO will provide plans for the halls, production facilities, and infrastructure for integrating the demonstrator into the existing sorting line. TXHO will draw up a specification sheet for the demonstrator and provide the machine builder with ongoing advice during the design phase.
TXHO will first draw up a diagram of this sorting process so that the mechanical engineers can work out the basic methods for automated sorting.
REFA will provide access to its automated sorting software system to detect various product attributes such as product type, wear condition, style, fabric construction, brand, colour, and size. Furthermore, the system will enable sorters to define feedstock requirements for each fraction via a management cockpit, allowing garments to be sorted according to a predefined ruleset. Additionally, REFA will provide the endpoint for the SKU retrieval system developed in WP2 for PSFS. The software system will also be capable of retrieving and processing DPP data.
SEM, TXHO, and REFA will conduct a feasibility study to explore how the feeding process of the sorting line, specifically for unfolding garments, can be accelerated. While automation of the feeding process is a possibility, it must not compromise the high sorting quality standards required by second-hand buyers. SEM, IRIS, and CEA will analyse the sorting scheme, jointly determine a suitable arrangement for the measurement technology, and develop the layout for the pilot plant. A suitable position for the test run in bypass operation can be determined based on the hall plans and the dimensions of the production systems installed there.

After successful testing, SEM and TXHO will integrate the pilot plant into the production line with its maintenance department and establish a connection to the higher-level plant control system.

The results of the Task 5.1 will be reported in D5.3.

T5.2: Upgrade, integrate and demonstrate the advanced sorting technologies for recycling into an operational automatised sorting line [TXHO, SEM, IRIS, CEA, IOM] (M25-M36)
The Multirange Spectroscopic Platform developed in WP3 by IRIS, which incorporates AI models from IRIS, IOM and CEA, will include the most relevant optical techniques for assessing the garments' composition. The device consists of a real-time monitoring system prototype that will be coupled to the current TXHO mechanical sorting system. SEM must only establish a data connection to the Supervisory Control and Data Acquisition (SCADA) systems. TXHO will support SEM, IOM, IRIS and CEA in integrating optical measurement technologies into the existing sorting plant and defining the measurement geometries. Additionally, CEA and IRIS will conduct a regression test to validate the performance of the demonstrators developed in WP2-3 on the sorting line.

The results of the Task 5.2 will be reported in D5.4.

T5.3: Validation of SORT4CIRC DPP blockchain for storage/retrieval of TVC [CU, SEM, TXHO, IOM, CEA, RESO] (M25-M34)
T5.3.1. TTRFID technology deployment in pilot demonstrator: CU will adopt TTRFID tags using the results of Del N°: 4.1, 4.2 and will deploy demonstration textile samples with attached TTRFIDs. The consolidation of TTRFID infrastructure with PSSR technologies will be assessed and validated by TXHO and deployed in demonstrator.
T5.3.2. DPP blockchain technology deployment and integration in pilot demonstrator: CU will adopt the guidelines for DPP blockchain architecture using the results of Del N°: 4.3, 4.4 and will deploy an eco-friendly PoS consensus and smart contract supporting structure. The blockchain architecture for DPP will be empowered by leveraging Semantic Web technologies to enhance data interoperability and machine readability across other technologies in the demonstrator. CU will define the data model using RDF and OWL and fulfil the standardised APIs and data formats to enable seamless communication between different technology systems within the TVC ecosystem. The DPP blockchain will be integrated with TTRFID, and the validation of seamless communication result will be a blockchain DPP token model integrated with TTRFID in the demonstration pilot. The DPP then will be validated for interoperability with the demonstrator pilot PSFS & PSSR technologies. SEM will test their sample queries for pre-sorting technologies in the SORT4CIRC demonstrator and validate the integrity of effective pre-sorting. REFA, IOM, CEA and IRIS will test their queries for sorting for recycling technologies in SORT4CIRC deployment embed and validate the effective distinguishing of the textile parts. RESO will test the DPP to recognise heat-dissolvable bio-based and synthetic stitching threads. SEM will integrate the results through the demonstrator pilot in terms of embedment of PSFS & PSSR queries with the DPP model and successful distinguishing of TTRFID-equipped textile products.
T5.3.3. DPP tokenisation realisation and validation in pilot demonstrator: CU will develop the token pilot model. The PSFS & PSSR technology bidirectional communication results will be used to shape the demonstration of how the distinguishing and separation of textile products will result in token circularity. The scenarios released in Del N°: 4.5 will simulated, and the token management system will be demonstrated. CU will present an autonomous and interoperable LCA assessment framework over the DPP tolerised demonstrator.

The results of the whole Task 5.3 will be reported in D5.1 and D5.2.

T5.4: Validation of SORT4CIRC sorted feedstock for high-value recycling at recyclers facilities [RECY, TXHO, PURFI] (M30-M36)
TXHO conducts sorting tests and makes the sorted materials available to all participating recycling companies for analysis and validation. The goal is to integrate the sorted feedstock into existing high-value recycling processes at the recyclers' facilities. TXHO will monitor and document the performance of the feedstock in various recycling operations, such as thermo-mechanical, chemical recycling, and fabric regeneration, and compare the performance of SORT4CIRC sorted feedstock with standard feedstock to identify differences in efficiency, yield, and product quality. The results from the performance validation will be analysed to pinpoint areas for improvement in the sorting process. Detailed feedback will be provided to the SORT4CIRC development team on the performance of the sorted feedstock. Additionally, TXHO will collaborate with the machinery builder, IOM, CEA, and IRIS to implement necessary adjustments and optimise the sorting parameters for better results.
Assuming the capacity of the current textile sorting plant is not expanded during the project period and the material composition of the garments for recycling remains the same, TXHO could deliver around 10% of the total sorted material to the recyclers for validation.
PURFI will run tests on 3 families of feedstock provided by TXHO (CO/PES, Cotton, Polyester). The first test will determine if the feedstock meets the requirements for closed loop mechanical recycling. Feedback will be provided to TXHO. After approval of test 1 PURFI can go into further testing to see if the desired outcome can be reached.

RECY will run text on 4 families of feedstocks provided by TXHO (PET/CO; PET/PA; PET/EA; multilayers with enough PET). For each family of feedstock, 2 test runs will be done by RECY at different scales (from lab to pilot or industrial demonstrator if relevant)1st: feedback to TXHO. 2nd test: 2nd runs to see if what does not work at the 1st has been corrected.) Analyses will be done on monomers and co-materials to validate that the sorted feedstocks fit with our recycling process.

The results of the Task 5.4 will be reported in D5.5.

Work package WP6 – Technical and economic feasibility, environmental and social assessment, design and efficiency improvement

Work Package Number	WP6	**Lead Beneficiary**	11 - PSB
Work Package Name	Technical and economic feasibility, environmental and social assessment, design and efficiency improvement		
Start Month	1	**End Month**	36

Objectives

• Determine the LCI of the whole value chain.
• Assess the economic and environmental impact of the textile sorting process automation.
• Identify and mitigate potential risks associated with the automation project, such as technical challenges, market demand fluctuations, and regulatory compliance.
• Evaluate the viability of the CE BM, industrial symbiosis among recyclers, along with social acceptance and consumer welfare.
• Optimise the design and operation of the automated sorting system.

Description

Description of work
T6.1: SORT4CIRC data collection [ARIS, PSB, ALL] (M3-M15)
T6.1.1: Process flow-sheeting, inventory data collection and preliminary economic potential and GHG emissions calculations: This task aims to conceptually determine and illustrate the SORT4CIRC process and its value chain. ARIS will develop a flowsheet depicting the sequence of process operations and equipment units, material streams (e.g., raw materials, products, by-products, wastes) and operating conditions (e.g., temperature, pressure, flow rate). The mass and energy balances (LCI) of both the existing (state-of-the-art) and the SORT4CIRC proposed process will be calculated based on data attained from WPs 2-5 and will provide the basis for estimating all the inventory data required for the techno-economic and environmental impact assessments (Tasks 6.2 and 6.3). A preliminary estimation of the economic potential and the environmental burden will also be performed, and the results will be used in T4.1.3 and T4.4.3, linked to D.4.2 and D.4.5, respectively, to facilitate a technology screening.
T6.1.2: Mapping the current textile forward and reverse SC network, physical, information, and financial flows across the EU, including demand and supply side specifications for each stakeholder and connection between them: PSB will (1) map the existing post-consumer value chain of the textile products by identifying the involved entities and stakeholders (consumers, collectors, recyclers, industrial users, second-hand markets, charity shops) as well as existing physical, information and financial flows and exploring potential supply points, and demand markets that can be possibly integrated into the value chain after introducing/implementing the new sorting system; and will (2) identify the existing BMs and practices in the textile reuse and recovery industry and their current effectiveness. All partners will provide necessary and detailed information concerning material, information, and financial flow, as well as existing supply and collection network.

The results of the whole Task 6.1 will be reported in D6.1.

T6.2: Environmental impact assessment [ARIS, ALL] (M15-M34)
To estimate the environmental performance of SORT4CIRC process, ARIS will initially perform an ex-ante, attributional, gate-to-gate LCA study, according to ISO 14040/44 and 14025, to assess the impact of the deployment of the new technologies on the existing sorting process. For this analysis, the system boundaries will include only the sorting processes. Subsequently, ARIS will perform an ex-ante, consequential cradle-to-cradle LCA study, according to ISO 14040/44 and 14025, to assess the consequential benefits imposed by the new technologies along the whole value chain of

the textile recycling process. For this analysis, the system boundaries will include all the foreground processes developed in WP2-4 of the SORT4CIRC and downstream (distribution) processes, including waste treatment and all the auxiliary (background) processes needed for the material/energy supply. LCI derived from Task 6.1, will be the basis for the LCA. The environmental impact, emphasising GHG emissions (kg CO2-eq) and water use (m3 depriv.), will be evaluated based on the Environmental Footprint v3.1 method of the EU Commission. Other relevant environmental impacts that are directly related to (i) minimisation of wastewater stream production (freshwater ecotoxicity and eutrophication), (ii) optimisation of resource use efficiency (resource use, fossils, minerals and metals); and (iii) volatile organic compounds avoidance (i.e. ozone depletion, photochemical ozone formation, particulate matter and human toxicity), will also be estimated to assess the global environmental burden of the SORT4CIRC. To determine the reliability of the obtained impact results, uncertainty analysis – Monte Carlo random sampling on inventory data probability distributions coupled with multiple iterations for LCA calculations – will be applied. All partners will validate the relevant assumptions concerning the system boundaries. The results will be summarised in D6.2, highlighting the environmental performance of SORT4CIRC and will be used for the benchmarking in T6.6.

The results of the Task 6.2 will be reported in D6.2.

T6.3: Cost-benefit analysis of SORT4CIRC [ARIS, ALL] (M15-M34)
Within this task the TEA of SORT4CIRC will be performed by ARIS. All partners will provide additional data and validate relevant assumptions concerning the pieces of equipment and the system boundary limits. ARIS will perform a cost-benefit analysis based on the LCI obtained in T6.1. Particularly, the operating cost of the sorting process (i.e., labour, capital depreciation, raw materials and utilities, etc.) will initially be estimated. Further, based on the anticipated sales revenue, the economic benefit of SORT4CIRC will be estimated in terms of profit margin, ROI, and payout time. The cost-benefit analysis will determine the SORT4CIRC financial viability and steer it towards a more cost-effective approach. The results will be summarised in D6.3 and benchmarked against the state-of-the-art process in T6.6.

The results of the Task 6.3 will be reported in D6.3.

T6.4: BM evaluation, industrial symbiosis, social acceptance and consumer welfare [PSB, ALL] (M1-M36)
This task evaluates the BMs for SORT4CIRC, focusing on industrial symbiosis, social acceptance, and consumer welfare. It aims to develop viable CE BMs that foster collaboration among producers, collectors, recyclers and other stakeholders. This task addresses the broader business and social implications of the project, as follows: CE Model: PSB evaluates BMs for the CE, with a focus on revenue generation strategies, SC management, and market demand for re-wearable and recycled textiles. Additionally, PSB assesses the impact of introducing DPP technologies in textiles, as well as the effects of EPR regulations on these areas. Collaboration for Efficiency: PSB analyses opportunities for industrial symbiosis, where different companies in the recycling sector can collaborate to enhance resource efficiency and potentially reduce costs. Public Perception: PSB assesses the social acceptance of using DPP technologies in clothes (e.g., TTRFID-equipped clothes) required as an input for WP4, and consumer attitudes towards re-wearable and recycled textiles. Consumer Well-being: PSB evaluates the impact of new processes developed by SORT4CIRC on consumer welfare. This includes the quality, availability, and affordability of re-wearable and recycled textile products. While the initial two points are addressed through analytical techniques, such as game-theoretical approaches, the remaining two are investigated mainly using empirical research methods. All partners provide additional data required for the modelling and survey design process, review and verify the validity of the models and surveys designed and assess the credibility of the results.

The results of the Task 6.4 will be reported in D6.4.

T6.5: Design and operation optimisation of the automated sorting system [PSB, ARIS, TXHO] (M12-M36)
This task aims to optimise the design of the sorting system and to ensure a smooth workflow and operations throughout the system. This task involves the development of mathematical models and heuristic approaches to tackle the main design problems, including equipment selection, staff requirements, facility layout planning, and line balancing. This task also includes the development and validation of a fine-grained simulation model using FlexSim© simulation software that allows us to imitate the behaviour of the real system considering detailed system characteristics and operations as well as dynamic and stochastic elements, which helps us measure the system's KPIs (e.g., throughput, time in the system, etc.), and identify potential bottlenecks, all lead to decide whether the designed system is efficient. We test the system's performance under various conditions (e.g., supply fluctuation, etc.), and also, based on the insights obtained from the simulation model, we test selected scenarios that could improve the system's efficiency using the developed simulation model. We also perform comparative studies using simulation modelling to evaluate the existing sorting system against the automated, technology-integrated system SORT4CIRC aims to create. ARIS provides the process flow data collected in T6.1.1. TXHO provides initial design and operations instructions as well as all necessary data that allows us to model the manual and automated sorting processes. TXHO also verifies the validity of models and results and provides input to create and test new scenarios for design and operations improvement.

The results of the Task 6.5 will be reported in D6.5.

T6.6: Technologies benchmarking [ARIS, WP Leaders] (M18-M36)
Herein, ARIS will benchmark the developing textile sorting process against its state-of-the-art counterparts with all WP leaders in support. It will indicate (i) the maximum sorting cost this new sorting process can reach and still be competitive and (ii) the positioning of the SORT4CIRC process against the competitors in terms of cost, environmental impact and technical performance.

The results of the Task 6.6 will be reported in D6.6 and D6.7.

Work package WP7 – Communication, Dissemination, and Exploitation

Work Package Number	WP7	**Lead Beneficiary**	7 - CEF
Work Package Name	Communication, Dissemination, and Exploitation		
Start Month	1	**End Month**	36

Objectives

• Raise awareness of the SORT4CIRC project
• Implement training sessions/webinars tailored to each WP theme, key challenges and needs
• Disseminate the results and lessons learnt to policymakers, project stakeholders and other interested parties to shape future policies and research agendas on circular textiles.

Description

T7.1: Communication, Dissemination and Exploitation activities [CEF, CGS, TRA, ALL] (M1-M36)
Together with all SORT4CIRC partners, CEF and TRA will develop a Communication, Dissemination and Exploitation (CDE) Plan to ensure the successful exploitation of the SORT4CIRC results and to identify opportunities for synergy and guide SORT4CIRC communication and dissemination activities relevant to the project, monitoring procedures and the sustainability of the project beyond its lifecycle. The Key Exploitable Results, as defined in Table 4, will be in responsibility of the individual partners or WPs and the strategy for utilization and exploitation of results and all resulting exploitation activities will be included in the CDE. Considering the three-year project duration, necessary adjustments will be made to the plan on a regular basis, taking into account the evaluation of outreach and communication analytics, as well as successes, lessons learnt and future opportunities that may have emerged since SORT4CIRC's inception as reported by all project partners. A monthly communications meeting between the partners will be held to keep each other up to date with communications activities and communications opportunities.
Along with the Plan, a Brand Book will be established by CEF at the project outset to articulate SORT4CIRC's visual identity, including logos, colour codes, images, templates (i.e. for web banners, social media visuals, flyers, posters, etc), key messages, and promotional text. CEF will prepare promotional and communication materials (e.g. SORT4CIRC brochures, visual materials for social media, short promotional videos, tailored presentations, flyers, etc.) to support the promotional and communication actions throughout the project. The format of the materials will be adapted to the diverse promotional and communication channels to be used. CGS will provide communications (i.e. social media content creation and management) and design support (i.e. visualisations, infographics, website design) to this task.
To ensure effective dissemination, the project will rely on continuous dissemination by maintaining an online presence and participating in existing events to regularly update external stakeholders on project progress, as well as punctual communications campaigns designed to give a boost to SORT4CIRC's dissemination efforts at crucial moments in the project. Project communication and dissemination activities will be implemented via different platforms and channels (i.e. SORT4CIRC website, social networks, media platforms, etc). Social media posts will be shared on SORT4CIRC and partners' social media accounts, while the news item will be pitched to relevant media and published on SORT4CIRC website. An opinion article will also be pitched to relevant media and/or published on SORT4CIRC website. Relevant media outlets focus on sustainability, SC, and recycling. For effective communication and advocacy, SORT4CIRC will produce a newsletter, and social media posts every year.
A dedicated website for SORT4CIRC will be set up in M3 by CEF, which is in line with the DMP. Project partners will develop regular content to be uploaded to the website through short blogs, news, key updates on the project, and other content types. The website will integrate a newsletter signup feature to give the public exclusive access to SORT4CIRC project updates. Following the project, the SORT4CIRC website will be assessed if potential improvements to its features are needed following the development of SORT4CIRC's activities.

Together with all SORT4CIRC partners, CEF and TRA will contribute to clustering activities (joint kick off, joint events or final events) with other EU projects selected under the same topic. Such activities will aim at identifying synergies, exchanging good practices and providing evidence for EU policy developments. The common activities might include joint dissemination and/or exploitation activities, as well as coordinated deliverables related to policy recommendations.

The results of the Task 7.1 will be reported in D7.1, D7.2, D7.3, D7.4, D7.6, D7.7, D7.8 and D7.11.

T7.2: Development of knowledge and policy support [CEF, CGS,ALL] (M1-M36)
To ensure that the outcome and impact of SORT4CIRC can also be translated beyond the project timeline, CEF will consolidate all content materials and lessons learnt with the inputs of all project partners throughout the project into a comprehensive, user-friendly toolbox. CGS will provide communications and design support to this tool. This toolbox will ensure that the best practices and approaches implemented in this project are transferable and can be applied by European businesses, start-ups, service providers, as well as academia and public institutions in the future when supporting the establishment and strengthening of a just and transparent CE for textiles in Europe. For future CE projects, the toolbox will be designed to help them avoid duplication of efforts, and build on existing best practices, speed up the knowledge translation process and contribute to the existing body of work on this subject matter as well as inform real-life circular practices. A launch event will be held to further disseminate the toolbox beyond the SORT4CIRC community. A set of policy briefing documents will be dedicated to European, national and regional policymakers with recommendations for support measures promoting the deployment of circular textile solutions. The recommendations will aim to improve existing policy frameworks that impact the CE and will be forward-looking to shape future research agendas and policy measures impacting circular solution implementation. The document will be anchored in results and findings from SORT4CIRC's WP2-6 activities.

The results of the Task 7.2 will be reported in in D7.5 and D7.9.

T7.3: Complementary knowledge transfer [CEF, CGS, TRA, ALL] (M1-M36)
Furthermore, a comprehensive knowledge sharing and training of five 90-minute webinars/training sessions will be developed by CEF in collaboration with CGS communication and design experts. The emphasis of each session will be based on needs expressed by project actors and the experience of project actors and include findings and learnings from WP2-WP6. While the content of the webinars will be tailored to the challenges and learnings/insights/interests expressed by SORT4CIRC project actors, other interested parties and other partners within the CCRI network are welcome to join and participate. The content materials used in the webinars will be shared with the project actors, and the webinar will be recorded, edited and shared as an additional resource uploaded to the SORT4CIRC website for future use. All content materials and webinars will be delivered in English. All project partners will support the capacity-building program.
CEF will coordinate with the CCRI to align communication efforts. To ensure SORT4CIRC's visibility, events where the project can be presented to encourage cross-promotion and cross-pollination with other actors within the CE sector will be identified in the Plan. Examples include the 'Textiles Ecosystem Stakeholders' event organised by the European Commission's DG GROW, the 'Global Fashion Summit' organised by the Global Fashion Agenda, or the European CE Stakeholder Platform's Textile Leadership Group events where CEF is a member and strategic partner. CEF is also an Associated Partner to CCRI's Coordination and Support Office, contributing to multiple Thematic Working Groups such as the Circular Resource Management group (which covers textiles), as well as a co-host of the World CE Forum. We expect to attend/participate in two external events per year (six events total) to increase SORT4CIRC's visibility and foster networking opportunities within the project partners' network and externally.
Create a standard of pre-sorting categories used by more operators with more than 20% market share. Using TRA's leverage with the CEO being the General Delegate of the Bureau of International Recycling's textiles division and a board member of the textiles division at the European Recycling Industries Confederation, TRA will work with WP partners first to deliver the framework for this categorisation, then using their extensive European network liaise with major European sorters to seek the necessary consensus on delivering this objective. Various cocreation/ participatory approaches will be adopted, including online and in-person workshops, one-to-one meetings, surveys and other appropriate means. TRA will also assist in providing WP partners with necessary garments to establish the data repository and facilitate improving garment recognition and worn-out garments where necessary.

The results of the Task 7.3 will be reported in D7.10.

STAFF EFFORT

LIST OF DELIVERABLES

Deliverables

Grant Preparation (Deliverables screen) — Enter the info.

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Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
D1.1	Project Management Plan	WP1	1 - CU	R — Document, report	SEN - Sensitive	3
D1.2	Data Management Plan	WP1	1 - CU	DMP — Data Management Plan	PU - Public	6
D1.3	Project meeting minutes and decision records M1-M18	WP1	1 - CU	R — Document, report	SEN - Sensitive	18
D1.4	Data Management Plan M18	WP1	1 - CU	DMP — Data Management Plan	PU - Public	18
D1.5	Project meeting minutes and decision records M18-M24	WP1	1 - CU	R — Document, report	SEN - Sensitive	24
D1.6	Project meeting minutes and decision records M24-M36	WP1	1 - CU	R — Document, report	SEN - Sensitive	36
D1.7	Data Management Plan M36	WP1	1 - CU	DMP — Data Management Plan	PU - Public	36
D2.1	Development of a complete machine for PSFS	WP2	9 - SEM	DEM — Demonstrator, pilot, prototype	SEN - Sensitive	12
D2.2	Sorting system for PSFS	WP2	9 - SEM	DEM — Demonstrator, pilot, prototype	SEN - Sensitive	20

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Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
D2.3	Automated detection for a PSFS garment sorting system	WP2	14 - REFA	DEM — Demonstrator, pilot, prototype	SEN - Sensitive	24
D2.4	Usage Guidelines	WP2	14 - REFA	R — Document, report	PU - Public	24
D2.5	Summaries of Learning and Deployment Insights	WP2	14 - REFA	R — Document, report	PU - Public	24
D3.1	Report on characterisation of textile samples	WP3	15 - TXHO	R — Document, report	PU - Public	6
D3.2	Report on the selection of best techniques for the recognition of complex garments	WP3	8 - CEA	R — Document, report	SEN - Sensitive	18
D3.3	Multirange Spectroscopic smart sensing system	WP3	13 - IRIS	DEM — Demonstrator, pilot, prototype	SEN - Sensitive	24
D3.4	Eco-design guidelines and report on the process to recognise DfD garments	WP3	5 - RESOR	R — Document, report	PU - Public	24
D3.5	Tests report	WP3	4 - RECY	R — Document, report	SEN - Sensitive	24
D4.1	TTRFID assessment	WP4	1 - CU	R — Document, report	SEN - Sensitive	6
D4.2	TTRFID design and selection guideline	WP4	1 - CU	R — Document, report	SEN - Sensitive	12
D4.3	DPP development guidelines	WP4	1 - CU	R — Document, report	PU - Public	12
D4.4	DPP smart contract guidelines for PSFS & PSSR compatibility	WP4	1 - CU	R — Document, report	PU - Public	24

Deliverables

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Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
D4.5	Tokenised Ecosystem framework analysis and development guideline	WP4	1 - CU	R — Document, report	SEN - Sensitive	30
D4.6	DfD solution development over DPP framework	WP4	5 - RESOR	DEM — Demonstrator, pilot, prototype	SEN - Sensitive	36
D4.7	Guidelines for recyclability and reusability	WP4	5 - RESOR	R — Document, report	PU - Public	36
D5.1	TTRFID technology deployment	WP5	1 - CU	DEM — Demonstrator, pilot, prototype	SEN - Sensitive	28
D5.2	DPP Blockchain demonstration deployment with tokenised LCA assessment dashboard	WP5	15 - TXHO	DEM — Demonstrator, pilot, prototype	SEN - Sensitive	34
D5.3	Automatised PSFS pilot	WP5	9 - SEM	DEM — Demonstrator, pilot, prototype	SEN - Sensitive	36
D5.4	Upgrade Automatised Sorting Line	WP5	15 - TXHO	DEM — Demonstrator, pilot, prototype	SEN - Sensitive	36
D5.5	Validation of Sorted Feedstock	WP5	4 - RECY	DEM — Demonstrator, pilot, prototype	SEN - Sensitive	36
D5.6	Successful recycling of sorted feedstock	WP5	4 - RECY	R — Document, report	PU - Public	36
D6.1	Data collection for SORT4CIRC	WP6	3 - ARIS	R — Document, report	SEN - Sensitive	15
D6.2	Report on environmental impact assessment	WP6	3 - ARIS	R — Document, report	SEN - Sensitive	34
D6.3	Report on cost-benefit analysis	WP6	3 - ARIS	R — Document, report	SEN - Sensitive	34

Deliverables

Grant Preparation (Deliverables screen) — Enter the info.

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Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
D6.4	Report on the BM settings	WP6	11 - PSB	R — Document, report	SEN - Sensitive	36
D6.5	Report on the efficient design and operation of the automated sorting system	WP6	11 - PSB	R — Document, report	SEN - Sensitive	36
D6.6	Holistic technology benchmarking among different textile sorting processes	WP6	3 - ARIS	R — Document, report	PU - Public	36
D6.7	Public report on environmental impact assessment	WP6	3 - ARIS	R — Document, report	PU - Public	34
D7.1	Communication, Dissemination, and Exploitation Plan M3	WP7	7 - CEF	R — Document, report	PU - Public	3
D7.2	SORT4CIRC Brand Book	WP7	7 - CEF	R — Document, report	SEN - Sensitive	3
D7.3	SORT4CIRC website	WP7	7 - CEF	DEC —Websites, patent filings, videos, etc	PU - Public	4
D7.4	Communication, Dissemination, and Exploitation Plan M12	WP7	7 - CEF	R — Document, report	PU - Public	12
D7.5	SORT4CIRC policy briefing document M18	WP7	7 - CEF	R — Document, report	PU - Public	18
D7.6	Communication, Dissemination, and Exploitation Plan M36	WP7	7 - CEF	R — Document, report	PU - Public	36
D7.7	SORT4CIRC circular textiles toolbox	WP7	7 - CEF	R — Document, report	PU - Public	36
D7.8	SORT4CIRC policy briefing document M36	WP7	7 - CEF	R — Document, report	PU - Public	36

Deliverables

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Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
D7.9	Report on capacity building webinars	WP7	7 - CEF	R — Document, report	SEN - Sensitive	36
D7.10	Summary of clustering activties	WP7	7 - CEF	R — Document, report	PU - Public	36

Deliverable D1.1 – Project Management Plan

Deliverable Number	D1.1	**Lead Beneficiary**	1 - CU
Deliverable Name	Project Management Plan		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	3	**Work Package No**	WP1

Description
Setting up a quality management policy covering all aspects related to practical management (document exchanges & internal workflows) of the project. (related to task 1.4).

Deliverable D1.2 – Data Management Plan

Deliverable Number	D1.2	**Lead Beneficiary**	1 - CU
Deliverable Name	Data Management Plan		
Type	DMP — Data Management Plan	**Dissemination Level**	PU - Public
Due Date (month)	6	**Work Package No**	WP1

Description
Establishment of procedures describing how project data will be handled and what data will be collected, processed or generated, methodologies and standards followed regarding sharing, duration, and preservation, ensuring full compliance with Open science policy (related to task 1.4).

Deliverable D1.3 – Project meeting minutes and decision records M1-M18

Deliverable Number	D1.3	**Lead Beneficiary**	1 - CU
Deliverable Name	Project meeting minutes and decision records M1-M18		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	18	**Work Package No**	WP1

Description
Preparation, distribution and approval of minutes of meetings incl. agenda, decision records, action plan (related to task 1.1-3).

Deliverable D1.4 – Data Management Plan M18

Deliverable Number	D1.4	**Lead Beneficiary**	1 - CU
Deliverable Name	Data Management Plan M18		
Type	DMP — Data Management Plan	**Dissemination Level**	PU - Public
Due Date (month)	18	**Work Package No**	WP1

Description
Establishment of procedures describing how project data will be handled and what data will be collected, processed or generated, methodologies and standards followed regarding sharing, duration, and preservation, ensuring full compliance with Open science policy (related to task 1.4).

Deliverable D1.5 – Project meeting minutes and decision records M18-M24

Deliverable Number	D1.5	**Lead Beneficiary**	1 - CU
Deliverable Name	Project meeting minutes and decision records M18-M24		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	24	**Work Package No**	WP1

Description
Preparation, distribution and approval of minutes of meetings incl. agenda, decision records, action plan (related to task 1.1-3).

Deliverable D1.6 – Project meeting minutes and decision records M24-M36

Deliverable Number	D1.6	**Lead Beneficiary**	1 - CU
Deliverable Name	Project meeting minutes and decision records M24-M36		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	36	**Work Package No**	WP1

Description
Preparation, distribution and approval of minutes of meetings incl. agenda, decision records, action plan (related to task 1.1-3).

Deliverable D1.7 – Data Management Plan M36

Deliverable Number	D1.7	**Lead Beneficiary**	1 - CU
Deliverable Name	Data Management Plan M36		
Type	DMP — Data Management Plan	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP1

Description
Establishment of procedures describing how project data will be handled and what data will be collected, processed or generated, methodologies and standards followed regarding sharing, duration, and preservation, ensuring full compliance with Open science policy (related to task 1.4).

Deliverable D2.1 – Development of a complete machine for PSFS

Deliverable Number	D2.1	**Lead Beneficiary**	9 - SEM

Deliverable Name	Development of a complete machine for PSFS		
Type	DEM — Demonstrator, pilot, prototype	**Dissemination Level**	SEN - Sensitive
Due Date (month)	12	**Work Package No**	WP2

Description
Development of an all-in-one solution for the new type of pre-sorting and advanced sorting technologies, integrating multirange spectroscopic smart sensing, data fusion capabilities, and automated material for optimised textile sorting process (related to task 2.3).

Deliverable D2.2 – Sorting system for PSFS

Deliverable Number	D2.2	**Lead Beneficiary**	9 - SEM
Deliverable Name	Sorting system for PSFS		
Type	DEM — Demonstrator, pilot, prototype	**Dissemination Level**	SEN - Sensitive
Due Date (month)	20	**Work Package No**	WP2

Description
Development of an automatic sorting system for pre- and fine sorting that uses camera technology and AI methods to determine the type and quality of garments and sort them into defined fractions using a suitable discharge module (related to task 2.2).

Deliverable D2.3 – Automated detection for a PSFS garment sorting system

Deliverable Number	D2.3	**Lead Beneficiary**	14 - REFA
Deliverable Name	Automated detection for a PSFS garment sorting system		
Type	DEM — Demonstrator, pilot, prototype	**Dissemination Level**	SEN - Sensitive
Due Date (month)	24	**Work Package No**	WP2

Description
Development of a sorting software system that automatically detects product attributes such as product type, wear condition, fabric construction, brand, colour and sizes, with a SKU retrieval system enabled by similarity detection for integrating DPP and e-commerce information into the sorting process (related to task 2.1).

Deliverable D2.4 – Usage Guidelines

Deliverable Number	D2.4	**Lead Beneficiary**	14 - REFA
Deliverable Name	Usage Guidelines		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	24	**Work Package No**	WP2

Description

A set of usage guidelines will be developed to provide instructions on system setup, configuration, and operation in a textile sorting plant. These guidelines will cover key steps for deployment, hardware and software requirements, and best practices for integration to enhance efficiency and accuracy while minimizing disruptions (related to task 2.1).

Deliverable D2.5 – Summaries of Learning and Deployment Insights

Deliverable Number	D2.5	**Lead Beneficiary**	14 - REFA
Deliverable Name	Summaries of Learning and Deployment Insights		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	24	**Work Package No**	WP2

Description
A report summarizing key findings from testing and deployment will be delivered. It will highlight performance benchmarks, discuss challenges encountered, and provide troubleshooting strategies and solutions to refine future iterations and optimize textile sorting (related to task 2.1).

Deliverable D3.1 – Report on characterisation of textile samples

Deliverable Number	D3.1	**Lead Beneficiary**	15 - TXHO
Deliverable Name	Report on characterisation of textile samples		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	6	**Work Package No**	WP3

Description
A report will be made on the characterisation of the textile samples that will be used for the development of recognition techniques (related to task 3.1).

Deliverable D3.2 – Report on the selection of best techniques for the recognition of complex garments

Deliverable Number	D3.2	**Lead Beneficiary**	8 - CEA
Deliverable Name	Report on the selection of best techniques for the recognition of complex garments		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	18	**Work Package No**	WP3

Description
Selection of best techniques for the recognition of complex garments for the identification of black textiles, low content material detection, chemical contaminants, coatings, fabric structure and multilayer garments (related to task 3.2).

Deliverable D3.3 – Multirange Spectroscopic smart sensing system

Deliverable Number	D3.3	**Lead Beneficiary**	13 - IRIS
Deliverable Name	Multirange Spectroscopic smart sensing system		

Type	DEM — Demonstrator, pilot, prototype	**Dissemination Level**	SEN - Sensitive
Due Date (month)	24	**Work Package No**	WP3

Description
Development of self-contained inline analyser and chemometric models for the recognition of PCT in real-time (related to task 3.3).

Deliverable D3.4 – Eco-design guidelines and report on the process to recognise DfD garments

Deliverable Number	D3.4	**Lead Beneficiary**	5 - RESOR
Deliverable Name	Eco-design guidelines and report on the process to recognise DfD garments		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	24	**Work Package No**	WP3

Description
This report will gather eco-design guidelines and a description of the process of recognising DfD garments in a relevant environment (related to task 3.4).

Deliverable D3.5 – Tests report

Deliverable Number	D3.5	**Lead Beneficiary**	4 - RECY
Deliverable Name	Tests report		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	24	**Work Package No**	WP3

Description
Report to present the tests carried out and the results obtained with the different feedstocks at RECY and PURFI facilities (related to task 3.5).

Deliverable D4.1 – TTRFID assessment

Deliverable Number	D4.1	**Lead Beneficiary**	1 - CU
Deliverable Name	TTRFID assessment		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	6	**Work Package No**	WP4

Description
Assessment of the eco-friendly, recyclable RFID tags customised for textiles (related to task 4.1).

Deliverable D4.2 – TTRFID design and selection guideline

Deliverable Number	D4.2	**Lead Beneficiary**	1 - CU
Deliverable Name	TTRFID design and selection guideline		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	12	**Work Package No**	WP4

Description
The design/procurement guidelines for TTRFID development, including technological characteristics, the economic and feasibility analysis and data privacy fulfilment criteria for consumers (related to task 4.1).

Deliverable D4.3 – DPP development guidelines

Deliverable Number	D4.3	**Lead Beneficiary**	1 - CU
Deliverable Name	DPP development guidelines		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	12	**Work Package No**	WP4

Description
Guidelines for developing scalable, eco-friendly blockchain architecture for DPP on platforms (e.g. Algorand, IOTA, Ethereum) as well as the requirements for data interoperability and API support over TVC. Moreover, API requirements for the interoperability of DPP with PSSR technologies will be developed (related to task 4.2).

Deliverable D4.4 – DPP smart contract guidelines for PSFS & PSSR compatibility

Deliverable Number	D4.4	**Lead Beneficiary**	1 - CU
Deliverable Name	DPP smart contract guidelines for PSFS & PSSR compatibility		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	24	**Work Package No**	WP4

Description
Integration of DPP data retrieval with PSFS & PSSR technologies to enable customised retrieval queries from DPP smart contract (related to task 4.3).

Deliverable D4.5 – Tokenised Ecosystem framework analysis and development guideline

Deliverable Number	D4.5	**Lead Beneficiary**	1 - CU
Deliverable Name	Tokenised Ecosystem framework analysis and development guideline		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	30	**Work Package No**	WP4

Description
Form a tokenised ecosystem over TVC for managing CO2 emissions, real-time tracking and incentivising sustainable

practices, including smart contracts. Moreover, the LCA and TEF interoperability framework will be created (related to task 4.4).

Deliverable D4.6 – DfD solution development over DPP framework

Deliverable Number	D4.6	**Lead Beneficiary**	5 - RESOR
Deliverable Name	DfD solution development over DPP framework		
Type	DEM — Demonstrator, pilot, prototype	**Dissemination Level**	SEN - Sensitive
Due Date (month)	36	**Work Package No**	WP4

Description
Set up DfD solutions over DPP seamless communication with PSSR technologies focusing on dismantling techniques (related to task 4.5).

Deliverable D4.7 – Guidelines for recyclability and reusability

Deliverable Number	D4.7	**Lead Beneficiary**	5 - RESOR
Deliverable Name	Guidelines for recyclability and reusability		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP4

Description
Guidelines for reusability and recycling (related to task 4.5).

Deliverable D5.1 – TTRFID technology deployment

Deliverable Number	D5.1	**Lead Beneficiary**	1 - CU
Deliverable Name	TTRFID technology deployment		
Type	DEM — Demonstrator, pilot, prototype	**Dissemination Level**	SEN - Sensitive
Due Date (month)	28	**Work Package No**	WP5

Description
Upgrading the pilot demonstrator with communication infrastructure with TTRFID (related to task 5.3).

Deliverable D5.2 – DPP Blockchain demonstration deployment with tokenised LCA assessment dashboard

Deliverable Number	D5.2	**Lead Beneficiary**	15 - TXHO
Deliverable Name	DPP Blockchain demonstration deployment with tokenised LCA assessment dashboard		
Type	DEM — Demonstrator, pilot, prototype	**Dissemination Level**	SEN - Sensitive

Due Date (month)	34	**Work Package No**	WP5

Description
The DPP Blockchain sample solution is interoperable with TTRFID and PSFS, as well as PSSR technologies in the demonstrator. Demonstrator LCA dashboard website, which will demonstrate the DPP tokenised capabilities to demonstrate the LCA assessment (related to task 5.3).

Deliverable D5.3 – Automatised PSFS pilot

Deliverable Number	D5.3	**Lead Beneficiary**	9 - SEM
Deliverable Name	Automatised PSFS pilot		
Type	DEM — Demonstrator, pilot, prototype	**Dissemination Level**	SEN - Sensitive
Due Date (month)	36	**Work Package No**	WP5

Description
Design and demonstration of an automatised pre-sorting pilot (related to task 5.1).

Deliverable D5.4 – Upgrade Automatised Sorting Line

Deliverable Number	D5.4	**Lead Beneficiary**	15 - TXHO
Deliverable Name	Upgrade Automatised Sorting Line		
Type	DEM — Demonstrator, pilot, prototype	**Dissemination Level**	SEN - Sensitive
Due Date (month)	36	**Work Package No**	WP5

Description
Upgrade, integrate and demonstrate the advanced sorting technologies into an operational automatised sorting line (related to task 5.4).

Deliverable D5.5 – Validation of Sorted Feedstock

Deliverable Number	D5.5	**Lead Beneficiary**	4 - RECY
Deliverable Name	Validation of Sorted Feedstock		
Type	DEM — Demonstrator, pilot, prototype	**Dissemination Level**	SEN - Sensitive
Due Date (month)	36	**Work Package No**	WP5

Description
Validation of SORT4CIRC sorted feedstock for high-value recycling at recycler's facilities (related to task 5.4).

Deliverable D5.6 – Successful recycling of sorted feedstock

Deliverable Number	D5.6	**Lead Beneficiary**	4 - RECY
Deliverable Name	Successful recycling of sorted feedstock		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP5

Description
SORT4CIRC sorted feedstock successfully used for high-value recycling at recycler's facilities (related to task 5.4).

Deliverable D6.1 – Data collection for SORT4CIRC

Deliverable Number	D6.1	**Lead Beneficiary**	3 - ARIS
Deliverable Name	Data collection for SORT4CIRC		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	15	**Work Package No**	WP6

Description
Report on data collection for the SORT4CIRC value chain, including mass and energy balances (related to task 6.1).

Deliverable D6.2 – Report on environmental impact assessment

Deliverable Number	D6.2	**Lead Beneficiary**	3 - ARIS
Deliverable Name	Report on environmental impact assessment		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	34	**Work Package No**	WP6

Description
Report on the environmental assessment of SORT4CIRC that emphasises GHG emissions along with other impact categories, determining their reliability (related to task 6.2).

Deliverable D6.3 – Report on cost-benefit analysis

Deliverable Number	D6.3	**Lead Beneficiary**	3 - ARIS
Deliverable Name	Report on cost-benefit analysis		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	34	**Work Package No**	WP6

Description
Report on the operating cost of the sorting process and profit margin estimation to determine the economic viability of the project (related to task 6.3).

Deliverable D6.4 – Report on the BM settings

Deliverable Number	D6.4	**Lead Beneficiary**	11 - PSB
Deliverable Name	Report on the BM settings		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	36	**Work Package No**	WP6

Description
Report describing the analysis and settings of the BM that secures the viability of the business in the long run (related to task 6.4).

Deliverable D6.5 – Report on the efficient design and operation of the automated sorting system

Deliverable Number	D6.5	**Lead Beneficiary**	11 - PSB
Deliverable Name	Report on the efficient design and operation of the automated sorting system		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	36	**Work Package No**	WP6

Description
Report including models and recommendations for the efficient design and operation of the automated sorting system (related to task 6.5).

Deliverable D6.6 – Holistic technology benchmarking among different textile sorting processes

Deliverable Number	D6.6	**Lead Beneficiary**	3 - ARIS
Deliverable Name	Holistic technology benchmarking among different textile sorting processes		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP6

Description
Comparison of SORT4CIRC against other textile sorting processes based on cost, environmental impact and technical aspects (related to task 6.6).

Deliverable D6.7 – Public report on environmental impact assessment

Deliverable Number	D6.7	**Lead Beneficiary**	3 - ARIS
Deliverable Name	Public report on environmental impact assessment		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	34	**Work Package No**	WP6

Description
Report on the environmental assessment of SORT4CIRC including only non-sensitive data. (related to task 6.2).

Deliverable D7.1 – Communication, Dissemination, and Exploitation Plan M3

Deliverable Number	D7.1	**Lead Beneficiary**	7 - CEF
Deliverable Name	Communication, Dissemination, and Exploitation Plan M3		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	3	**Work Package No**	WP7

Description
CDE plan detailing strategy for activities to be used by the project and partners involved, including activities, strategic events and a tracker to measure and verify KPIs (related to task 7.1).

Deliverable D7.2 – SORT4CIRC Brand Book

Deliverable Number	D7.2	**Lead Beneficiary**	7 - CEF
Deliverable Name	SORT4CIRC Brand Book		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	3	**Work Package No**	WP7

Description
Visual identity guidelines, logo, and templates for communication outputs to be used by partners in their respective channels (related to task 7.1).

Deliverable D7.3 – SORT4CIRC website

Deliverable Number	D7.3	**Lead Beneficiary**	7 - CEF
Deliverable Name	SORT4CIRC website		
Type	DEC —Websites, patent filings, videos, etc	**Dissemination Level**	PU - Public
Due Date (month)	4	**Work Package No**	WP7

Description
SORT4CIRC project website with pages on the project, updates, partners, deliverables and contact details (related to task 7.1).

Deliverable D7.4 – Communication, Dissemination, and Exploitation Plan M12

Deliverable Number	D7.4	**Lead Beneficiary**	7 - CEF
Deliverable Name	Communication, Dissemination, and Exploitation Plan M12		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	12	**Work Package No**	WP7

Description

Compilation of communication activities and materials (i.e. newsletter and social media post) used for dissemination and updates on exploitation strategy (related to task 7.1).

Deliverable D7.5 – SORT4CIRC policy briefing document M18

Deliverable Number	D7.5	**Lead Beneficiary**	7 - CEF
Deliverable Name	SORT4CIRC policy briefing document M18		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	18	**Work Package No**	WP7

Description
Policy recommendations for regional, national, and European policymakers (related to task 7.2).

Deliverable D7.6 – Communication, Dissemination, and Exploitation Plan M36

Deliverable Number	D7.6	**Lead Beneficiary**	7 - CEF
Deliverable Name	Communication, Dissemination, and Exploitation Plan M36		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP7

Description
Compilation of communication activities and materials (i.e. newsletter and social media post) used for dissemination and updates on exploitation strategy (related to task 7.1).

Deliverable D7.7 – SORT4CIRC circular textiles toolbox

Deliverable Number	D7.7	**Lead Beneficiary**	7 - CEF
Deliverable Name	SORT4CIRC circular textiles toolbox		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP7

Description
Best practices toolbox for project stakeholders to exploit (related to task 7.1).

Deliverable D7.8 – SORT4CIRC policy briefing document M36

Deliverable Number	D7.8	**Lead Beneficiary**	7 - CEF
Deliverable Name	SORT4CIRC policy briefing document M36		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP7

Description

Policy recommendations for regional, national, and European policymakers (related to task 7.2).

Deliverable D7.9 – Report on capacity building webinars

Deliverable Number	D7.9	**Lead Beneficiary**	7 - CEF
Deliverable Name	Report on capacity building webinars		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	36	**Work Package No**	WP7

Description
Report outlining the sessions, insights and participant results of webinars held (related to task 7.3).

Deliverable D7.10 – Summary of clustering activties

Deliverable Number	D7.10	**Lead Beneficiary**	7 - CEF
Deliverable Name	Summary of clustering activties		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP7

Description
Clustering activities with sister projects and initatives such as ECOSYSTEX is summarised (related to task 7.1).

LIST OF MILESTONES

Milestones *Grant Preparation (Milestones screen) — Enter the info.*					
Milestone No	**Milestone Name**	**Work Package No**	**Lead Beneficiary**	**Means of Verification**	**Due Date (month)**
1	Completion of mechanical design and validation testing at SEM's facilities	WP2	9 - SEM	Successfully finalise the mechanical design and conduct comprehensive validation tests at SEM's facilities to ensure all design specifications and performance criteria are met	10
2	Real-time monitoring system for textiles sorting process available for integrating in demonstrator	WP3	13 - IRIS	The prototype should be capable of detecting materials with abundance <3%, detecting fabric structures with an accuracy >95%, and identifying the material with carbon black pigments. Affordable configuration of spectroscopic techniques integrated into compact hardware based on PoC work with VISUM Fusion platform and data fusion strategy	24
3	Recognition process to identify eco-design products	WP3	5 - RESOR	Recognise with an accuracy of >95% of the garments made following eco-design guidelines at a speed of 1 tonne/hour to enable automatic disassembly and sorting.	24
4	Eco-Friendly DPP Blockchain-based Architecture Framework	WP4	1 - CU	Design documents, architecture diagrams, initial setup demonstration and white paper. Comprehensive design documents and technical specifications for the scalable and eco-friendly blockchain architecture, incorporating Semantic Web technologies for data interoperability. Releases of Del N° 4.3	12
5	Integration of DPP Data Retrieval System with PSSR	WP4	1 - CU	Integration test results, IoT sensor deployment reports, and data retrieval demos. Release of Del N° 4.4	24

Milestones *Grant Preparation (Milestones screen) — Enter the info.*					
Milestone No	**Milestone Name**	**Work Package No**	**Lead Beneficiary**	**Means of Verification**	**Due Date (month)**
6	First validation at lab scale of SORT4CIRC sorted feedstock for high-value recycling at recycler's facilities	WP5	4 - RECY	Feedstocks provided by TXHO meet the recyclers' specifications, confirmed by analyses of the recycled textile components	30
7	TTRFID-enabled textile feeding and DPP blockchain interoperability	WP5	1 - CU	Recognition of TTRFID in demonstrator. Interoperability of PSSR technologies with DPP blockchain	32
8	Gathering inventory data for mass and energy balances is complete	WP6	3 - ARIS	Collection of mass and energy balances. Data collection from primary and secondary data sources through site visits, interviews, questionnaires, desk research, industrial and commercial reports, and academic and grey literature. Completion of D6.1.	15

LIST OF CRITICAL RISKS

Critical risks & risk management strategy *Grant Preparation (Critical Risks screen) — Enter the info.*			
Risk number	**Description**	**Work Package No(s)**	**Proposed Mitigation Measures**
1	SKU retrieval algorithm cannot distinguish between generic products and specific products due to a large gap between the training data set and data recorded in the industrial process (Likelihood: High, Severity: Medium)	WP2	The dataset used for training should be pre-filtered, specifically crafted, and annotated for that purpose.

Critical risks & risk management strategy *Grant Preparation (Critical Risks screen) — Enter the info.*			
Risk number	**Description**	**Work Package No(s)**	**Proposed Mitigation Measures**
2	Lack of possibility to analyse some of the textiles' parameters of interest due to the high complexity of samples (Likelihood: Medium, Severity: Low)	WP3	A wide range of optical techniques will be considered for the analysis, and simultaneous use of these techniques will be made. Many partners of this consortium are experienced in the use of optical techniques. Moreover, the creation of an extensive database of samples will contribute to mitigating this risk.
3	Limited sorting rate due to acquisition and analysis (Likelihood: Medium, Severity: Medium)	WP3	Optimise material recognition algorithms to have a near real-time recognition speed
4	Over cost in computing power (for developing the ensemble models) (Likelihood: Medium, Severity: Medium)	WP3	Migration (if required) to the cloud to use outsourced parallel computing resources.
5	Limited sensitivity of the thread recognition (Likelihood: High, Severity: Low)	WP3	An identification system at the thread or garment level will be implemented.
6	Delays in TTRFID tag development due to technical challenges (Likelihood: Low, Severity: High)	WP4	Conduct preliminary research, benefiting from RTL 04 based technologies and experts in the team, and set realistic timelines with buffer periods
7	Blockchain architecture not meeting scalability or eco-friendliness requirements (Likelihood: Low, Severity: High)	WP4	Focusing on compatible blockchain platforms (e.g., Algorand) with scalability characteristics, conducting testing, consulting with blockchain experts and benefiting from the EBSI infrastructure.
8	Integration difficulties between DPP data retrieval system and PSSR technologies (Likelihood: Low, Severity: High)	WP4	Fulfilling standards IEC 62264: (ISA-95) and IEC 62443: (ISA -99) and protocols like Collaborate with IoT and automation experts, conducting phased integration, and performing extensive testing and benefiting from CIRPASS-2 outcomes.
9	Interoperability Challenges (Likelihood: Medium, Severity: High)	WP4	Difficulty in achieving seamless integration with legacy systems, other blockchains, and different IoT devices will be fulfilled with tasks 4.2.2-4
10	Validation of sorted feedstock for high-value recycling at recycler's facilities fails (Likelihood: Medium, Severity: High)	WP5	Carry out regular tests, investigate the causes of deviations, draw up plans to optimise the production process and implement them in practice.
11	Delayed data mining due to limited experimental progress or confidentiality reasons (Likelihood: Medium, Severity: High)	WP6	Use of process simulation tools or literature available data to determine mass & energy balances for implementation in the cost-benefit analysis and LCA model.

Critical risks & risk management strategy *Grant Preparation (Critical Risks screen) — Enter the info.*			
Risk number	**Description**	**Work Package No(s)**	**Proposed Mitigation Measures**
12	Scarce data of capital expenditure estimation (Likelihood: Medium, Severity: Medium)	WP6	A factorial approach will be used: the cost objects are estimated as a function of the purchased equipment cost; the parameters and factors used in those cost objects-purchased equipment cost correlations are obtained from suppliers, vendors or manufacturers and published articles or white papers.
13	Access to valid and reliable data sources for value chain mapping and system modelling (Likelihood: Low, Severity: Medium)	WP6	Triangulation and cross-verifying the acquired data from multiple sources. These sources may include primary ones, derived from unstructured interviews, and secondary ones, sourced from literature, technical reports, and grey literature.
14	Low attendance at the online circular toolbox launch event (Likelihood: Low, Severity: Low)	WP7	Start promotion early, leveraging SORT4CIRC and CEF's connection with the CCRI community to explore potential synergies with other initiatives to organise joint events and maximise the attendance of key players. Relevant presentations of in-person events will also be available online for interested participants who could not attend.
15	Lack of relevant content in for the webinars/training sessions for participants (Likelihood: Low, Severity: Low)	WP7	Alignment and collaboration with other CCRI networks, ensuring our content is relevant and distinct from existing ones. Leverage existing expertise, networks, partner organisations, and other channels to gain insights into participant needs, validate the relevance of topics for the webinars, and tailor each session to relevant WPs.
16	Low awareness of the SORT4CIRC project (Likelihood: Low, Severity: Low)	WP7	Leverage connections with initiatives such as the CCRI, DEFINITE-CCRI, CircularInvest, CCRI CSA, SOLSTICE and other projects to build awareness of the project. Participation in external events will also help build the project's profile.

PROJECT REVIEWS

Project Reviews *Grant Preparation (Reviews screen) — Enter the info.*			
Review No	**Timing** (month)	**Location**	**Comments**
RV1	21	To be confirmed	To be confirmed. After the submission of the mid-term technical report.
RV2	36	To be confirmed	To be confirmed.

TABLE OF HISTORY OF CHANGES		
		Version (date)
Annex 1 Part A		
Beneficiaries	• The Rennes Business School former Beneficiary 11, has been replaced by Paris School of Business (PSB) PIC 944056866, due to the leading researcher (Reza Zanjirani Farahani) changing position to this new institution and Rennes Business School does not have the qualified expertise to carry out the activities • An affiliated entity - Circularity Gap Solutions B.V. PIC 877344631 (CGS) - has been added on beneficiary 7 CEF	15/1/25
	• Following the change in ownership of SOEX, the legal entity of the project coordinator has been changed to Interzero Circular Solutions (IZCS), PIC 890257368. In addition, an affiliated entity to the project coordinator has been added IZ Circular Textiles GmbH (IZCT), PIC 873039092.	24/2/25
	• Interzero Circular Solutions Germany GmbH (IZCS) has handed over the project coordination to Constructor University Bremen GmbH (CU). • Interzero Circular Solutions Germany GmbH (IZCS) has been terminated. • IZ Circular Textiles GmbH (IZCT) has been terminated. • Textile House for EURO TRADE, s.r.o. (TXHO) PIC 870613413 has been added to the Consortium and will take over the responsibilities of IZCT.	16/7/25
Financial information	• The project costs of affiliated partner 7.1 CGS have been transferred from beneficiary 7 CEF • ARIS will declare some SME owner unit costs	15/1/25
	• The former SOEX budget has been split between IZCS and IZCT	24/2/25
	• The full budget allocation for IZCS has been transferred to CU • The full budget allocation for IZCT has been transferred to TXHO	16/7/25
Researchers	• The researchers' table has been revised, due to a partner change of beneficiary 11 and the two people belonging to beneficiary 10 PURFI are not qualified as researchers.	15/1/25
Workplan	• WP7 – affiliated partner 7.1 CGS role has been added to tasks 7.1, 7.2 and 7.3	15/1/25
	• Task 7.1 has been revised to also include clustering activities	24/1/25
	• Task 1.2 has been revised to also include reporting on the activities of the ethics mentor.	7/2/25

	• The role of SOEX has been split between IZCS and IZCT
	  ○ IZCS will carry out WP1
	  ○ IZCT will take over all other tasks formerly assigned to SOEX in the other WPs
		24/2/25

	• Links between tasks and deliverables have been added in the task description	24/03/25
	• All WP1 Tasks have been handed over from IZCS to CU. • All WP tasks for IZCT have been handed over to TXHO	16/7/25
Effort	• The person months have been revised for beneficiary 7 CEF and affiliated partner 7.1 CGS • This makes a total of 71,1 PM for WP7 and a total of 574.51 PM for the project	15/1/25
	• The former PM distribution of SOEX has been split between IZCS and IZCT	24/2/25
	• The PM allocation for WP1 has been transferred from IZCS to CU.	16/7/25
Deliverables	• One mid-term and final data management plan has been added (D1.7 at M18 and D.11 at M36)	15/1/25

	• One mid-term policy briefing document has been added at M18 (D7.5) • The dissemination level of D1.3, D1.7, D1.11, D3.1 and D3.4 has been changed to public	
	• Type of D1.2, D1.4 and D1.7 have been changed to DMP • 2 public deliverables have been added to WP2 (D2.4-5) • Dissemination level of D4.3 and D4.4 has been changed to public • 1 public deliverable has been added to WP5 (D5.6) • 1 public deliverable has been added to WP6 (D6.7)	24/1/25
	• Numbering of WP1 deliverables has been revised due to the grouping of deliverables • Deliverable type changed to Data Management Plans has been revised to DMP (new D1.3, D1.7 and D1.11	7/2/25

	• The deliverables of SOEX have been split between IZCS and IZCT
	  o IZCS will be responsible for all WP1 deliverables
	  o IZCT will be responsible for all other former SOEX deliverables.
		24/2/25

	• The Communication, Dissemination, and Exploitation Plan at M24 has been deleted • The Communication, Dissemination, and Exploitation Plans deliverables have been changed to public (D7.4 and D7.6)	02/04/25
	• The Communication, Dissemination, and Exploitation Plans deliverable at M3 have been changed to public (D7.1)	04/04/25
	• The deliverables of IZCS have been transferred to CU and therefore CU will be responsible for all deliverables in WP1. • The deliverables of IZCT have been transferred to TXHO and therefore TXHO will be responsible for all deliverables in WP5.	16/7/25
Annex 1 Part B		
All parts	• SOEX has been changed to IZCS or IZCT in relevant parts.	24/2/25
	• IZCS has been replaced by CU in all relevant parts. • IZCT has been replaced by TXHO in all relevant parts.	16/7/25
Section 1		
1.1.3	• Ambition 1 has been revised to stress better why manual feeding and unfolding of the garments at the first stage of the pilot system is needed. As the first stage focuses on sorting for the reuse market, the efficiency of AI algorithms for type and quality detection is vital. Although the manual feeding mechanisms are proposed, due to the efficiency of quality detection for the next stages, the rework for manual inspection of clothes will be avoided. So, the throughput of the system will maintain both higher speeds and quality in comparison with other literature solutions like falling models	15/1/25
1.2.1	• The technical robustness of AI-based systems has been detailed in Challenge 1 and Challenge 2. • Figure 2 has been modified for replacing the layout of TXHO in replacement of IZCT • Figure 3 has been cited for the reference. • Figures 3, 4, 5, 6 has updated to Figures 4, 5, 6, 7 accordingly after updating the figures' ordering and numbering.	15/1/25
1.2.2	• TXHO engagement in initiatives and relations with other projects have been added.	16/7/25
Section 2	• Figure 7 has updated to Figure 8 and Figure 9 has been added for "*Communication & dissemination tools and proposed dissemination*" accordingly after updating the figures' ordering and numbering.	16/7/25
Section 3		
3.1.1	• The figure 8 has been updated to consider the partner change of beneficiary 11	15/1/25
3.2.1	• The Gantt chart has been revised to consider new deliverables and change of partner for beneficiary 11	15/1/25
	• The Gantt chart has been revised to consider new deliverables	24/1/25

	• The Gantt chart has been revised to consider the change in the project coordinator short names	24/2/25
	• The Gantt chart has been revised to consider revised deliverables	7/2/25
	• The Gantt chart has been corrected to consider links between tasks and deliverables	24/3/25
	• The Gantt chart has been corrected to consider the erased deliverable	02/04/25
	• The Gantt chart has been corrected to consider the change of coordination from IZCS to CU. • The Gantt chart has been corrected to consider the change of responsible tasks of IZCT to TXHO. • The Figure caption for Gannt chart has changed from Figure 8 to Figure 11 after ordering the former figures accordingly. • Figure 11 has been modified for replacing the layout of TXHO in replacement of IZCT	16/7/25
Table 3.1h	• Reference to Wolfen has been replaced by the term "sorting plant" • Purchase costs of CEF have been revised	15/1/25
	• Amount of travel costs of CEF have been corrected	7/2/25
	• Purchase costs of IZCT has been revised	24/2/25
	• The following statement has been added "The equipment costs reported are depreciation costs only". • Travel costs of IZCT for face-to-face meetings with each individual Consortium partner has been clarified. These are technical meetings at partner locations and are related to WP3, WP4 and WP5	10/03/25
	• The budget of 1.1 IZCT has been transferred to TXHO. • The Budget of CU has been revised to include the WP1 tasks budget from IZCT former PM tasks. The travel costs have been added to CU table with more detailed information. • The Budget for CEA has been corrected for Travel C.1 Travel and subsistence and C.3 Other goods, works and services	16/7/25
3.2.1	• Figure 9 has been updated to consider change of partner for beneficiary 11 • The affiliated entity Circularity Gap Solutions B.V. has been added in beneficiary 7 CEF description	15/1/25
	• The text describing the composition of the partnership has been revised to include the affiliated entity.	7/2/25
	• The text describing the composition of the partnership has been revised to include the additional affiliated entity. • The figure 8 and 9 have been updated	24/2/25
	• The text describing the composition of the partnership has been revised	10/03/25
	• Figures 11(old version 8) and 12 (old version 9) have been updated to consider the change of project coordination from IZCS to CU and also IZCT to TXHO. • The Figure caption for Gannt chart has changed from Figures 8,9 to Figures 11,12 after ordering the former figures accordingly.	16/7/25
Section 4		
4.1	• Addition of ethics section	15/1/25
	• Statement has been added in ethics section	24/1/25
	• Role of Ethics Mentor has been described	7/2/25
	• The following statement has been added "The Consortium acknowledge the ethics issues identified in the Summary Report related to HUMANS, PERSONAL DATA, ENVIRONMENT, HEALTH AND SAFETY, and ARTIFICIAL INTELLIGENCE"	24/3/25

Table of contents

1 Excellence 6

1.1 Objectives and ambition 6

1.1.1 Main objective / overall purpose 6

1.1.2 Specific objectives 7
1.1.3 Main innovations and ambitions / progress beyond the state-of-the-art (SOTA) 9
1.1.4 TRL assessment 12
1.2 Methodology 13
1.2.1 Overall methodology 13
1.2.2 Relations with other projects 21
1.2.3 Do Not Significant Harm Principle 22
1.2.4 Interdisciplinary 22
1.2.5 Gender Dimension 22
1.2.6 Open Science Practices 22
1.2.7 Research Data Management 23
2 Impact 25
2.1 Project's pathways towards impact 25
2.1.1 Outcome 1 and 2: Roll-out of systemic solutions for textile sorting, using innovative digital technologies and facilitated disintegration to be incorporated in product design as an enabler for recycling ... 25
2.1.2 Outcome 3: Increased uptake of mechanical recycling solutions that deliver competitive, high-quality secondary materials 25
2.1.3 Outcome 4: Roll-out of thermo-mechanical, chemical and other (e.g., enzymatic) recycling solutions that are sustainable from a zero-pollution, circular material and energy efficiency perspective 26
2.1.4 Target group 26
2.1.5 SORT4CIRC's unique contribution to wider scientific impacts 26
2.1.6 SORT4CIRC's unique contribution to wider economic and technological impacts 26
2.1.7 SORT4CIRC unique contribution to wider societal and environmental impacts 27
2.1.8 Barriers and requirements 28
2.2 Measures to maximise impact - Dissemination, exploitation and communication 29
2.2.1 Planned measures to maximise the impact of SORT4CIRC 29
2.2.2 Intellectual property management 31
2.3 Summary 32
3 Quality and efficiency of the implementation 33
3.1 Work plan and resources 33
3.1.1 Overall structure of the SORT4CIRC work plan 33
3.1.2 SORT4CIRC Planning 34
3.2 Capacity of participants and consortium as a whole 36
3.2.1 Consortium value chain overview 36
3.2.2 Consortium description and complementarity 37
3.2.3 Resources distribution 38
3.2.4 Consortium critical infrastructures 39
3.2.5 Industrial involvement 39
4 Ethics 40
4.1 Ethics Issues Table 40
4.2 Ethics Self-Assessment 42

Abbreviation	Definition
AI	Artificial Intelligence
API	Application Programming Interface
BM	Business Model
CCRI	Circular Cities and Regions Initiative
CE	Circular Economy
DfD	Design for Disassembly
DPP	Digital Product Passport
EBSI	European Blockchain Services Infrastructure
EPR	Extended Producer Responsibility
ERP	Enterprise Resource Planning
FAT	Factory Acceptance Tests
FMEA	Failure Mode and Effects Analysis
FTIR	Fourier Transform Infrared
IoT	Internet of Things
LCA	Life Cycle Assessment
LCC	Life Cycle Cost
LCI	Life Cycle Inventory
LIBS	Laser Induced Breakdown Spectroscopy
ML	Machine Learning
NIR	Near InfraRed

Abbreviation	Definition
OWL	Web Ontology Language
PCT	Post-Consumer Textile
PET	Polyethylene terephthalate
PET/CO	Polyethylene terephthalate/Cotton
PET/EA	Polyethylene terephthalate/Elastane
PET/PA	Polyethylene terephthalate/Polyamide
PoC	Proof of Concept
PoS	Proof of Stake
PSFS	Pre-Sorting and facilitating Fine Sorting
PSSR	Pre-Sorting and Sorting for Recycling
RDF	Resource Description Framework
RFID	Radio Frequency IDentification
ROI	Return on Investment
SC	Supply Chain
SKU	Stock Keeping Unit
SWIR/MWIR	Short/Mid-Wave InfraRed
TEA	Technico-Economic Assessment
TEF	Technical & Economic Feasibility
TTRFID	Textile customised Tokenised RFID
TVC	Textile Value Chain

1 Excellence

#@REL-EVA-RE@#

1.1 Objectives and ambition

#@PRJ-OBJ-PO@#

1.1.1 Main objective / overall purpose

The EU generates 12.6 million tonnes of textile waste per year. Currently, only 22% of Post-Consumer Textile (PCT) waste is collected separately for reuse or recycling, while the remainder is often incinerated or landfilled[1]. However, with the updated legislative directives on separate textile collection across the EU by 1 January 2025[2], the amount of textile waste collected will increase. To ensure the sustainable recovery of this material, upscaling sorting and recycling capacity and efficiency must go hand in hand. Sorting for reuse and recycling is the critical step of the value chain that decides the next best use and enables textile resources to cycle repeatedly through the supply chain (SC). Textile reuse clearly offers greater environmental and economic benefits than recycling, but its sorting is carried out manually and is associated with high labour and energy costs in the EU. Several brands and manufacturers are

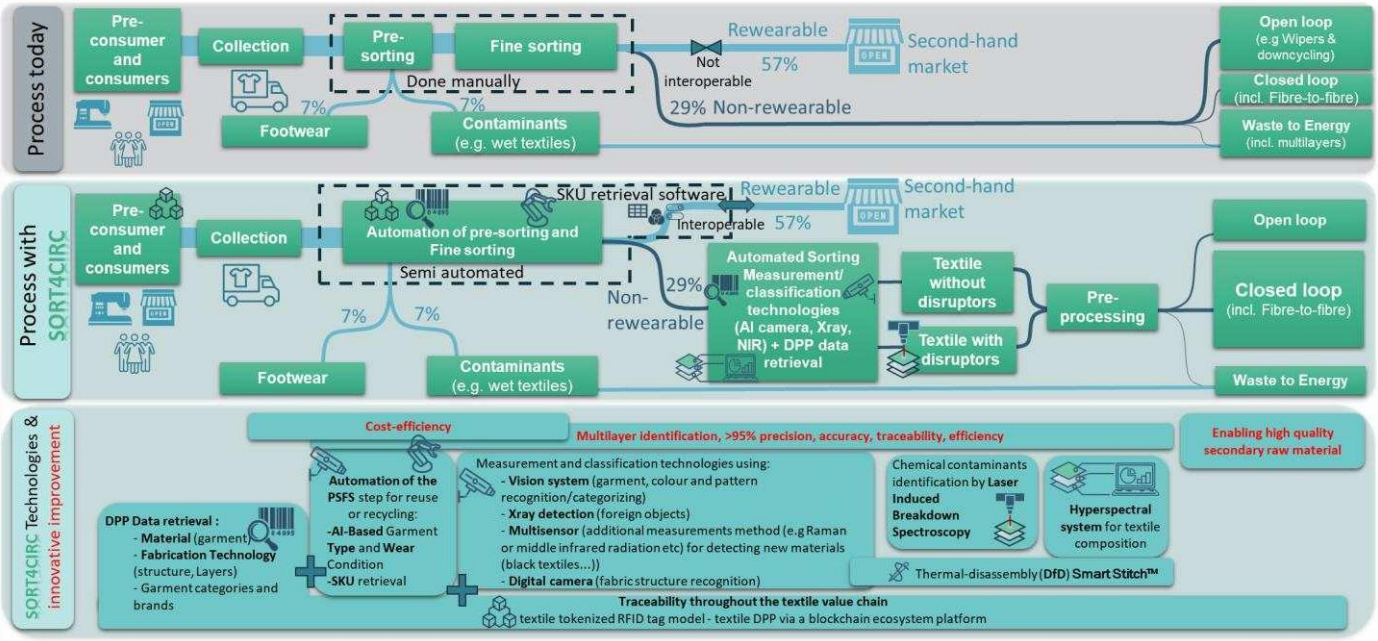

Figure 1 - Overall concept of SORT4CIRC

already incorporating recycled content in their collections; however, most are not using PCTs. To date, recycled fibre and fabrics made from PCTs are priced higher, intimately related to the high costs required to process them.

SORT4CIRC's overall ambition is to improve value creation and cost efficiency through **automated sorting and innovative recycling**. This will generate new revenue streams for sorters and recyclers by enhancing sorting for reuse and turning non-re-wearable textiles into valuable feedstock for high-value recycling. The project will strive to **unlock the current technological gaps for textile identification and value-chain traceability** while **providing** an **overall sound business case** driving circularity within the Textile Value Chain (TVC).

The use of **modern digital technology in the sorting process** will make it easier to achieve comprehensive information on the materials and fabrication process step. This information can then be **integrated** into automated Pre-Sorting and facilitating Fine Sorting (**PSFS**) and Pre-Sorting and Sorting for Recycling (**PSSR**) processes. The different complementary technologies are interlinked with interoperable architecture to achieve complete traceability. The results generated are cross-checked to ensure reliability and consistency, enabling informed **decisions** about the **optimal channel for reuse or recycling** besides ensuring high quality input to **reuse** and **recycling**.

In response to this challenge, SORT4CIRC will develop a **systemic, transformative solution** by pioneering, scalable solutions for **PSFS** & **PSSR**, coupled with traceability and 'Design for Disassembly' (DfD). An innovative solution that can be developed and deployed across the industry actors, maximising the impact on waste reduction and resource conservation. Fostering a more resilient and sustainable textile sector, reducing the industry's reliance on natural resources and decreasing emissions, aligning with the objectives of the EU Green Deal Industrial Plan[3]. The overall scope is described in **Figure 1**.

[1] https://ec.europa.eu/commission/presscorner/detail/%20en/ip_23_3635
[2] Directive - 2018/851 - EN - EUR-Lex (europa.eu)
[3] https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A52023DC0062

SORT4CIRC gathers all relevant expertise to achieve the four strategic objectives in the following section: industrial collector and sorter, technology developers (machine learning (ML), Artificial Intelligence (AI), hyperspectral & spectroscopy solutions, Design-for-Disassembly (DfD) solutions, recyclers (chemical and mechanical), along with specialists in SC tracking and tracing, digital transformation tools, digital passports, blockchain technology and Internet of Things (IoT), textile Business Models (BMs), trade and Life Cycle Cost/Assessment (LCC/LCA). Industrial partners will ensure the effective scaling and implementation of the various solutions at the demonstration and commercial levels.

1.1.2 Specific objectives

Objective 1: Revolutionize textile sorting by employing innovative, intelligent, interconnected technologies within an automatised system to ensure an efficient and optimal channel for reuse or recycling.

Pre-sorting classification (reuse or recycling):

- Develop a system based on computer vision categorisation that provides a smart and efficient automated solution to power the earlier stages of sorting, focused on sorting for reuse (waste hierarchy).
- Around 15 categories are used in the sorting line as a pre-sorting step; the digital technology will automatically identify parameters such as the product category and condition (e.g., damp and soiled).
- Increase the productivity, profitability, and sustainability of textile waste management by automating PSFS and increasing the fixed throughput of sorted textiles.
- Develop a Stock Keeping Unit (SKU) retrieval software to visually identify PCTs using AI-based computer vision, allowing access to product data for those without a readable identifier.
- Set up generic/scalable/rapid/robust classification techniques using visible and NIR (Near InfraRed) imaging and suitable algorithms.

Textile recognition in sorting for recycling:

- Leverage a novel comprehensive Multirange Spectroscopic Platform and hybrid ensemble modelling to efficiently provide an affordable and compact real-time monitoring system capable of tacking the complex monitoring requirements of the textile items.
- Develop a semi-industrial real-time recognition system to identify the features of multilayer materials in a single textile item (currently non-identifiable with the existing NIR spectroscopy system and, therefore, sent to waste).
- Develop a new, more efficient recognition system based on a set of complementary recognition techniques, which unlock the current limits of NIR identification and increases the reliability of measurement technologies for (a) identification of fibre material < 3%; (b) identification of textiles containing black carbon-based pigments; (c) Detection and identification of finishing agents and coatings; (d) PA6/PA66 and cotton/linen differentiation; (e) identification of chemical contaminants; (f) identification of multilayer materials.
- Integrate and combine different technologies into an efficient and replicable system, enabling the management of raw material flows that meet standardised requirements and sorting criteria for recycling technologies.

DfD for recycling:

- Assess DfD solutions using a Digital Product Passport (DPP) for identification and innovative technology to identify and disassemble textile products using heat-dissolvable bio-based and synthetic stitching threads and thermal disassembly systems.

Key Performance Indicators

Pre-sorting classification (reuse or recycling):

- ☑ *Create a standard of pre-sorting categories the sorting operators use representing>20% of the EU market*
- ☑ *Precision levels for accuracy of garment type recognition ≥ 90%, wear conditions (soiling) classification ≥ 75%*
- ☑ *By using the automated PSFS, it will lead to a reduction of costs for the sorting process step by 20-30%*
- ☑ *Detection and retrieval of product data within 800 milliseconds for SKU retrieval*
- ☑ *Enriched data repository including ≥20 tonnes of various garments as ML reference models*

Textile recognition for recycling:

- ☑ *A new hyperspectral dataset of garments for new code developments*
- ☑ *Decrease the composition estimation error to a value of less than 5%*
- ☑ *Enable to estimate the presence of one material with less than 3% content*
- ☑ *Allow the detection of black textiles (cont. carbon pigments), chemical contaminants, and foreign materials*
- ☑ *Fabric structure recognition accuracy >95%*
- ☑ *Efficient separation of PET (Polyethylene terephthalate) fibres from multilayer textiles with >80% of PET and no metallic parts*

Multilayer disassembly for recycling:

- ☑ *Increase from 40-60% to 90-95% of recovered textile products from disassembly process*
- ☑ *At pre- and post-disassembly stage, eco-designed garments enable >95% of accurate identification/sorting.*
- ☑ *Successfully identified DfD smart stitch via DPP Radio Frequency IDentification (RFID) tags.*

Objective 2: Increase traceability of pre-owned clothes with integrity and confidentiality of information for more efficient sorting and recycling.

- Develop a textile customised tokenised RFID (TTRFID) tag model for autonomous/interoperable data communication
- Design a textile DPP via eco-friendly blockchain architecture for traceability of data over the TVC ecosystem
- Integrate the DPP data retrieval with the automatised PSFS & PSSR solutions in *Objective 1* for seamless and interconnected textile product data exchange
- Establish a tokenised ecosystem over TVC to enable DfD via information sharing of TVC stakeholders

Key Performance Indicators

- ☑ *Compatibility >98% with design for recycling of the developed TTRFID tags*
- ☑ *Compatibility >95% with standardised data communication protocols of RFID gates and related EU projects*
- ☑ *Autonomous compatibility of DPPs for interoperability >80% of APIs (Application Programming Interfaces) of Enterprise Resource Planning (ERP) systems used in TVC*
- ☑ *100% success rate for DPP in fulfilling the confidentiality of data over the blockchain platform like European Blockchain Services Infrastructure (EBSI).*
- ☑ *Successful, accurate data retrieval >95% of DPP-equipped textile products for garment, technology and layer*

Objective 3: Design and demonstrate automatised PSSR system, integrate and combine new technologies into an operational and industrial environment, ensuring high quality and economically viable feedstock for existing and emerging mechanical/chemical/thermo-mechanical recycling.

- Design and demonstrate at a pilot scale, an automated pre-sorting line that categorises items according to the product type and condition collected from the commonly used market
- Upgrade textile sorting systems with suitable measurement technology to adapt the quality of the output material to the current and future requirements of mechanical, chemical, and thermo-mechanical recycling companies
- Integrate the advanced sorting technologies into an operational automatised sorting line
- Consolidation for technologies' interoperability in the form of DPP data retrieval and AI-based detection models
- Set up innovative smart modular sorting technology for single-unit operations to enhance the sorting process faster by embedding the latest AI detection technologies developed within SORT4CIRC
- Integrate and demonstrate the systemic solution into an operational automatised sorting system, incl. automated pre-sorting, measurement technologies for textile classification/recognition and DPP traceable identifiers.
- Validate the SORT4CIRC DPP blockchain for the storage/retrieval of TVCs regarding garments, technology, layer, and recyclability of developed TTRFIDs
- Validate SORT4CIRC sorted feedstock for high-value recycling at recyclers facilities focusing on fibre-to-fibre requirements: (a) "Soft mechanical recycling" (i.e., cotton, cotton/polyesters, and acrylic); (b) "Pulping" chemical recycling (i.e., cotton and cellulosic); (c) Methanolysis chemical recycling for "complex textiles" (separate polyester from co-materials (i.e., cotton, polyamide, and elastane.); (d) Chemical recycling (Polyamid6); (e) Thermo-mechanical recycling (Polyester high purity)

Key Performance Indicators

*Successful integration of recognition & automatised system into **PSSR** for the recycling process*

- ☑ *Cost savings of up to 30% by reducing labour costs, contributing to a positive Return on Investment (ROI) on the investment in automation **PSSR***
- ☑ *Precision level >85 % for garment classification and conditions dryness/soiling classification at pilot-scale*
- ☑ *Keeping the same accuracy results (mentioned in objective 1) at 500 kg/hour throughput.*
- ☑ *Unchanged throughput and productivity objectives via automatisation*
- ☑ *Less than 1% of material losses and sorting speed of >1m/s.*
- ☑ *Success rate >95% for DPP seamless data exchange with AI-based technologies for **PSSR***
- ☑ *Bring non-identifiable textiles, mixed & multi-material textiles to the recycling stream*
- ☑ *From 60% to 98% of high-quality textiles meeting recyclers requirements*
- ☑ *Improved process steps documentation and continuous quality assurance of the end product*

Objective 4: Technical & Economic Feasibility (TEF) including LCC/LCA perspective & viability of circular economy (CE) BMs

- Validate TEF of upscaled processes along with workplace implications
- Assess the environmental benefits of the textile sorting process automation through a detailed LCA based on ISO standards (14040-44, 14025)
- Develop a plant-wide process model that can calculate the Life Cycle Inventory (LCI) of the whole value chain
- Identify and mitigate potential risks associated with the automation project, such as technical challenges, market demand fluctuations, and regulatory compliance
- Evaluate the CE BM viability, industrial symbiosis among recyclers, and social acceptance-consumer welfare

Key Performance Indicators
☑ *Environmental:* ***20% lower GHG emissions*** *than the state-of-the-art waste textile sorting process.*
☑ *Economics:* ***A positive ROI of up to 26%*** *at the end of the project.*
☑ *Sorting Efficiency by 40% compared to manual, resulting in higher throughput rates and processing speeds.*
☑ *Ability to increase throughput by 30-50% without significant additional investment.*
☑ *Social welfare:* ***>25 % of consumers*** *in the EU consider buying preowned clothes (currently, it varies between 18% and 21% depending on consumer behaviour/ culture).*

1.1.3 Main innovations and ambitions / progress beyond the state-of-the-art (SOTA)

Ambition 1 – Automated PSFS sorting, recognition monitoring, measurement and classification technologies

SOTA. Textile PSFS systems still rely heavily on manual decision-making processes. While some innovations to automate these processes have been announced and are in development, they have not yet been fully developed or launched[4]. Moreover, the PSFS needs to ensure effective mechanisms for high quality of cloth wear condition aligned with second-hand market and reuse market pipelines. The SOTA shows the gap for effective SKU based cloth categorization and high accuracy wear condition for reuse market[5].

Bottlenecks. Current manual PSFS processes make sorting subjective, focusing primarily on product type and quality. This limits resale options to traditional second-hand shops and export markets. To unlock new resale channels and prevent re-wearable items from being recycled, especially as separate textile collection in all EU member states, legally mandatory from 2025 on, will increase collection volumes by more than 80%, improved sorting quality and detection of more relevant product attributes are needed. Innovative computer vision solutions, including those from REFA, are being developed alongside free-fall detection approaches.

To reliably assess the quality of a garment, more than 95% of its surface must be visible, which requires the garment to be fully unfolded. There are two distinct approaches to robotic handling systems: free-fall systems with separation and flat presentation systems. Both have significant limitations.

1. **Free-fall systems require robotic separation**, where individual garments are extracted from a pile and placed onto a conveyor in random orientations. While this step alone already takes more than two seconds per garment, it does not solve the issue of unfolding. In a free-fall state, garments are not sufficiently flattened, making it impossible to derive detailed product attributes or assess the garment's quality with the required precision. This leads to unreliable detection and necessitates additional manual sorting, reducing the overall utility and cost-effectiveness of the system.
2. **Flat presentation systems, on the other hand, ensure garments are fully unfolded** and laid flat on a conveyor belt. This approach enables accurate quality assessment, as the entire surface of the garment is visible. However, achieving this flat presentation with robotic systems requires significant processing time. Current state-of-the-art robotic solutions need upwards of 30 seconds, often approaching a full minute, to unfold and flatten a single garment. These long cycle times are far slower than manual handling and make such systems economically unfeasible for industrial-scale sorting operations.

While free-fall detection eliminates the need for unfolding, it struggles with image resolution for garments larger than accessories or kidswear. Tests by REFA revealed incomplete unfolding, leading to unpredictable flight paths incompatible with the limited depth of field of high-resolution imaging systems. High-resolution images are crucial for detecting defects and avoiding costly false positives, where reusable items are mistakenly recycled, impacting sorting plant economics and undermining waste hierarchy principles. Furthermore, due to the limitations in detection capabilities of free-fall systems, an additional fine-sorting step will still be required. In comparison, the fine sorting process can occur automatically when a garment is unfolded on a conveyor belt, making the free-fall solution economically less attractive.

In summary, free-fall systems suffer from inadequate quality detection due to incomplete visibility of garments, while flat presentation systems face prohibitive processing times and costs. A semi-automated solution, where garments are manually unfolded and placed on a conveyor belt for automated quality assessment, offers a practical compromise. This approach is 30% more efficient than purely manual sorting and combines human dexterity with the precision of digital technologies. While robotic systems may improve in the future, their current state is insufficient to meet the demands of large-scale garment sorting. Any advancements in robotic unfolding could still be integrated into existing workflows, but for now, semi-automation remains the most reliable and cost-effective option.

[4] Rehman, M. M., Exploring the Link Between Artificial Intelligence and Circular Economy in the Fashion Industry: an example of Lennol Oy, https://www.theseus.fi/handle/10024/865358

[5] Herzberg, P. H., Pirrwitz, F., Analysing the EU textile and clothing material flow with an emphasis on reuse-based value chains, https://www.diva-portal.org/smash/record.jsf?pid=diva2%3A1802563&dswid=-7763

Innovation. Automated PSFS System to Unlock New Reuse Markets: The PSFS system integrated into this project builds on REFA's automated sorting technology. Sorters unfold the garments on the conveyor belt, allowing the system to capture high-resolution images from both sides. The system utilises AI-based computer vision models to identify key attributes, such as clothing type, degree of wear or soiling, style, brand (if visible), fabric construction (e.g. knit, woven), colours, and size. Additionally, the system can access DPP data, providing essential information for reuse sorting decisions, such as brand, original sales prices, and material composition. A similarity detection AI algorithm, developed in this project and combined with a database of image feature vectors, will enable the retrieval of product data from DPPs and e-commerce platforms without needing a visible or accessible data carrier. This allows for detecting valuable garments, even for products without a DPP or visible data carrier. The system facilitates sorting curated feedstock for new resale channels, such as retailers of new goods, ultimately leading to increased volumes for second-hand markets instead of recycling.

Ambition 2 – Automated PSSR sorting, recognition monitoring, measurement and classification technologies

SOTA. The existing processes for disassembling textiles are either manual or mechanical, but both involve a high percentage of losses ranging from 20-60% and are economically not viable[6]. PSSR processes have been done manually. In the recent years, several NIR spectroscopy solutions have been launched for a more accurate material composition detection due to new requirements from innovative recyclers[7].

Bottlenecks. Today, the pre-sorting stage is fully manual and labour-intensive. Manual **PSSR** chemical recycling requires precise identification and strict specifications. To supply these markets, it is necessary to develop automated or semi-automated sorting of the non-re-wearable fraction by fibre type and colour. Some integrators have started to use the **hyperspectral technique** for recycling. However, it still needs to be improved to determine the composition of garments with two or more mixed materials. It needs to be combined with other optical/photonic technologies to provide additional recognition functions like identifying black-dyed clothes and garments with low content (<3%) of elastane. The existing NIR technology faces limitations in accurately sorting textiles due to issues like the difficulties of penetration of multilayer textiles and the presence of carbon-based pigments. Integrating innovative techniques as a pre-recycling process and scalability to the available recycling technologies is currently lacking. **Mechanical disassembly does not offer a solution** to **multi-material/multi-layer products** and reduces the fibre quality and recyclability rates.

Innovation. New multimodal automatic PSSR monitoring: SORT4CIRC will provide an innovative and cost-effective textile analysis system for automated **PSSR** (for recycling using a co-integration of multimodal imaging/photonics technologies (Laser Induced Breakdown Spectroscopy (LIBS), hyperspectral imaging, Vis-NIR and short/mid-wave infrared (SWIR/MWIR))).While **automated PSSR systems** will record image data of garments to detect the presence of finishing chemicals, coatings or other undesirable components on the surface of functional textiles, enabling the separation of unidentified components and accessories. It will detect and identify the characteristics of multi-layer materials in a textile article. Automatic sorting will classify garments and fabric structures and recognise textile materials. It will identify single or multi-component materials with the capacity to detect < 3% of other types of materials in the textile, even in the presence of black dyes or wear conditions or soiled textiles. **Algorithmic strategies** will be developed to closely match the actual composition of the garment as measured in the laboratory or to give better compositional estimates in the presence of low-content materials. The monitoring system will use advanced **AI algorithms** and **deep ML** to constantly increase its capacity for improving consequently the current technology will be the robustness and accuracy of the **classification algorithm** for complex compositions or new unknown compositions. **A feedstock with a well-defined composition ready for recycling** (mechanical, chemical, enzymatic fibre-to-fibre recycling) will be sorted and sent to recyclers for evaluation.

Ambition 3 – Textile DPP

SOTA. EC has proposed the **DPP** as a tool to create transparency and unlock circularity for sharing product information across the entire value chain, including data on raw material procurement and logistics, production and manufacturing processes and recycling[8]. DPP is part of the **Eco-design for Sustainable Products Regulation** and one of the key actions under the **CE Action Plan,** which aims to establish an introduction of a DPP in at least 3 key market,including textiles, by 2027. Moreover, as an objective for the **EU Commission's 2030 vision for textiles**, the textiles sector should move toward competitiveness, resiliency and innovation along with the producers by taking responsibility for their products along the value chain with sufficient capacities for **recycling**

[6] Isatou Ndure, Fashion for Good to sort garments with AI for re-wearable textiles, https://www.just-style.com/news/fashion-for-good-to-sort-garments-with-ai-for-re-wearable-textiles/

[7] Sorting and Composition Analyses of Discarded Textiles, https://www.taylorfrancis.com/chapters/edit/10.1201/9781003044413-11/sorting-composition-analyses-discarded-textiles-kirsti-cura-mikko-m%C3%A4kel%C3%A4 Cura, K., Mäkelä, M., Sorting and Composition Analyses of Discarded Textiles, https://www.taylorfrancis.com/chapters/edit/10.1201/9781003044413-11/sorting-composition-analyses-discarded-textiles-kirsti-cura-mikko-m%C3%A4kel%C3%A4

[8] "Circular economy action plan - European Commission." https://ec.europa.eu/environment/strategy/circular-economy-action-plan_en.

and minimal incineration and landfilling[9]. The requirements for supporting the store and sharing comprehensive data of TVC through whole textile ecosystems from raw material procurer to retailer highlight the importance of **decentralised technologies**. In recent years, **blockchain technology** has gone well beyond its original association with cryptocurrencies (e.g. 1st and 2nd CIRPASS projects[10,11], **EBSI** platform[12]). It is essentially a **secure** and **decentralised digital ledger** that records transactions across multiple computers in a transparent, tamper-resistant, and efficient way[13]. Features like **smart contracts** for self-executing contracts with predefined rules and enabling automated and trusted agreements trigger the tokenisation idea of DPP-based ecosystems; data traceability[14] also makes it a suitable candidate for DPP. It has shown evolving potential to address relative concerns for environmental aspects of energy usage and scalability. It has been suggested that we advance to be more secure, efficient, and versatile, opening possibilities beyond what was initially imagined[15].

Bottlenecks. DPP complexity for seamless and interoperable data communication through TVC; Scalability and confidentiality of DPP for textile data over the TVC, including raw material procurer, producers, retailers, and consumers; Lack of recyclable IoT mechanisms for textile products for autonomous data communication over the TVC; Integration of DPP capabilities with the current sorting solution; Lack of mechanisms for using DPP historical data and enhance DfD.

Innovation. SORT4CIRC will contribute to the SOTA of TTRFID tags for an IoT-enabled textile DPP focusing on recyclable chip-less technologies. SORT4CIRC will develop an eco-friendly blockchain architecture for the textile DPP model for **traceability** over TVC and **integrate** it with its proposed technologies in *Ambition 1&2*. SORT4CIRC will design the blockchain backbone of textile DPP with capabilities like an environment-friendly consensus mechanism, scalability for comprehensive TVC data, security and confidentiality of data, interoperability, and autonomous data exchange. This traceable and comprehensive information exchange model of TVC will be supported by a **tokenised architecture**, enabling analysis of textile DPP-related operational PSSR historical data for DfD and providing the basis for paradigms like CO_2 market exchange in TVC.

Ambition 4 – Creating value to trigger circularity: satisfying the overall sustainable TVC

SOTA. The sorting and recycling companies across the EU deal with stakeholders such as consumers, producers of new clothes, partner collectors, and buyers of reusable clothes and recycled raw materials. Most existing **CE BMs** focus on the pre-consumer stage, and only a limited number are proven in the post-consumer stage (furniture, carpets, durable products, and luxury clothes). A few BMs exist that operate as partnerships between a firm, its partner recycler, and charity recyclers for the collection and resale of pre-owned clothes.

Bottlenecks. Recyclers manage clothing from numerous unknown suppliers, each with unique characteristics, rendering standard partnerships, collaborations, and cost/profit-sharing models impractical. **Innovative BMs** are essential in the apparel industry, particularly because it is dominated by fast fashion, which is characterised by low prices, unlike luxury clothing. The abundance of these clothes makes sorting exhaustive, and their constant updating creates competition with new garments. Another challenge is consumer behaviour; while some are sustainability-conscious and interested in second-hand products, others prefer new items, disregarding second-hand options. Additionally, circularity may generate **less short-term value**. For instance, materials from recycled cotton clothes can be sold for a higher price when used to produce wipes rather than for clothing production. Another issue is that some circular BMs need to be **sustainable**; for example, it has been shown that collecting and shredding clothes like jeans is not environmentally sustainable in terms of CO_2 emissions. The current situation reveals that sorting capacity is limited, and its marginal cost is high, presenting a significant challenge.

Innovation. SORT4CIRC impact is supposed to increase the **capacity** and **value** of generating clothes and materials with pros and cons from competing stakeholders. So, **innovative BM**s are required to protect the EU stakeholders regarding various **sustainability metrics** in the **long term**. **Circularity** in textiles can only be reached if strong business cases are behind it. SORT4CIRC circular BM will capitalise on (i) higher sorting capacity, (ii) lower sorting and recycled material costs, (iii) a higher supply of reusable products and high-quality feedstock tailored according to relevant materials and subsequent spinning/weaving/knitting/dying process, and (iv) consider consumer behaviour and market demands. To maintain a balance among all stakeholders, the BMs will include various combinations of **revenue model**, **supply model** and appropriate markets to sell reusable products and recycled materials, along with the assessment of trade-offs. SORT4CIRC scenarios will be mathematically formulated, analytically examined or simulated and validated using empirical data to suggest 'if-then' rules to prescribe how to prescribe control mechanisms (e.g., commissioning and incentivising) to ensure all stakeholders' benefit. It will

[9] https://eur-lex.europa.eu/legal-content/EN/TXT/HTML/?uri=CELEX:52022DC0141
[10] https://cirpassproject.eu/
[11] https://cirpass2.eu/
[12] Home - EBSI - (europa.eu)
[13] https://www.ibm.com/topics/benefits-of-blockchain
[14] https://digital-strategy.ec.europa.eu/en/policies/blockchain-strategy
[15] "Energy Efficiency of Blockchain Technologies", www.eublockchainforum.eu

also include scenarios that can service the existing (France) or imminent textile Extended Producer Responsibility (EPR) in EU countries such as the Netherlands.

1.1.4 TRL assessment

Technology	*TRL at the beginning of the project*	*TRL at the end of the project*
Textile tokenised RFID (TTRFID) tags	TRL4: Chipless tag for RFID application prototype developed and validated in the lab setting	TRL6: TTRFID tags prototype demonstration in a demonstration environment
DPP blockchain platform	TRL4: SC traceability over value chain in the form of lab-validated demonstration	TRL6: Textile DPP blockchain platform for data tracing and feeding the proactive sorting in an integrated sorting environment
DPP smart contracts tokenisation	TRL4: Prototype Proof of Concept (PoC) for the smart contract in the SC validated in a lab environment	TRL6: validated models for appraisal of sustainability contribution over the TVC environment will be demonstrated in a relevant environment
Establishment of a tokenised ecosystem over TVC	TRL 3: Initial concept and feasibility of tokenisation models over SC models are demonstrated	TRL 6: Tokenisation token scenarios integrated with the TTRFID system will be demonstrated in the relevant environment
Vision system for garment attributes recognition such as wear condition	TRL 3: Classifications of laid-out garments are tested. Need to refine AI to improve the classification of garments.	TRL 6: The vision system should offer garment classification based on various attributes, including wearability and soiling conditions, product type, style, brand, colour, fabric construction and size.
SKU retrieval for garment classification ontology model	TRL 2: The need has been identified, and the core functions of the technical concept are understood. Some functions of the existing neural networks can be reused.	TRL 5: By the end of the project, a TRL 5 level shall be achieved. The technology must function and be validated in a sorting machine.
Xray detection of foreign objects	TRL5: System able to detect metallic parts within the garment	TRL6: It will be able to detect a wider range of accessories
Detection of garment composition by infrared vision	TRL4 - All compositions must be known for correct detection, and difficulties remain in detecting the composition of garments containing a low material content and textiles dyed black.	TRL6 – The composition determination of all garments, including black textiles and textiles with a small amount of material. Statistical models of spectral similarities can improve AI models to determine the composition of any unknown blend.
Garment composition accuracy	TRL4: The composition estimate is based on the analysis of the surface image. Composition discrepancy with laboratory measurements over 10% can be detected.	TRL6: New algorithms that consider the weaving pattern can reduce the composition estimation error to below 10%
Recognition of multilayer garments	TRL4: Recognition prototype based in NIR-HSI successfully validated at lab-scale for detecting some multilayer compositions including PET, cotton and PA.	TRL6: Upscale the prototype, integrate it in-line, improve robustness, and increase the number of compositions recognised.
Classification of unidentifiable materials	TRL 3: New infrared technology seems to identify differences between PA6 and PA6.6	TRL6: This technology is integrated with other conventional technologies to classify these polyamide textiles

#§PRJ-OBJ-PO§#

1.2 Methodology

#@CON-MET-CM@# #@COM-PLE-CP@#

1.2.1 Overall methodology

Due to the lack of the SOTA automated textile sorting technologies available for PSFS and the limitations of current NIR technologies used for PSSR such as identification of multilayers, material composition below 3%, black textiles based on carbon pigments, coatings and finishing chemicals, nylon6 and nylon66 identification, cotton and linen differentiation and fabric structure recognition, sorting systems must be upgraded to guarantee a reliable, affordable and accurate feedstock for second-hand markets and feedstock for recycling. With the learnings of using NIR spectroscopy for sorting PCTs at an industrial scale, the most significant challenges to ensure a reliable, high-quality feedstock allowing fibre-to-fibre recycling are **precision in composition and colour recognition, automated categorisation and quality assessment of output material.** Moreover, the heterogeneity of PCTs poses several challenges for textile disassembly and pre-processing due to the complex nature of textile structures and the diverse range of materials commonly used for performance and/or aesthetic reasons. Multi-material products represent 78% of the market (9% are multilayer, 69% are single layer with disruptors[16]), requiring specialised sorting to identify, separate and recover individual components before recycling. While the second-hand market in the EU is growing, limitation in the current sorting technologies makes most second-hand clothes exported to developing nations, creating small value. This project addresses challenges in material composition, colour recognition, and multi-material textiles, increasing the availability of reliable textiles for the second-hand market and supporting its growth.

While the recycling technologies need to scale up and further develop their processes to face the textile waste complexity, automated sorting using suitable robotics, sensors and AI technologies is required to meet their various requirements and specifications and thus facilitate the incorporation of textile waste into new yarns or products, increasing the circularity of textile products and limiting the use of natural resources. The lack of traceability of most textiles carries the risk of re-introducing unwanted textiles into the system, preventing a true circular value chain. Tracking and tracing technologies such as blockchain integrated with TTRFID tags can support addressing this issue. Therefore, structured and accessible data on material composition and substances of concern are important information for appropriate sorting and recycling. The practical application of tracking and tracing technologies is thus of utmost importance to sorters and recyclers. Also, to ensure the efficient implementation of innovative solutions, it is essential to analyse the TEF and LCA of upscaled processes in conjunction with developing viable BMs. SORT4CIRC overall methodology has been illustrated in **Figure 2**, and it will tackle the following challenges:

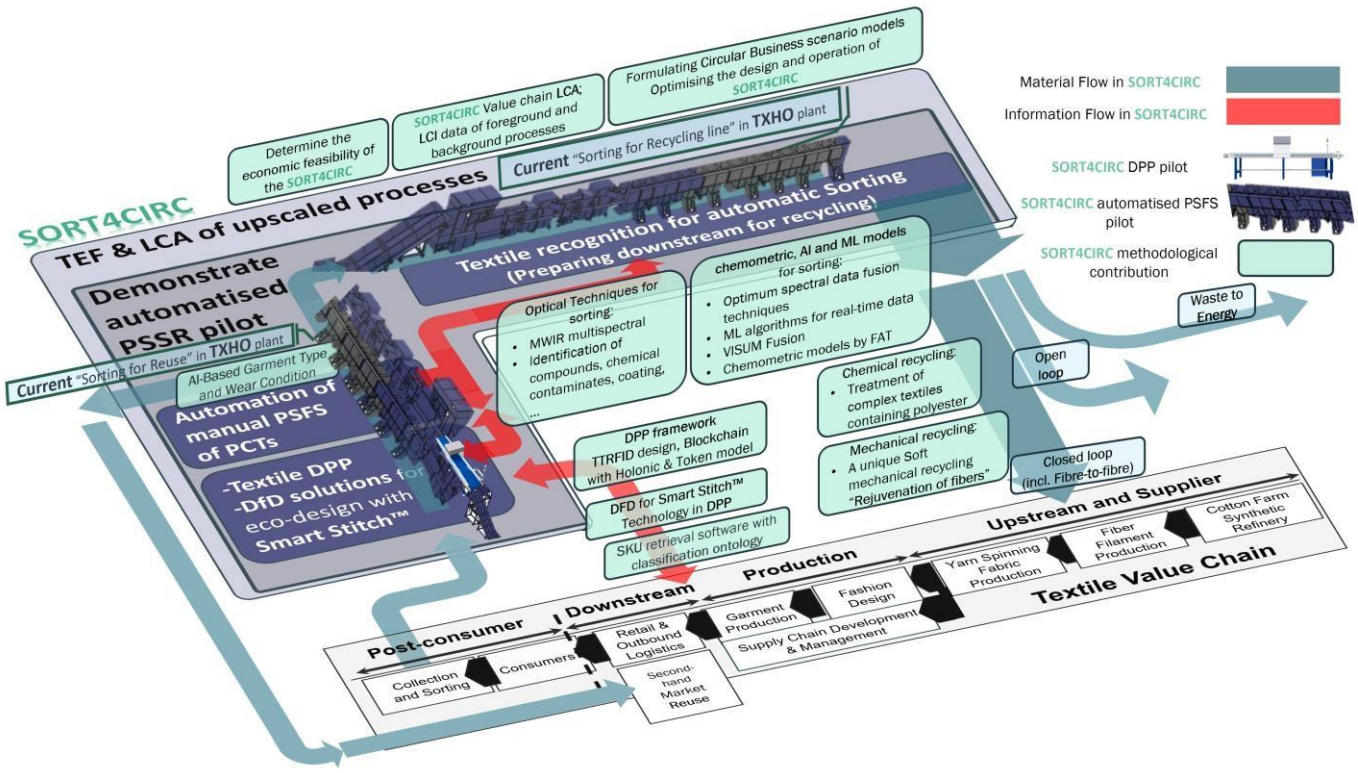

Figure 2 - Overall Methodological framework of SORT4CIRC

[16] https://refashion.fr/pro/sites/default/files/rapport-etude/Synth%C3%A8se_Caract%C3%A9risation_Refashion_2023_EN.pdf

Challenge 1 / Automation of the previously manual PSFS process for PCTs – Tackled in WP2

Garment type and wear condition determination:

The current process for determining garment type and wear condition is fully manual and labor-intensive. Particularly for products entering recycling streams, revenues often do not cover sorting costs. Automating this process is essential to establishing a cost-effective circular economy (CE) model. The proposed AI-based technology supports both pre-sorting and fine sorting by automatically detecting product types and wear conditions (e.g., holes, pilling, and discoloration). Garments are spread on a conveyor belt, scanned from both sides with high-resolution imaging, and sorting software assigns them to appropriate categories. The items are then automatically directed into the correct bins via a switch system.

To ensure technical robustness, the AI systems utilize machine learning models trained on diverse, high-quality datasets that account for the variability in garment types, wear conditions, and additional features such as style and fiber composition. High-resolution imaging is rigorously tested under industrial conditions to maintain consistent performance despite challenges like dust, vibrations, and lighting fluctuations. Quality assurance measures include ML pipelines for continuous monitoring, automated retraining based on new data, and lifecycle management for sustained reliability. Additional mechanisms such as periodic calibration, benchmarking against industry standards, and fallback strategies (e.g., manual sorting or reclassification) enhance robustness in large-scale operations.

Retrieval of product data without data carrier:

Product data from e-commerce platforms and, in the future, from Digital Product Passports (DPPs), can support sorting decisions. The connection between a product to be sorted and its data can be established via data carriers embedded in the clothing item. However, many clothing items currently lack such carriers, and it is unclear whether all DPP-enabled products will integrate easily readable carriers in the sorting process. For example, hidden QR codes may not be readable under high-speed sorting conditions.

To overcome this challenge, existing AI algorithms will be enhanced to recognize individual products and infer their SKUs using image similarity techniques. These systems will leverage advanced computer vision models and continuously updated product databases to match garments with their digital profiles. To ensure reliability, the models will be subjected to stress testing for edge cases, such as heavily worn garments or partial visibility. The integration of this product data into the sorting process will significantly improve sorting quality and efficiency.

Facilitating the sorting for reuse:

Current buyers of wearable clothing are mainly second-hand shops in Europe and exporters to the Global South. However, as collection volumes increase in Europe, second-hand markets risk becoming saturated, potentially leading to wearable products being recycled prematurely. To address this, new channels for reuse must be unlocked. Suitable options include retailers that currently only purchase new goods, as well as efficient repair and upcycling services.

Challenge 2 / Enable automatic sorting in real-time through the development of a textile recognition system – Tackled in WP3

MAIN SUB-CHALLENGE 1: EXPANSION OF CONVENTIONAL TEXTILE SORTING SYSTEMS: To increase the reliability of the identification of black textiles, multilayers and interfering material, it is critical to enrich the conventional textile sorting systems with appropriate new measurement technologies. If different materials, impurities or production methods of garments cannot be adequately detected using conventional measurement technology, such textiles will inevitably harm the purity of the end product. SORT4CIRC will develop textile identification using a vision system to recognise and classify a broader range of textiles, including several optical techniques offering new detection functions (e.g. low-content materials, black C-based textiles, PA6/PA6.6, cotton and linen differentiation, multilayers, etc.). A demonstrator equipped with AI-based algorithms capable of classifying into 15 categories at an industrial speed will be tested first out of line using hyperspectral imaging systems and other methods.

For **optical techniques**, experimental approaches will be mainly based on NIR imaging or spectroscopy. Still, the characterisation of black textiles, particularly those with carbon-based pigments, is very difficult due to the extremely low reflectance of such materials. This issue affects all experimental techniques based on reflection (including NIR spectroscopy/imaging & Raman). The potential of other techniques to solve specific difficult analytical problems will be tested. For example, CEA will perform a spectral analysis ranging from visible to long wave infrared using a Fourier transform infrared (FTIR) to define the best bands to classify carbon black textiles. This will be followed by mounting an adapted MWIR multispectral imaging to classify black textiles. CEA and IOM will also deal with (i) the identification of certain compounds with a minor concentration in the textile, such as elastane (content < 3wt%), (ii) the detection and identification of chemical contaminates such as oils, greases, varnishes, rust, etc., but also moisture (wet textiles), (iii) the detection and identification of finishing agents (e.g. flame retardants) and coatings.

For **chemometric, AI and ML models,** using the selected optical techniques, different types of models based on AI and/or chemometric and statistical approaches will be developed for extracting actionable information to accurately

identify complex textile waste (multilayer structures, various fabric structures such as woven knitted or non-woven patterns, black textiles, chemical contaminants, coatings and minor components). The models developed will be embedded into an efficient and replicable textile real-time recognition multi-sensor platform. By leveraging high-resolution imaging and AI, the identification system aims to enhance the precision and efficiency of detection, contributing to improved sorting and processing in textile applications.

In a first stage (offline), the technical robustness of the AI models (supervised machine learning such as SVMs (Support Vector Machines with different kernels), NNs (Neural Networks) and Decision Trees (e.g. Random Forest) will initially be validated by using relevant benchmark offline train and test datasets including different boundary conditions for which the outcomes are known. This will be combined with a *consensus approach* to cross-validate using different contrasting techniques: statistical (e.g. PLS-DA, Regression) *vs* AI *vs* human expert. This will allow the quantification of the false/true positives/negatives under different conditions using realistic/representative train and test data. In a second stage (online, once integrated in the multi-sensor platform), the field experiment results will be evaluated by the *consensus approach*, including human experts in which the automatically generated output labels will be contrasted with expert assigned labels. This will be combined with a DoE (Design of Experiment) to test robustness and reliability under different boundary conditions and combinations (different types of textile materials and properties) which are expected in the real operations environment.

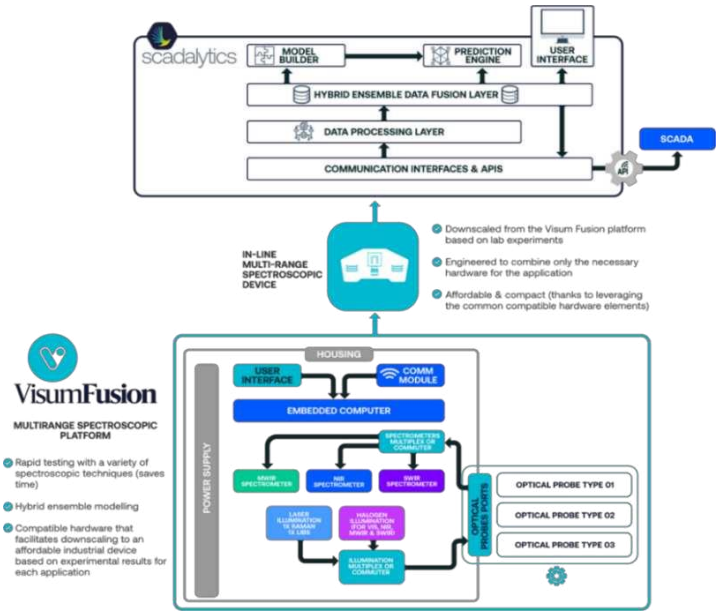

Figure 3 - Downscaling concept: from the Fusion Test Bench to a case-wise optimum set-up

Based on the preliminary SORT4CIRC results, the optimum spectral data fusion techniques as described in **Figure 3** will be defined using innovative hybrid assembling for their configuration into a reliable and cost-effective in-line monitoring solution based on a smart sensing device with automated data acquisition system and ML algorithms to enable real-time data analysis and predictive systems. A leveraged novel Multirange Spectroscopic Platform (VISUM Fusion - IRIS) will measure the predefined critical parameters and indicators. This platform was designed and engineered as a flexible test rig to deliver the most suitable real-time monitoring solution from the following sensing techniques in one comprehensive platform: SWIR spectroscopy, MWIR spectroscopy, NIR spectroscopy, VIS spectroscopy, Raman spectroscopy, Fluorescence and LIBS spectroscopy. This comprehensive experimental unit enables rapid testing with a variety of spectroscopic techniques to save time in the experimental phase to allow the most suitable techniques to be identified, informing the 'downscaling' (**Figure 3**) to a compact and affordable industrial inline analytical device that benefits from the common hardware compatible ('Fusion Core') of the VISUM Fusion. To further extend the monitoring capabilities, data from off-the-shelf compatible univariate sensors can be fused with the spectroscopy and imaging data related to textile parameters to provide a more robust mathematical model of automatic control of the sorting process.

As such, the VISUM Fusion multirange spectroscopic platform will facilitate the effective 'downgrading' to the most successful configuration of techniques that will be engineered into a compact, reliable and affordable self-contained inline device for seamless integration into the automatic sorting process line at TXHO (including the mechanical requirements coupling it with the feeding and sorting system currently available at TXHO). The recognition speed of textiles will be around 750kg/h. The chemometric models will be tested and validated through Factory Acceptance Tests (FAT). In addition to multivariate data from spectroscopy analysers, univariate data from available traditional sensors installed in the pilot unit will also be assessed to explore the data fusion opportunities.

MAIN SUB-CHALLENGE 2: IDENTIFICATION OF CONTAMINANTS, COATINGS OR CHEMICAL CONTAMINANTS USED FOR FUNCTIONAL TEXTILES: According to the needs imposed by recyclers and, consequently, by the sorting process, categorisation and quality assessment must be carried out objectively. SORT4CIRC will test the LIBS technology, Raman and hyperspectral imaging to identify chemical contaminants or unknown surface substances in the pre- or post-sorting step because such additives could interfere with certain chemical recycling processes. LIBS will be used to identify metal objects and sharp accessories to direct them out of the loop.

MAIN SUB-CHALLENGE 3: OPTIMISE THE SORTING AND DISASSEMBLY PROCESS FOR ECO-DESIGNED PRODUCTS WITH DISASSEMBLY CAPABILITIES: The pre-sorting step identifies garments made for disassembly and separates them from the waste stream to enable a subsequent disassembly before recycling. Seamlessly integrate pre-processing

and post-sorting technologies into a continuous industrial process by using heat-dissolvable bio-based and synthetic stitching threads and thermal disassembly system to enhance the disassembly process, boost recycling yields and unlock new recycling opportunities for previously unattainable textile products. Combining sorting and disassembly technologies will demonstrate higher efficiency and purity levels in the pre-processing of textile products. The automatic disassembly and sorting will be tested on various garment types.

MAIN SUB-CHALLENGE 4: BETTER DOCUMENTATION OF THE PROCESS STEPS AND CONTINUOUS QUALITY ASSURANCE OF THE END PRODUCT: Automation has also introduced digital technology into the sorting process. SORT4CIRC will enable recording relevant parameters inline and introduce automatic, continuous documentation of process data and quality assurance. A data management platform will be developed, and data will be acquired from the Multirange Spectroscopic smart sensing system and traditional sensors. In addition, other types of information from the garment (i.e. market demand data, DPP and traceability data) will also be collected. All the data will be gathered in a data repository and fused for feeding predictive models developed in this task to obtain actionable information and decision-making concerning the actions to be taken for each garment.

Challenge 3 / Increase the information and its traceability about the origin and content of post-consumer clothes, designing and assessing a tokenisation model for stakeholders' engagement– Tackled in WP4

MAIN SUB-CHALLENGE 1: DEVELOP AN ECO-FRIENDLY & RECYCLABLE CUSTOMISED TTRFID TAG MODEL: The textile DPP methodological implementation will cover all the TVC data from upstream to the end consumer. **TTRFID tags** with textile products compatibility will be accomplished by developing a specialised RFID tag model tailored for textile applications that is both eco-friendly and recyclable with the help of SCATTERERID project. The RFID tags will store and transmit comprehensive data about each textile item, such as material composition, lifecycle information, and CO_2 emissions. The chosen RFID technology will be designed to withstand typical textile processes, including washing and heat exposure. In that sense, there are 7 possible TRL 4 alternatives: a bi-layer of fleece/non-woven[17], an embroidering polyamide fully silver-coated HC12[18], 2 distinct PCB types (Rogers 3003 and Rogers 58)[19], a Taconic board on fleece substrate[20], a Rogers4350 dielectric substrate[21], an acrylonitrile butadiene styrene (ABS) substrate[22] and a nickel-copper-coated ripstop[23]. There will be 3 main key points to check: (i) contaminant detection on the final product; even a small amount of TTRFID chip embedded in the fabric could alter the recycling process; (ii) **thermal resistance**, methanolysis processes require high temperatures (>150°C) and can affect and degrade chips, this needs to be checked; (iii) **RFID chip components recycling,** to be fully integrated in the recycling process, materials used for RFID manufacturing should be recyclable and do not affect the recycling process.

MAIN SUB-CHALLENGE 2: ENABLE TRACEABILITY OF TEXTILE DATA OVER THE TVC ECOSYSTEM THROUGH A BLOCKCHAIN PLATFORM: **The blockchain platform** enables the traceability of comprehensive TVC data for concrete analysis and interpretation of the characteristics of textile products for its raw material, production and manufacturing processes and collection and transfer operation to the sorting and recycling points as illustrated in **Figure 4**. A tokenised architecture will help realise the sustainable enabled exchange market for textile products. The pre-sorting area will be equipped with suitable reading devices to incorporate relevant information into the decision-making process and process documentation. The architecture will use Algorand, IOTA or Ethereum for its eco-friendly Pure Proof of Stake (PoS) consensus and support for smart contracts over the **EBSI platform**.

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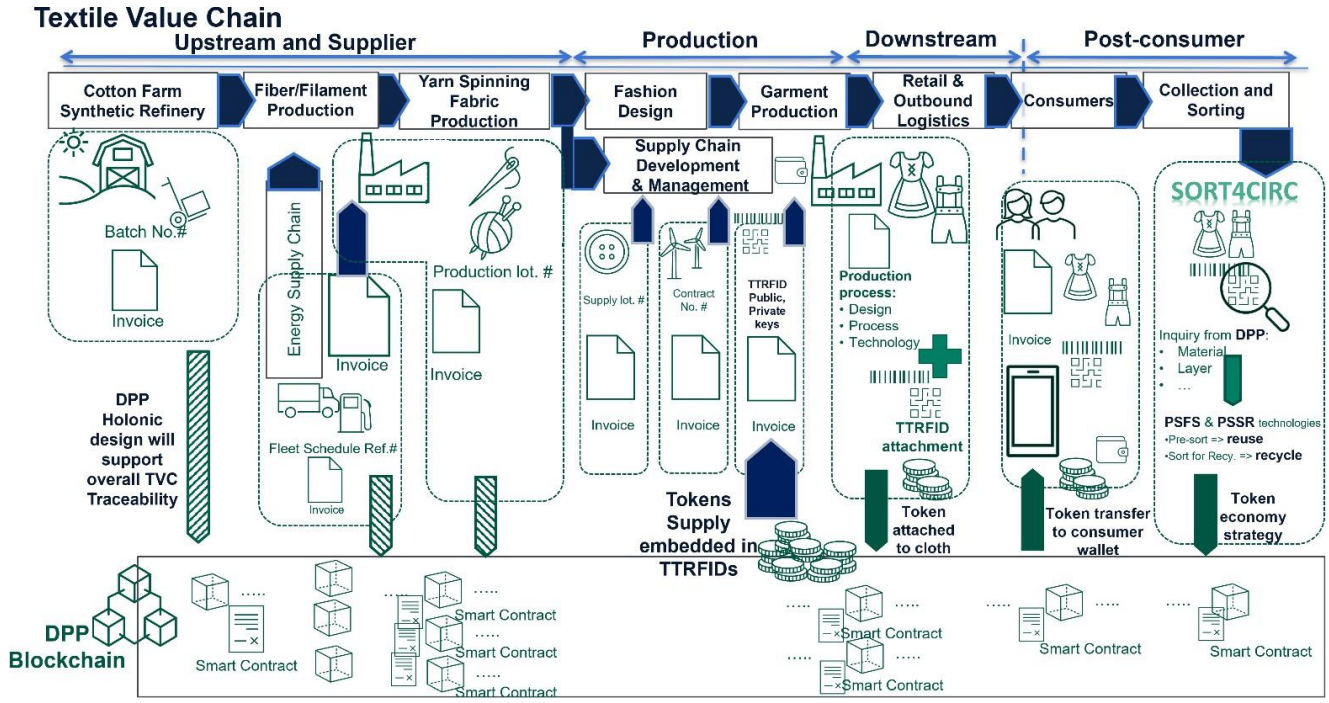

Figure 4 – Traceability of TVC ecosystem scope in SORT4CIRC and the tokenisation architecture

To achieve this, the blockchain architecture will leverage Semantic Web technologies to enhance data interoperability and machine readability across different platforms. This will involve using Resource Description Framework (RDF) and Web Ontology Language (OWL) to create a shared vocabulary and data model that all participants in the TVC ecosystem can understand and use. By employing these technologies, we ensure that textile data is uniformly interpreted and utilised across various systems and complying with standards like IEC 62264: (ISA-95) and IEC 62443: (ISA -99) and ISO 27001 Certification. APIs and standardised data formats will facilitate communication between systems, allowing for real-time updates and retrieval of textile data. **Blockchain Architecture on Generation 2 or 3 Blockchains.** Second-generation blockchains focus on enabling smart contract functionality and decentralised applications (dApps). Key requirements include robust smart contract capabilities, efficient consensus algorithms to reduce energy consumption, high transaction throughput to handle frequent data updates, and secure and scalable infrastructure to support large-scale operations. Third-generation blockchains aim to solve scalability and interoperability issues. Key requirements include advanced scalability solutions to support a high volume of transactions without compromising performance, enhanced interoperability to facilitate seamless integration with other blockchain networks and legacy systems, improved security features to protect against emerging threats, and sustainable and energy-efficient consensus mechanisms, which are also emphasised through other DPP related initiatives like CIRPASS & CIRPASS-2 projects.

MAIN SUB-CHALLENGE 3: ENABLE TRACEABILITY OF TEXTILE DATA OVER THE WHOLE TVC ECOSYSTEM CONSIDERING THE RAW MATERIAL AND VERTICAL AND HORIZONTAL INTERACTIONS: Holonic View and Merging Process. Incorporating the Holonic view ensures a holistic approach to data integration and management, facilitating seamless communication and interoperability between diverse systems within the TVC ecosystem. This perspective emphasises the importance of a unified data model and consistent data exchange protocols, which are crucial for successfully implementing a blockchain-based traceability system over the whole TVC, as illustrated in **Figure 4**. The merging process involves integrating data from various sources through the Holonic structure of tokens, enabling the TVC coverage for elements and even Energy and CO_2 emissions through the whole TVC and ensuring consistency and accuracy across the blockchain. This includes aligning data formats, standardising protocols, and implementing robust data validation mechanisms. Integrating IoT and blockchain technologies will enable real-time data capture and transmission, ensuring that all relevant information is accurately recorded and shared across the ecosystem. The system will (i) enhance transparency and trust, (ii) enhance security and privacy, and (iii) enable comprehensive data tracking. **Developing interfaces** to connect the DPP data retrieval system with automated sorting machines, ensuring seamless data exchange. This will be achieved using APIs and standardised data formats, such as JSON or XML, to facilitate communication between systems, allowing for real-time updates and retrieval of textile data. IoT sensors and readers will be deployed at various stages of the sorting process to automatically capture and transmit data from TTRFID tags to the DPP system. The data retrieved from the DPP will inform sorting decisions, such as identifying textiles suitable for recycling, reuse, or disposal. The data from the DPP will be accessible to all relevant stakeholders within the TVC ecosystem.

The integrated system will ensure compliance with environmental regulations and sustainability standards, providing

detailed reports on textiles' lifecycle and environmental impact. It will also track and document the reduction of CO_2 emissions and other environmental benefits achieved through the automated sorting and recycling processes and scale with increasing volumes of textiles, ensuring it can handle large amounts of data and high throughput of items. Flexibility in the system design will allow for easy updates and adaptations to accommodate new sorting technologies, regulatory changes, and evolving industry practices. For example, if pre-sorting is required to distinguish specific categories of garments, this query can be defined over the DPP smart contract. Another query could be designed for sorting in recycling to distinguish between multi-layer and single-layer garments or to identify different percentages of cotton and synthetic blends. This could be integrated into a **token economy** to incentivise sustainable practices and track environmental impacts within the TVC ecosystem. In the tokenised model, each participant within the TVC must purchase tokens to gain permission for CO_2 emissions and energy consumption. These tokens function similarly to credits that must be authenticated and verified through blockchain transactions. The process begins with each stakeholder buying an initial set of tokens proportional to their projected emissions and energy usage. These tokens are stored in a digital wallet and managed on the DPP blockchain, providing a transparent and immutable record of all transactions and balances. The PoS consensus mechanism will be employed to enhance the eco-friendliness and efficiency of the DPP blockchain. Stakeholders will stake their tokens, and validators will be chosen to add new blocks based on the number of tokens they hold. This method reduces the computational power required compared to traditional proof-of-work systems, aligning with the environmental goals of the project. The tokenised system also includes a PoC phase to validate the practical implementation of this model. During the PoC, selected participants will use the system in a controlled environment to test the accuracy of IoT data integration, the functionality of smart contracts, and the overall user experience. Feedback from this phase will be crucial in refining the system before a full-scale rollout. Smart contracts will automate the management and verification of tokens. For example, when a firm emits CO_2 or consumes energy, smart contracts automatically deduct the appropriate number of tokens from their digital wallets. **DPP enabler for DfD solutions** to facilitate recycling and reuse of textiles. This includes the development of guidelines and technologies that enable easy disassembly of textile products using insights from DPP TVC data. Implementing layering strategies where different materials can be easily separated ensures that each layer can be processed individually during recycling. DPP model should enable labelling and information inclusion. TTRFID and the QR codes should be integrated to provide detailed information about the materials and construction techniques used in the product. Developing user manuals that include instructions for efficient and safe disassembly further supports this initiative. In terms of technology, analysing the DPP historical data over TVC should highlight the DfD principles, including the self-disassembling materials, which change properties under specific conditions and can also facilitate easier disassembly.

Challenge 4 / Demonstrate the automatised PSFS and PSFR pilot system – Tackled in WP5

MAIN SUB-CHALLENGE 1: EXISTING SYSTEM IMPLEMENTATION: Once the test phases of the pre-sorting pilot and the extended sorting system for recycling have been successfully completed, they must be implemented in the existing production lines. This requires both the connection of the control technology to the higher-level system control and to the material feed system. In addition, the new sorting unit must be integrated into the output of the pre-sorting pilot, with which the garments classified into 15 categories can be specifically sorted into 16 different containers (**Figure 5**).

MAIN SUB-CHALLENGE 2: DEMONSTRATE THE OPERATIONAL CAPABILITIES OF THE ADVANCED SORTING AND PRE-SORTING LINES: The existing pre- and sorting line will be expanded and adapted with suitable additional measurement technology and sensors with integrated monitoring to demonstrate automatic high-volume textile treatment (speed 1m/s) with increased reliability of black clothing identification and disruptive materials and the ability to adapt the quality of the source textile material to the future requirements of mechanical and chemical/enzymatic recycling companies. The integration could be carried out in a two-stage process. First, the pilot plant is set up near the existing manual pre-sorting system and supplied with material offline via a bypass. In this way, the old production process is not hindered, tests can be carried out, and improvements can be made independently at the pilot plant. The pre-sorting system will identify the material, sort it into 15 defined

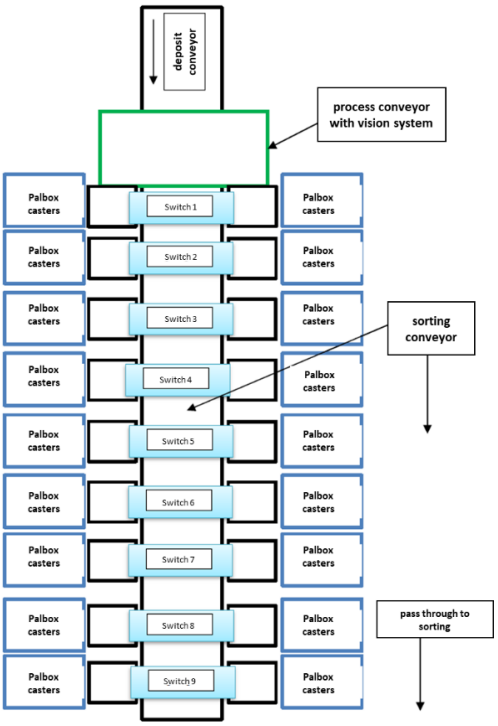

Figure 5– New sorting unit to classify garments into 18 containers

material types (fitting the actual categories at TXHO facilities) and fill it into 18 separate containers (including a waste/unidentified container and 2 "buffer" containers to facilitate container exchange without stopping sorting -

Figure 5). Technical drawings and design documents for the sorting plant, as well as hall plans, are already available

in full. The system can also be equipped with modules containing conveyor belts for material transport and stable frames for the mechanical mounting of measuring devices or RFID detectors. After the successful test phase, the manual pre-sorting system could be replaced by the pilot system and automated sorting process. TTRFID will be deployed in the demonstrator, including various categories like textile products with different materials (cotton, synthetic and DfD-enabled heat-dissolvable bio-based stitching thread) and production technologies (one-layer or multi-layer). The required infrastructure will include gates and readers for communicating data with TTRFID.

MAIN SUB-CHALLENGE 3: VALIDATION OF ALL TECHNOLOGIES BY RECYCLERS:

Chemical Recycling: The chemical recycling process developed by RECY allows the treatment of complex textiles containing polyester. This process selectively depolymerises (recycles) polyester without damaging the other co-materials present (elastane, polyamide, cotton) (**Figure 6**). Several tests will be run during the project to assess that the quality of the sorted feedstock reaches the recyclers' requirements: at laboratory phase: tests on 10-100g of textile waste /batch and at pilot scale with different sizes: (i) 30L: tests on 1-2kg of textile/batch; (ii) 100L: tests on 6-10kg of textile/batch and (iii) 1000L: tests on 50-80kg of textile/batch. At the end of the process, the final products obtained are monomers from the polyester part that could be repolymerised to obtain recycled polyester pellets and separated pure co-materials that are reusable/recyclable. A larger test scale (1000L) will provide results closer to those of an industrial-scale process.

Figure 6– Chemical recycling process

Mechanical Recycling: Purfi developed a unique recycling technology, soft mechanical recycling, that offers a solution to conventional recycling technologies. The process is called the rejuvenation of fibres, as Purfi tries to restore them to parameters likQqaa 11B virgin materials. Rejuvenated fibres can be reused to manufacture high-quality clothing. With an existing line of 150m and a capacity of up to 8 tonnes daily, it is ready to be scaled up to industrial levels. In-house tests on different parameters help maintain the requested parameters. The overall SORT4CIRC sorting process will give access to high-quality feedstock for all kinds of recycling (**Figure 7**).

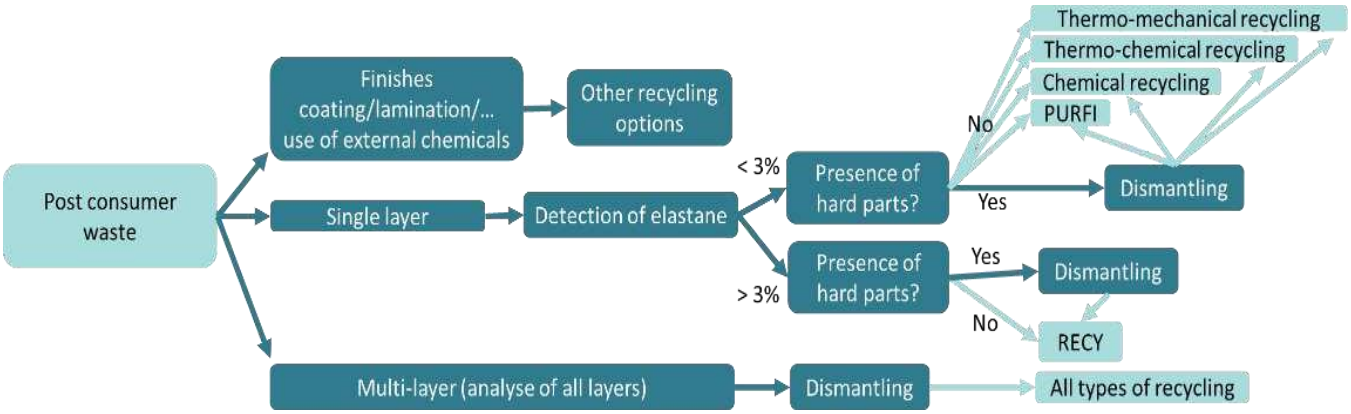

Figure 7 – Pathway to all recycling options

Challenge 5 / Analysing TEF and LCA of upscaled processes along with BMs to ensure actual implementation of the innovative solutions – Tackled in WP6

MAIN SUB-CHALLENGE 1: LCI DETERMINATION AND ECONOMIC AND ENVIRONMENTAL IMPACT ASSESSMENT:

All the inventory data required for the techno-economic and environmental impact calculations for both the existing and the SORT4CIRC process will be based on the mass and energy balances depicted on the corresponding process

flow diagrams (section 2.1.7- **Figure 9**). **MAIN SUB-CHALLENGE 1: LCI DETERMINATION AND ECONOMIC AND**

ENVIRONMENTAL IMPACT ASSESSMENT: All the inventory data required for the techno-economic and environmental impact calculations for both the existing and the SORT4CIRC process will be based on the mass and energy balances depicted on the corresponding process flow diagrams (section 2.1.7 - **Figure 9**). The Technico-Economic assessment (TEA) will determine the economic feasibility of the SORT4CIRC process. The equipment cost estimation will be based on data provided by the respective partners. An extrapolation of the calculated equipment cost to the year of the unit construction (using the appropriate cost indices) will be followed. CAPEX and OPEX (considering the applied operating conditions) will be estimated for both SORT4CIRC and the conventional sorting process. The findings will also serve as input to the DPP token economy model.

To estimate the SORT4CIRC value chain environmental performance, a LCA will fully comply with ISO14040 and 14044 standards. The boundary limits determine the applied methodology for the environmental assessment. Initially, an ex-ante, attributional, gate-to-gate approach (starting from the point of receiving the textile – and ending at the point that the textile is ready to leave the sorting facility) will be employed to estimate the direct impact of replacing the existing sorting process by the innovative SORT4CIRC process. Subsequently, an ex-ante, consequential cradle-to-cradle LCA study will be performed to estimate the sequential impact and track the benefits derived from the implementation of the SORT4CIRC process to the whole value chain of the textile recycling process. Thus, the boundary limits will be expanded, including all the upstream (collection and transportation) and downstream (distribution, recycling, etc.) processes. The downstream processes depend on the further processing the sorted textiles (upcycling/downcycling/burning). The LCI data of all foreground and background processes will be modelled using the *ecoinvent* database v3.10. To quantify the environmental impacts and be aligned with the European Directives, the EU recommends applying the Environmental Footprint v3.1 method. Uncertainty analysis will also be applied to examine the accuracy of the environmental impact assessment. The purpose is to quantify the output values' fluctuation (quantification of the error margin), which derives from the volatility or lack of knowledge of the input data, by developing non-linear uncertainty models and estimating statistical quantities of interest. Therefore, the 1st step will identify potential errors and pitfalls propagated to the output calculated values.

The economic and environmental analysis will eventually enable fair technology benchmarking of the developing SORT4CIRC process against other state-of-the-art sorting processes, including the existing one deployed by TXHO. Data about relevant technologies will be gathered through a literature review, and the key comparators will be defined. This comparison will help us to determine the breakeven sorting cost, identify the competitive position of the SORT4CIRC sorting process within the market and define the areas of improvement.

MAIN SUB-CHALLENGE 2: DESIGNING INNOVATIVE BMS SECURING VIABLE CIRCULARITY OF COLLECTED TEXTILES OVER LONG-TERM THROUGH ECONOMIC EFFICIENCY: The TVC involves many stakeholders, including producers of new clothes, consumers, collectors, sorters, and recyclers. SORT4CIRC introduces new technology-driven features into this value chain, enhancing the efficiency of circularity by improving the sorting process for collected textiles. In turn, this supplies the post-consumer TVC with higher-quality and greater quantities of materials suitable for reuse or recycling while also opening new business opportunities that require precise configurations and strategic decision-making. Addressing this business challenge effectively requires the development of innovative BMs that establish the framework for stakeholder interaction and coordination throughout the value chain, which are essential for business success and viability. A key challenge to address in the future is determining the quantity of sorted re-wearable clothing to be distributed in the market considering the market demand, establishing appropriate pricing, and identifying the most effective sales channels. Another critical issue is assessing the financial viability of selling second-hand clothing alongside new collections for companies adhering to the EPR policy and evaluating how this policy might impact profitability. Understanding and analysing consumer behaviour regarding new collections and second-hand clothing is crucial, which necessitates further investigation. To overcome these challenges and to develop robust BMs, SORT4CIRC will apply analytical techniques (e.g., game-theoretical model) considering critical factors (e.g., market demand, market competition, consumer behaviour, price, etc.), which enables us to discover an efficient setting for the circularity of the sorted textile.

MAIN SUB-CHALLENGE 3: OPTIMISING THE DESIGN AND OPERATION OF THE AUTOMATED SORTING SYSTEM FOR ENHANCED EFFICIENCY: Capacity planning of the new sorting system is challenging and crucial to ensure the system can handle expected volumes without bottlenecks. This includes accurately determining the number of facilities and staff, designing a practical layout, ensuring smooth human-machine interaction, and optimising task assignment and line balancing. To overcome this challenge, SORT4CIRC will apply a systematic, stepwise, and recursive approach, where each relevant facility design problem, including equipment selection, staff requirements, facility layout planning, and line balancing, is first dealt with separately in a reasonable sequence, mainly through the optimisation modelling techniques (e.g., mathematical programming). Collecting and combining the results from solving these problems leads to an initial system design. In the second stage, a fine-grained simulation model of the initially designed sorting system will be developed using FlexSim© Simulation software to assess the system's operational efficiency. This model can mimic the behaviour of a real-life sorting system by capturing detailed characteristics and incorporating dynamic and stochastic elements. The developed simulation model enables us to

primarily verify whether the designed sorting system operates smoothly and efficiently and to reveal any existing bottlenecks. Thus, if the designed system does not meet the design requirements and key performance indicators (KPIs), we can adjust the model parameters directly in the simulation model based on the experts' opinion or by tuning the model's assumptions serving the simulation model. This recursive loop continues until we develop a system that fits requirements and satisfies the KPIs. The application of the simulation model is polyvalent in this project and is not limited to bottleneck identification and system efficiency checks. This model is also employed in comparative studies to evaluate the existing sorting system against the automated, technology-integrated system SORT4CIRC aims to develop. The visual and dynamic aspects of the simulation model are also precious in making complex systems more straightforward to understand, facilitating communication with stakeholders, and gaining insights into the system's behaviour under various conditions.

1.2.2 Relations with other projects

Part.	Project – year	Link to SORT4CIRC
TXHO	Refashion 2024 - 2025	The "Material Sorting & Preprocessing and Recycling Experiments" is a collaborative initiative led by Refashion, French national EPR organization, that aimed at accelerating the development of industrial-scale recycling solutions for post-consumer textiles and footwear, in line with broader circular economy goals. The project brings together actors from across the textile value chain to test and refine innovative approaches—such as optical sorting, advanced preprocessing and quality control—that improve the alignment between collected materials and recyclers' technical requirements. Since 2024, TXHO has taken part in the project and contributed to 31 experiments. These experiments have focused on evaluating the recycling potential of optically sorted fractions, including polyester, cotton, viscose, polyamide, rubber and synthetics. TXHO is an active member of Refashion and will share the knowledge for enriching the textile value chain systematic points in SORT4CIRC.
RECY	REGIOGREEN TEX 2023-2026	Promoting collaboration in research and development for the textile industry to establish a systematic CE BM across the EU.
CEA	MISTERY 2022-2023	It aimed to find out if hyperspectral imaging in the NIR or SWIR domain can improve textile material identification and assess this technique's potential to identify the blend composition, detect low percentages of elastane and differentiate between PA6/ PA6.6.
CU	CIRPASS & CIRPASS-2 2022-2027	CIRPASS prepared the ground for the gradual piloting and deployment of the DPP and developed roadmaps for prototypes in three value chains: electronics, batteries and textiles. CIRPASS-2 will demonstrate functioning DPP in real settings through circular pilot deployments and use cases in textiles, electrical and electronic equipment, tyres and construction value chains. SORT4CIRC will benefit from the resources provided in these projects and validate its outcomes based on the use cases for the textile industry.
REFA	Fashionsort.ai 2023-2026	The project partners TU Berlin and circular.fashion develop a sorting machine that enables automated sorting for reuse (including quality). We are preparing the IP transfer from these two project partners to reverse.fashion, the IP transfer should be finalised by the start of SORT4CIRC. The project's developments are the basis for the sorting developments described in this project.
PURFI	tExtended 2022-2026	It aims to provide feedstock for circularity and demonstrate how a CE can only be made possible with automated sorting. In addition, it aims to increase the adoption of mechanical and chemical recycling through innovative approaches.
IRIS	WHITECYCLE 2022-2026	It will introduce a two-fold process that will allow the conversion of PET waste into high-added-value products. It will be demonstrated in a series of real cases in preparation for market uptake for high-added-value applications (including multi-material textiles). IRIS is working on recognising PET within multilayer textiles and obtaining successful results for specific garments.
CEF	Fibersort: Closing the loop in the textiles industry 2016 - 2019	The two main challenges are the environmental need to reduce the impact of virgin textile materials and the development of new BMs and open markets for the growing amounts of recyclable textiles in North-West Europe. It expects to realise the implementation of Fibersort technology as the new industry standard and key value-adding step to enable high-value textile-to-textile recycling in the region.
	SOLSTICE 2024-2027	Demonstration of 4 innovative chemical recycling technologies (TRL7) able to recycle efficiently, in mild conditions, part of the major streams of polymers in textiles, and subsequent transfer of the by-products between demonstrators for relevant integration.
CU	SCATTERERID 2018-2024	It seeks to eliminate the human presence in the product identification process. One of its primary objectives is to automate the product identification process by developing RFID tags. In the next phase, the project aims to create environmentally compatible tags while also reducing costs and power

		consumption through design improvements. A key outcome of this project is the development of chipless passive RFID tags. SORT4CIRC will benefit from the chipless RFID models that enable the TTRFID model.

1.2.3 Do Not Significant Harm Principle

SORT4CIRC will adhere to **the 'do no significant harm' principle of the EU Taxonomy Regulation (EU) No 2020/852 by promoting the automation of textile sorting,** allowing a circularity in this sector, reducing the environmental impact of textile manufacturing, in terms of resource consumption, and impact on biodiversity.

1.2.4 Interdisciplinary

SORT4CIRC **is based on strong interdisciplinary cooperation** between all partners, with expertise in sorting, recycling, reuse, technology development (AI, NIR, DPP, hyperspectral system), and environmental/economic/societal impact assessment. SORT4CIRC counts on a strong involvement of industry (3 including the coordinator) and SMEs (5) among the 14 partners. In SORT4CIRC, data mining, knowledge modelling, and multicriteria optimisation with uncertainty will used by applying existing AI techniques or developing new algorithms. Industrial involved partners have the necessary computational intelligence competencies to apply **AI technologies**. They will comply with the underdeveloped legal framework of the AI Act in the EU to guarantee the safety and fundamental rights of people and businesses in TVC.

1.2.5 Gender Dimension

The SORT4CIRC project consortium understands the gender dimension to be a key part of the definition of our vision of diversity and equal opportunities. The consortium understands, values, and promotes diversity as a matter of principle, pursuing the goal of equal opportunities. For this reason, the SORT4CIRC project organisation, execution, and results are suitable for all individuals regardless of their gender identity, sex, age or ethnicity. Technology expertise, academic and professional backgrounds, and relevant job skills are the only criteria for being part of the SORT4CIRC consortium, which comprises both men and women. Moreover, the gender dimensions of the project have been further assessed using the Engineering Checklist from Gendered Innovations[24]. The end users of SORT4CIRC results will be textile sorters. Traditionally, textile sorting often employed a significant number of women. Automation could lead to job displacement, disproportionately affecting women holding these positions (80% of the world's garment workers are women[25]). However, automation will require new skills, such as operating and maintaining automated systems. This shift could create opportunities for both men and women, regardless of gender, to upskill and transition into more technical roles. All end users alike expect CE in textiles to be easily accessible. All innovations in this area will be equally relevant and available to both men and women. The main application area is textile sorting in the EU. As explained above, the textile sector is female-dominated. However, SORT4CIRC will ensure technological development suitable for all individuals regardless of gender identity, sex, age, or ethnicity. Finally, it is important to restate that for the SORT4CIRC consortium, the diversity issue goes beyond gender equality. SORT4CIRC strives for the goal of equal opportunities without room for any discrimination. For this purpose, an interdisciplinary team has been assembled, and a cooperative work process is strived for as the basis for achieving successful project work.

1.2.6 Open Science Practices

An important role is **open access and dissemination of the main experience and knowledge** analysed by the consortium and the replicability of the results of SORT4CIRC. The FAIR Research Data Management approach outlined in the following section will foster open, cooperative work and more efficient knowledge exploitation during and after the project. That will be carried out by sharing as much as possible collected data and knowledge within the project with the consortium members and actively participating in panel discussions, workshops, and conferences, where appropriate, respecting the sensitive nature of some of the project outcomes. SORT4CIRC will specifically link to any relevant activity within the ECOSYSTEX community to share best practices along the sustainable TVC. Data available on the project website will be stored in the most open and common formats possible. **Scholarly publishing** – collected results and findings within the SORT4CIRC project will be available online to anyone interested via i) publishing in scientific journals and ii) disseminating the results via social networks (LinkedIn) when this complies with IPR. **Accelerate innovation & ensure market uptake:** Overcome the barriers that innovative companies might have in adopting SORT4CIRC technologies. For instance, sharing openly crucial data such as scientific publications and white papers for policymakers.

[24] http://genderedinnovations.stanford.edu/methods/engineering_checklist.html
[25] https://www.fairtrade.net/news/the-women-who-make-our-clothes-are-invisible-its-time-to-change-that

1.2.7 Research Data Management

SORT4CIRC will ensure the data is managed responsibly and in line with the **FAIR** – **Findable** – unique DOI per dataset, as well as the use of keywords. Research data will be searchable online (public reports, project website, communications, etc.); **Accessible**- SORT4CIRC's SharePoint will allow accessibility for at least 20 years, together with corresponding sources of that data and contact information from the data contributors; all data described in the Table 1 will be in open access. **Interoperable**- The published data will be mainly in open or widely used formats (Excel, Word, PDF, etc.) to boost interoperability (formats: Code files, PDF, Word, Excel). The developed DPP platform in data management and sharing will comply with EU software development legal and non-legal mandatory requirements and issues like privacy in compliance with the General Data Protection Regulation (GDPR) and the ePrivacy Directive. The DPP platform will be shared via open repository platforms for public usage and extensions, fulfilling the European Commission Open-source software strategy 2020-2023. **Reusable**- Facilitate knowledge transfer among multidisciplinary groups to promote maximum replication on a European scale and beyond. The main data types of the project are experimental data from laboratory experiments and pilot set-up, LCA reports, guidelines for LCA studies, publications with supplementary data, reports and demonstrators/pilot lines, and AI-driven prediction models. During the project duration, all data will be stored on a centralised file storage system (SharePoint®), which has daily automated backups and can be accessed via a secure VPN. Different levels of access rights and security measures applying to different drives will be granted according to data sensitivity. General guidelines for data management will be available within each relevant WP.

Table 1 – Research data management (F=Findability; I=Interoperability; R= Reusability; C=Curation)

Type	F	I	R	C
Numerical collected/generated data as research input/output	Patents and publications	The data will be released according to the standards set by the publishers	Can be re-used by researchers and practitioners. Such data are included in the articles or can be found through a provided link on publishers' websites.	Institutional repositories improve their discoverability and guarantee long-term access.
RGB images of garments for type identification (<10GB)	Online repository	General image format PNG with metadata		Institutional repositories
Spectral dataset of garment types (<200MB)		Spectra data in a JSON or CSV format with metadata		
TTRFID assessment and technology data		Work with industry-specific standards for RFID in textiles, such as those from GS1 or EPCglobal, to ensure alignment between different stakeholders.		
Blockchain architecture design and performance metrics		Leverage interoperability frameworks; smart contracts capable of operating across multiple blockchain networks; Use standardised abbreviations across all blockchain networks.	Regularly perform compatibility testing to ensure that your blockchain systems remain interoperable and that any issues are promptly identified and addressed.	Specialised blockchain repositories such as Hyperledger and Ethereum's Swarm.
DPP data retrieval integration results	Online repository (e.g. IEEE Xplore and SpringerLink)	Harmonisation of data protocols ensures that data can be accurately retrieved and integrated across different platforms without losing integrity.	The modular design and interoperability frameworks facilitate seamless integration and reuse across different data protection systems.	Data will be deposited in institutional repositories, such as LinkedIn, and project-specific platforms like the European Product Database for Energy Labelling (EPREL)
Tokenised ecosystem development data	Online repository (e.g. SSRN and arXiv)	Use of collaborative platforms like GitHub. Regular compliance checks with financial regulations across different jurisdictions.	Adhesion to tokenisation protocols, maintenance of comprehensive development documentation, and implementation of open access policies.	Repositories like IPFS and Arweave. Public repositories and financial data repositories like CoinMarketCap.
Detection of Elasthane and other components with low content, chemical	Publications with DOI	Word, PDF, PowerPoint file	Open data	Secure digital storage (internal), repositories

contaminates, finishing agents, and coating.				

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2 Impact

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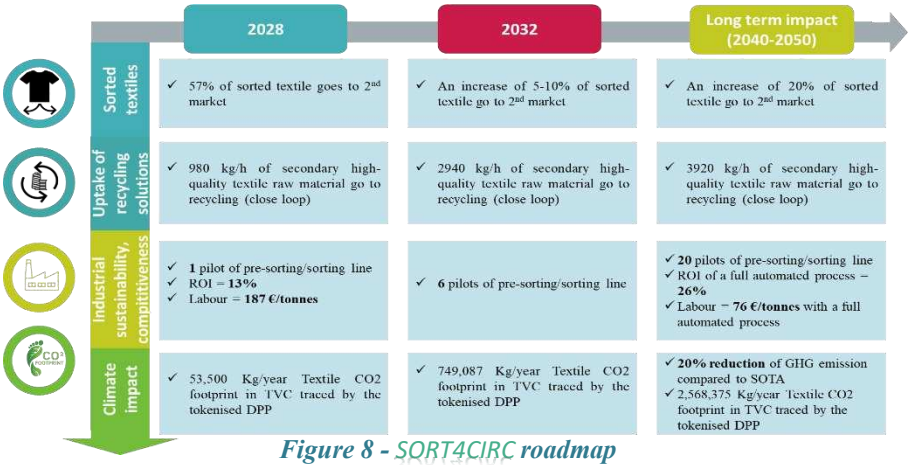

Figure 8 - SORT4CIRC roadmap

SORT4CIRC roadmap (**Figure 8**) is established to quantify the project impacts. By the end of the project in 2028, SORT4CIRC technologies will have reached TRL 5-6 and, thanks to TXHO involvement, will be ready to be validated, scaled up and replicated. It is estimated that by 2032, the TRL of SORT4CIRC technologies will have reached TRL9 and will be ready to be commercially available. SORT4CIRC's innovations will have spread across Europe, having a concrete impact on the textile sector by increasing the recycling of PCT, decreasing sorting and recycling costs, as well as CO_2 emissions by decreasing the culture and exploitation of new fibres and using recycling ones.

2.1 Project's pathways towards impact

2.1.1 Outcome 1 and 2: Roll-out of systemic solutions for textile sorting, using innovative digital technologies and facilitated disintegration to be incorporated in product design as an enabler for recycling

SORT4CIRC delivers an automated sorting system, proper detection of materials and blends in textiles, in single and multilayer textiles, a high precision level for garment classification and pre- and post-disassembly sorting processes, with a DfD. Today, the project's sorter can sort 5.25 tonnes/day (operating time ≈ 7.5 h/day, 750 kg/h) with current settings and manual sorting. SORT4CIRC will contribute to improving the business case by offering an automated sorting and identification of non-wearable textiles fitting current/future (mechanical, thermo-mechanical, chemical/enzymatic) recyclers specifications and quality requests and ultimately increase the quantity and added value of fibre-to-fibre transformation of PCT waste. SORT4CIRC will transform PCT waste into high-quality secondary raw materials, even if the quality of products put on the primary market decreases due to fast fashion. From the total sorted textile, considering manually and automatically, the amount of secondary high-quality raw material is expected to reach 980 kg/h by the end of the project. SORT4CIRC will use its textile DPP as a revealing spotlight on the textile production path, making it transparent and traceable. DPP yields two pivotal impacts: DPP streamlines 90% of the sorting process during the collection of waste textiles, freeing up existing devices for subsequent sorting stages; the comprehensive data provided by DPP covering the textile production process, energy consumption, CO_2 emissions, and raw materials usage, fuels detailed analyses. These insights play a crucial role in reshaping the TVC, promoting a more recyclable/sustainable approach to industry practices, and enabling more effective collection for sorting strategies. The blockchain backbone provides a token economy model for the circularity of textile ecosystems.

Scale and significance:	2027(project end)	2030	2035	2040
Secondary high-quality textile raw material (kg/hour)	980	1960	3920	3920
Number of pilots (Europe/Worldwide)	1	2	10	20
Total (tonnes/year considering 7h/day - 312 days/year)	2,140.32	8,561.28	85,612.8	171,225.6
Textile Sorting & recycling Tokenised economy market scale (K€/year)[26]	1,222	5,137	51,367	102,735
Textile CO_2 footprint in TVC traced by the tokenised DPP (Kg/year)[27]	53,500	214,000	1,284,175	2,568,375

2.1.2 Outcome 3: Increased uptake of mechanical recycling solutions that deliver competitive, high-quality secondary materials

SORT4CIRC will unlock the current technological gaps in textile sorting through automation of textile classification, categorisation and composition definition, decreasing the overall sorting cost and giving access to recyclers to affordable raw materials, thus making recycled textiles competitive and high-quality at the same time. An input of highly pure textiles of good quality will be a first step to obtaining a fraction of spinnable fibres used in a new textile product at a low cost and a lower environmental footprint. Using higher-quality textile waste as feedstock by mechanical recyclers will lead to higher-quality and longer recycled fibres without additional fibres (e.g., virgin cotton or polyester recycled) for creating new yarn. SORT4CIRC will greatly impact mechanical recyclers, providing them a substantial volume. Recycler sorting lines must increase their capacity and multiply the number of pilots

[26] McKinsey & Company, https://vb.nweurope.eu/media/19019/2207-scaling-textile-recycling-in-europe-turning-waste-into-value.pdf
[27] https://www.eea.europa.eu/publications/textiles-in-europes-circular-economy

across Europe and Worldwide to respond to this upcoming stream. **Scale and significance:** By closing the loop, SORT4CIRC will allow mechanical recyclers to access a mature technology with favourable economics, **up to 50% total cost reduction**, delivering competitive and high-quality secondary raw materials.

2.1.3 Outcome 4: Roll-out of thermo-mechanical, chemical and other (e.g., enzymatic) recycling solutions that are sustainable from a zero-pollution, circular material and energy efficiency perspective

SORT4CIRC aims to revolutionise textile recycling and will be the major game-changer by delivering well-defined, disassembled and sorted raw materials as input to all kinds of recyclers. This will close the loop of the textile industry, provide environmentally friendly materials, reduce dependence on non-renewable resources, and minimise water and energy consumption. Globally, the fashion industry consumes an estimated amount of 79 billion cubic metres of fresh water annually. Water is used for growing and producing fibres (the greatest quantity of water) and when dyeing, finishing and washing clothes[28]. In addition, high volumes of water are contaminated with chemicals throughout these operations. Today, most of the recyclers can only recycle pure products. SORT4CIRC will lead to a wide range of recycled raw materials and optimising their use. Indeed, the project's partners can separate contaminants, dyes and then polyester or cellulose from mixed materials (cotton, polyamide, elastane, etc.) using soft conditions without degrading them, providing clean materials to other recyclers that could handle them. It could give them access to a large amount of high-quality feedstock, particularly for chemical/enzymatic recyclers they need to become cost-competitive. Indeed, unlocking the potential of clearly identified blended textiles empowers recyclers to efficiently separate diverse materials, yielding pure, high-quality resources ready for recycling. **Scale and significance:** On average, approximately 200 L of water are required to produce 1 kg of textiles[8]; based on this, we can approximate the water consumption avoided by using SORT4CIRC technologies to sort and recycle waste textiles: 428,064 m^3/year by 2027, 1,7M m^3/year by 2030, 17M m^3/year by 2035, 34M m^3/year by 2040. Considering today's textile waste (92M tonnes/year), 38% is treated in the valorisation process, 28% is incinerated with recovered energy, and 34% is landfilled. Based on the average carbon impact of landfill and incineration (+0.741 kg $CO_{2\,eq}$/kg of fibre), 68M tonnes $CO_{2\,eq}$/year could be avoided using SORT4CIRC technologies.

	Natural fibres (58%)	Artificial fibres (42%)	Average
Landfill (kg $CO_{2\,eq}$/kg of fibre) – 55%	+1.38	+0.12	+0.75
Incineration (kg $CO_{2\,eq}$/kg of fibre) – 45%	-0.12	+1.91	+0.73

2.1.4 Target group

Brands/manufacturers: (≈143,000 EU companies) Sustainability is a key concern for brands in the textile industry. Automated sorting systems can help efficiently recycle/reuse textile materials, thereby reducing waste and environmental impact. They are committed to sustainability initiatives and can leverage automated sorting to support their goals. **Collectors/sorters:** (≥ 9 main EU actors + charities) Automated sorting systems can significantly increase productivity and efficiency. They allow them to handle larger textile volumes and potentially increase their earning through performance-based incentives. Sorting technologies will turn waste into valuable feedstock. **Recyclers:** (≥ 84 EU companies) Chemical, thermo-mechanical and mechanical recyclers would be interested in getting well-defined sorted feedstocks for their recycling processes. Disassembled feedstocks (no more hard points and multilayers) will make recycling easier. **Policymakers/regulators:** Local, national, and international policymakers will advocate for policies supporting the responsible development/deployment of SORT4CIRC technologies. **Consumer associations**: SORT4CIRC promotes sustainable product recycling; consumer associations will raise awareness, empower consumers to make more informed purchasing decisions and participate in recycling initiatives.

2.1.5 SORT4CIRC's unique contribution to wider scientific impacts

Creating high-quality new knowledge and improving innovation capacity: Increase the quality and quantity of upcycling fibre-to-fibre recycling. *Strengthening human capital in research and innovation:* Process automation is a great opportunity to add value to human capital, switching from repetitive manual handling/sorting to better-qualified process operators. Such jobs will require higher qualifications/diplomas, increasing the education of EU citizens and promoting EU training programs to master such skills. The increase in the textile waste stream leads to the development of new recycling methods and resulting products, ultimately creating new jobs. *Fostering diffusion of knowledge and Open source:* Partners will push for collaborations with entities interested in co-products generated by recyclers and with other funded projects under this call to promote a sustainable CE.

2.1.6 SORT4CIRC's unique contribution to wider economic and technological impacts

Develop the CE: Thanks to SORT4CIRC technologies, textiles will be sorted more efficiently, allowing better resource utilisation as textiles can be directed towards the most appropriate recycling or reuse pathways. EPR legislation for clothing is currently mandated in France and is imminent in other EU countries, such as Sweden and the Netherlands. Any changes in sorting practices must be prepared to support the upcoming EPR requirements. Therefore, SORT4CIRC will work closely with projects under "CL6-2024-CIRCBIO-01-2" to contribute to recommendations on design for

[28] https://publications.parliament.uk/pa/cm201719/cmselect/cmenvaud/1952/report-files/195206.htm#footnote-172-backlink

recycling for textile products to target EPR schemes. Additionally, there are standards and certifications not exclusive to textiles. Still, they offer guidelines that SORT4CIRC will address, such as the Global Recycled Standard (GRS), Cradle to Cradle Certified™, ISO 14001, Zero Discharge of Hazardous Chemicals (ZDHC), and Bluesign. SORT4CIRC will also work closely with professional organisations that provide best practices in CE and industrial symbiosis and offer BMs for various industries. *Unlock the full potential and benefits of the CE through wide deployment, market uptake of innovative technologies and other solutions:* It could enable other recyclers to have an "extra" feedstock they can't use till now. It could enable collectors/sorters to have an outlet for complex textiles. *European industrial sustainability, competitiveness and resource independence are strengthened:* Some aspects could possibly also be used in other applications in the textile industry, such as textile finishers, laundries and the clothing industry. The project will ensure that the proportion of the down-cycling category decreases and that more non-wearable clothing leads to new high-quality secondary materials. Advanced automation technologies use significantly reduces costs and enhance EU ability to compete globally as it improves efficiency and reduce operations costs. Using waste as a resource within the EU will reduce dependence on external or global SC.

The economic potential of SORT4CIRC is depicted and benchmarked against conventional and other semi-manual and automated counterpart technologies in Table 2. Compared to manual and semi-manual (manual + FTIR) sorting processes, automated processes require significantly lower manpower. Thus, the labour cost is considerably mitigated (up to 67% cut-off). However, automation of the sorting process requires a certain capital investment. Although the capital investment is depreciated over the lifespan of the sorting facility, annual depreciation constitutes the cost driver of the sorting activity. Moreover, the running cost also increases since the demand for electricity rises. Yet, i) the higher volume of textile waste treated per unit of time; ii) the higher efficiency of the sorting process, which results in higher quality sorted textile streams (e.g. by prevention of contamination with elastane) and the lower amount of waste produced resulting in lower charges for waste disposal and management services have a positive collective impact on the global economics of the sorting process. Specifically, the profit can be even four times higher than the conventional manual sorting process. Comparing the SORT4CIRC to other automated processes, it is found that SORT4CIRC requires a lower amount of capital investment. In contrast, the rest of the costs of the sorting activity remain comparable, rendering the SORT4CIRC the most promising and economically affordable sorting alternative.

Table 2 - Preliminary economic analysis and benchmarking against the SOTA waste textile sorting processes.

Economic parameter	Sorting technology				
	Manual	**Manual & FTIR**	**RFID (automated)**	**Barcode (automated)**	**SORT4CIRC (automated)**
Labour, €/tonne	234	187	82	82	76
Capital investment, k€	N/A	496	754	1310	660
Depreciation, €/tonne	N/A	10	15	26	13
Running cost, €/tonne	N/A	6	8	7	9
Profit, €/tonne	35	66	164	154	171
Margin on sales, %	4	7	18	17	19
ROI (%)	N/A	13%	22%	12%	26%
*Economic parameters (applied to all sorting technologies): capacity: 5 Ktonnes/a; feedstock cost: 643 €/tonne; product price 912 €/tonne; equipment lifetime: 10 years					

2.1.7 SORT4CIRC unique contribution to wider societal and environmental impacts

SORT4CIRC's technologies will contribute to **SDG9 (Industry, Innovation, and Infrastructure)** by developing quality, reliable, sustainable and resilient infrastructure promoting inclusive and sustainable industrialisation and fostering innovation; **SDG12 (Responsible Consumption and Production)** by improving the efficiency of the sorting chain through automation and digitalisation. Leading to a reduction of new textile use and an increase in recycled textile use, contributing to sustainable consumption and production patterns; **SDG13 (Climate Action):** Textile recycling will play a significant role in reducing the environmental impact of the fashion and textile industry, by reducing transport and the use of chemicals; **SDG14 (Life Below Water)** by reducing pollution, decreasing chemical and microplastic release, and minimising the environmental impact on marine ecosystems. *Prevent and remove pollution:* SORT4CIRC promotes textile recycling through automated textile sorting, reducing the need for incineration and virgin material production, thereby mitigating harmful air pollution and greenhouse gas emissions. SORT4CIRC will consider the Safe and Sustainable by Design (SSbD) approach within DPP and subsequently by offering high-quality recycled textiles. Also, increasing the use of recycled textiles will decrease demand for new raw materials, which often involve toxic substances in their production. *The benefits for consumers and citizens, including those in rural areas, are improved:* SORT4CIRC will increase productivity and improve the quality of the sorted recycling material, giving consumers access to high-quality recycled textiles and garments.

To evaluate the impact of SORT4CIRC on the environmental burden of waste textiles management technologies, a preliminary estimation of the GHG emissions was performed and compared to incineration, as the conventional waste textile management process and to the SOTA waste textile sorting process (**Figure 9**). The consequential LCA was considering (i) waste textile collection and transportation to the sorting facility; it was previously found that the impact of the waste textile transportation is not significant[29], (ii) conventional sorting process and (iii) NIR sorting as boundary limits, while the LCI data were modelled using the Ecoinvent database v3.10. The breakdown of the impact of climate change indicates that avoiding virgin (primary) fibre production and incineration are the main benefits. Moreover, the more efficient recognition system produces less waste, reducing SORT4CIRC GHG emissions by approximately 20%, compared to the SOTA waste textile sorting process.

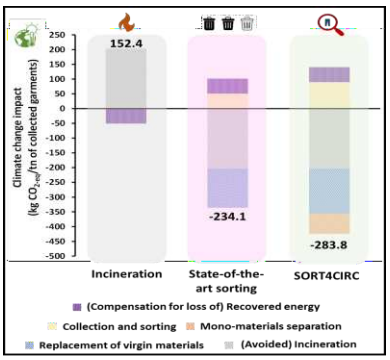

Figure 9 - Preliminary consequential LCA. Mono-materials: textiles sorted out of the waste and consist of one raw material; Replacement of virgin materials: the end products of the waste textiles sorting processes will replace virgin materials that would be used for new textiles production.

2.1.8 Barriers and requirements

Table 3 - Potential barriers and proposed mitigation measures for SORT4CIRC

Potential barriers (effect)	Proposed mitigation measures by SORT4CIRC
Regulatory	
Lack of stakeholder engagement and collaboration in the policy-making process can lead to policies that must adequately reflect key stakeholders' needs, perspectives, and expertise.	SORT4CIRC will include policymakers as soon as possible in its technology development to ensure the adoption of best practices and regulations in textile recycling.
DPP Adoption Resistance by stakeholders	Resistance from stakeholders to adopt and adapt to new technologies and processes is addressed in sub-tasks with dedicated dissemination initiatives like sub-tasks **4.2.6** and **4.4.4.**
Lack of adoption of tokenised ecosystem by stakeholders due to complexity or lack of incentives	Design user-friendly interfaces, provide clear incentives, offer training sessions, and gather continuous feedback for improvement. WP6 will conduct a simulation over strategies. WP7 will ensure effective stakeholder engagement.
Regulatory Compliance of DPP	Regulations related to data protection, environmental standards, and industry-specific rules will be fulfilled like ISO 14040 and ISO 14044 LCA; IEC 62264 (ISA-95) and IEC 62443 (ISA -99) and ISO 27001 Certification; ISO 27001 Certification.
Social	
Some consumers may prioritise convenience and price over sustainability	Automation can reduce recycled textile prices; SORT4CIRC will promote its technologies as sustainable fashion practices.
TTRFID Acceptance (Privacy concerns), durability, production cost	Societal analysis and assessment will be considered with a special focus on consumers, and surveys and sentiment analysis will be conducted to analyse consumers' perception of technology, besides using economically efficient materials and economic approaches in massive volume production. The recycling assessment, besides the factors mentioned for the adoption of the TTRFID with textile product compatibility, like resistance to washing and difficult conditions, will be tested. Motivating events to introduce the new eco-system and making effective marketing campaigns will be organised.
Risk of tokenisation method incentivising overconsumption of textiles by consumers	The token system will be designed to reward sustainable practices such as recycling and reuse rather than increased purchasing by simulation in T4.4.3 and complying with the requirements of ISO 14040 and ISO 14044 LCA and will be communicated with consumers and stakeholders.
Economical	
The economics of emerging technologies are only sometimes the most encouraging ones.	New adjustments to the technology or new BMs may be applied to turn the technology into a profitable solution. The social and environmental gains should be taken into account and monetised to improve the economy.
Implementing automation and digitalisation technologies in textile sorting and recycling	SORT4CIRC will push Public-Private Partnerships after the project (public agencies, private investors, industry partners, and research institutions) to pool

[29] https://www.ivl.se/english/ivl/publications/publications/siptex-wp5-report-life-cycle-assessment-of-textile-recycling-products.html

requires significant upfront investment. The costs associated with purchasing and installing advanced sorting equipment, integrating digital platforms, and upgrading existing infrastructure can be substantial.	resources, share risks, and mobilise funding, enabling the up-scaling and deployment of SORT4CIRC's technologies
Environmental	
Automation and digitalisation processes often require significant energy inputs, particularly for powering machinery, sensors, and data processing systems. High energy consumption can contribute to environmental pollution and GHG emissions.	SORT4CIRC will prioritise energy-efficient technologies, renewable energy sources, and optimised operational practices for energy inputs
Integrating diverse technologies such as sensors, robotics, AI, ML, computer vision, and data analytics into a cohesive and interoperable system presents technical challenges. Ensuring compatibility, scalability, and reliability of integrated technologies is crucial for effective automation and digitalisation.	SORT4CIRC interdisciplinary collaboration is bringing together diverse stakeholders, including researchers, engineers, technicians, software developers, and industry professionals, to exchange knowledge, share best practices, and co-create innovative solutions to ensure proper integration of SORT4CIRC's technologies into the current recycling and sorting chain
Technological	
DfD solutions not being effective or practical for large-scale implementation.	Pilot DfD solutions in controlled environments, gather data on effectiveness, and iterate based on feedback (from **WP5**).
Long-term sustainability of blockchain	A focus on the scalability of the blockchain architecture for DPP will be conducted. The holonic approach of SORT4CIRC will ensure the scalability of DPP for encompassing the whole TVC.
DPP Network Congestion, Data Overload and Cybersecurity Threats	The focus will be on the scalability of the DPP over TVC. The holonic approach and the fulfilment of Web 3.0 will enable the effective extension and interoperability of the DPP with other operational systems. The fulfilment of the ISO 27001 Certification for data security will be considered in DPP guidelines.
The unknown composition and presence of different materials require different recycling processes to allow the rollout of recycling solutions.	SORT4CIRC will deliver an efficient way to analyse and give the composition of each material with an accuracy of >95%, allowing access to a wide range of high-quality feedstock for recyclers

2.2 Measures to maximise impact - Dissemination, exploitation and communication

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2.2.1 Planned measures to maximise the impact of SORT4CIRC

Communication, dissemination and exploitation plans will be prepared in WP7, and all actions will be closely monitored throughout the project's progress. CEF will coordinate dissemination, communication and exploitation activities. The intellectual property management activities and definition of the strategy for result protection and exploitation will be coordinated by CU and CEF. To properly and efficiently develop these actions, the first step is to define key messages to be passed and then adequate measures and methodology to pass to the target groups.

2.2.1.1 *Relevant target stakeholder for communication and dissemination activities*

Main message and results to be shared: **(i)** Increase cost-efficiency and automation in sorting processes, achieve savings from improved inventory control and reduced waste and train the workforce in advanced textile technologies; **(ii)** Development of sorting technologies compatible with recyclers' expectations and **(iii)** Lower carbon footprint through sustainable practices, enhancing sustainability through better resource management and reducing energy use and environmental impact of textile sector. Target groups for dissemination and exploitation: to maximise the impact of SORT4CIRC outcomes during and after the project, the effort shall be put into engagement with key stakeholders, market players and end-users in the EU textile recycling and manufacturing value chain. For each category, adequate messages associated with appropriate communication and dissemination means (direct contacts, website, social media… etc.) will be designed within WP7. SORT4CIRC will involve specific stakeholders within their technical advisory boards and will address the following target groups: **Sorters** (TXHO, Hubency); **Textile manufacturers** (Inditex, LVMH, Salvatore Ferragamo, Koninklijke Ten Cate NV, etc.); **Recyclers** (BASF, Infinite Fiber, Renaissance textile, Carbios, SABIC) and **Regulators** (Refashion, Euratex, etc.).

2.2.1.2 *Communication and dissemination strategy and timeline*

To ensure the successful exploitation of the SORT4CIRC results, a Communication, Dissemination, and Exploitation (CDE) Plan (D7.1) will be developed by CEF to identify opportunities for synergy and guide SORT4CIRC's communication and dissemination activities relevant to the project, monitoring procedures and the sustainability of the project beyond its lifecycle. Considering the three-year project duration, necessary adjustments will be made to

the plan regularly, considering the evaluation of outreach and communication analytics, as well as successes, lessons learnt and future opportunities that may have emerged since SORT4CIRC's inception as reported by all project partners. SORT4CIRC CDE Plan will also follow the guidelines published by the EC on social media: 'Social media guide for EU funded R&I projects.'[30] A monthly communications meeting between the partners will be held to keep each other updated with communications activities and opportunities.

The activities described in **Figure 10** aim to achieve impact and exchange knowledge and innovations generated during the project with the already identified target stakeholders. The plan will also set objectives for project impact performance through ad-hoc KPIs. The objectives are **Disseminating information** regarding the opportunities and advantages reached during the project and onwards (lasting effects).

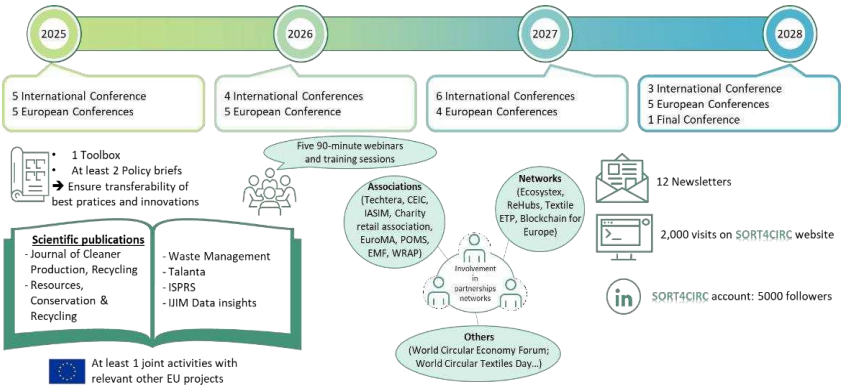

Figure 10 - Communication & dissemination tools and proposed dissemination activities

Boosting the transferability of knowledge, exchanging experiences for the increment of learning (R&Ds towards industrial partners and end users). **Gathering helpful feedback from all the stakeholders** (researchers, industries, policymakers, industrial associations, standardisation bodies and committees, other running projects and so on). **Replicate SORT4CIRC concepts** to other European processes and geographical areas (diversification).

Targeted International Conferences: WHISPERS, World Circular Textiles Day, ICTRD… **Targeted EU Conferences:** EURIC Textiles, Circular Cities and Regions Initiative (CCRI) Working Group, ECESP, TechTextil... SORT4CIRC aims to collaborate through projects in the frame of stakeholder engagement and communication, as well as prolong the impact and lifetime of project outputs beyond the end date of projects. SORT4CIRC will create a core for the textile recycling and reuse cluster and will stay active to contribute to further expansion. Moreover, SORT4CIRC will actively seek to collaborate with other relevant funded projects under "HORIZON-CL6-2024-CIRCBIO-01-2". The SORT4CIRC consortium will leverage our existing involvements in other CCRI projects such as DEFINITE-CCRI, CircularInvest, CCRI CSA, SOLSTICE, other relevant EU projects such as Circular City Centre (C3), RESYNTEX, REGIOGREENTEX, MISTERY, WHITECYCLE, to offer SORT4CIRC project actors' new channels to exchange project learnings, participate in industry events and other opportunity-based activities.

2.2.1.3 *Preliminary description of Key Exploitable Results (KER) and Exploitation plans*

The Plan for Exploitation of Results will describe KER and its owners, potential end users, and the pathways to exploitation foreseen to ensure the results are taken up and further used by the partners or by third parties.

Table 4 - Preliminary identification of Key Exploitable Results of SORT4CIRC

	KER	*Background (IP)*	*Protect*	*Exploitation route and benefits*
TXHO	Complete pre-sorting and sorting semi-automatised pilot line	Fully automatic textile sorting system that sorts used textiles by material and feeds them into chemical recycling.	Confidential	1 pilot by the end of the project, 20 pilots of pre-sorting/sorting line by 2040.
CU	TTRFID Technology	Technical papers on RFID technology and data management linked to existing patents in RFID technology.	Patents and trademarks	Use in further research studies to develop new RFID-based services for textile industries. Sales of new products/services. First installation, replication to other sites.
	Blockchain-based DPP System	Background in blockchain technology and data security, with references to patents in blockchain systems like EBSI.	Patent and copyright	Development of secure data processing products, enhancing data traceability and security. Sales of Secure Blockchain-Based DPP Systems. Integration with Existing SCs.
	Tokenized Ecosystem for TVC	Experience with token economy and digital asset management, referencing existing token systems	Industrial design and trademark	Development of token-based products and use in standardisation activities. Sales of Token-Based Transaction and Validation Services. Launch Initial Token Ecosystem.
	DfD Solutions	Knowledge in sustainable design and recycling, linked to patents in sustainable product design	Trade secret and industrial design.	Recyclable products development and sustainable design guidelines. DfD-compliant products sales and consulting services. DfD principles in product lines implementation.

[30] https://ec.europa.eu/research/participants/data/ref/h2020/other/grants_manual/amga/soc-med-guide_en.pdf

RESO	Align pre-processing output with feedstock (input) requirements of recyclers	The internal know-how and the patented knowledge BE1029856 (WO2023/062190) will be used.	Secret know-how	The results will be exploited for further research activities in developing eco-design products.
	Recognition technologies validation for eco-design products (pre-disassembly)	The internal know-how	Secret now how	The results will be exploited to develop new processes
	Disassembly integration with post-disassembly sorting processes	The internal knowhow and the external testing done with other companies	Secret know-how	The results will be exploited to develop new processes
PURF|RE	The new source of feedstock efficiently sorted	Tests at a lab scale with collectors/sorters in the framework of experimentations organised by Refashion,	Confidential	Textile types are more precisely defined to be collected and sorted that will match our process. Know better what kind of textiles sorters could provide. Develop the recycling chain. Develop partnerships with sorters.
ARIS	Emerging textile sorting processes and textile recycling processes value chain	Access to databases (e.g. Ecoinvent).	Open access	Knowledge
	Flow sheeting of textile recycling processes	In-house developed economic models	Open access	Know-how
CEA	AI algorithms for material composition	Analytical methods were utilised to address this task.	Open-Access	Knowledge Apply it in future collaborative projects
	Identification solution of carbon-based black garments	The test used in SWIR and NIR sensors.	Patent	Patent – exploit them in further research and projects.
CEF	Circular Textiles Toolbox	CEF has previously developed a Circular Toolbox, an open-access step-by-step guide for apparel brands to design and launch. A rental or resale pilot.	Techno. Know-how, policy diss.	Integrating new methodologies/technologies, and experiences in other location. Using for further research activities, developing new products/processes, creating services, using in standardisation activities
IOM	AI technologies in NIR spectroscopy and hyperspectral imaging	AI approaches that build on chemometric models, internal know-how	Open access	Exploitation of these approaches in other sectors of industry (other research activities)
	IR technologies in other spectral ranges (SWIR, LWIR)	IR spectral ranges potential outside of classical NIR spect., internal know-how	Open access	Exploitation of these new analytical approaches for other research activities
	Improved analytical detection and identification methods for finishing agents and coatings	NIR hyperspectral imaging for use in textile production has been developed.	Open access	Exploitation of these new analytical approaches for process control in the textile manufacturing industry.
IRIS	Several detection techniques incorporation in a common hardware platform to perform in-line analysis of garments' parameters in real-time	Know-how regarding chemometrics and patent pending for textiles multilayer recognition through NIR hyperspectral device	Trademark and patent	Licensed to 3rd parties. Scale up and improve the in-line device throughput for real-time recognition of textiles and its detection speed to have an operational system that can be sold. Carry out in future projects to achieve TRL9.
REFA	SKU retrieval system that allows to retrieve product data online without a data carrier and upgraded product attributes detection AI neural network	The product attributes detection builds upon results from other projects.	Patent	New developments integration into automated textile sorting lines. Reach out to further sorting companies to sell this product.

2.2.2 Intellectual property management

The IP strategy is based on principles included in a Consortium Agreement (CA) and is "proactive". It consists of identifying all relevant outputs during the tasks implementation and establishing, with the partners having generated the results, the most appropriate strategy to protect (IP, non-disclosure), to exploit (further research project, commercial agreements) or to disseminate (open access publications, international conferences, website) the results. For any protectable result that may arise during the project, the following steps are carried out: *(i) t*he partner notifies the consortium about the technical contents to be protected & related ownership rights; *(ii)* in the case of jointly owned new IP, the procedures for IP protection, use, and licensing will comply with the rules set in the Grant Agreement and described in the CA; *(iii)* Filing the related IP protection application is followed directly by the relevant partner(s), according to the perimeter agreed with other partners, and the perimeter of innovation. We have already identified in Table 4 results that could generate IP, as well as internal know-how, knowledge, expertise and IP that can be accessible for the project or further used to exploit the results. #§COM-DIS-VIS-CDV§#

2.3 Summary

Specific needs	Expected results	D & E & C Measures
Systemic solutions for textile sorting using innovative digital technologies Feasible solutions for facilitated disintegration to be incorporated in product design as an enabler for recycling Increase the quality and the stream of feedstock for recyclers Ensure the viability of CE BMs in the textile sector	• Composition estimation error < 5%. • Detect a material under 3%. dentification of black textiles; Detection of chemical contaminants and foreign materials. • Precision level >85 % for garment classification and conditions dryness/soiling classification. • Success rate >95% for DPP seamless data exchange with AI-based technologies • ***20% lower GHG emissions*** • *Positive ROI, 26% targeted.* • *Increase of sorting efficiency by 40%.* • *Increase throughput by 30-50%.*	**Media and Websites:** at least 12 newsletters. Presence on LinkedIn with more than 5,000. A dedicated website with 2,000 visits over the 3 years. **Event and conferences:** 5 webinars, at least 30 conferences and X multi-actor workshops. **Scientific publications, white papers and conference papers:** at least 12 publications and 1 conference paper. 1 toolbox. At least 2 policy briefs
Target groups	**Outcomes**	**Impacts**
The results of SORT4CIRC will impact the whole TVC: • **Brands and textile manufacturers**: Better resource management, enhancing sustainability. • **Sorters**: cost-efficiency and automation in sorting processes will be increased (TXHO, Hubency) • **Recyclers**: Sorting technologies meeting recyclers' specifications - high-quality feedstock. (BASF, Infinite Fiber.) • **Regulators**: Enhanced sustainability through better resource management. Reduce waste generation and the environmental footprint of textile products. • **2nd hand markets**: Improved access to high-quality feedstock.	Increase sorting speed from 500 kg/h to 3920 kg/h. Deployed the pilot across Europe and the World, reaching 20 sorting lines by 2040 Increase textile sorting and recycling tokenised economy market scale from 1 to 102 K€/year by 2040.	• Scientific: Creating high-quality new knowledge and improving innovation capacity, strengthening human capital in research and innovation, fostering the diffusion of knowledge and Open source. • Economic: Develop the CE, decrease labour from 234 to 76 €/tonne, increase profit from 35 to 171€/tonne, increase ROI up to 26% thanks to automated sorting. • Technologic: unlock the full potential and benefits of the CE through wide deployment, market uptake of innovative technologies and other solutions. • Environmental: A more efficient recognition system produces less waste, reducing GHG emissions by approximately 20%. • Societal: increase productivity and quality of the sorted recycling material, giving access to high-quality recycled textiles and garments.

#§IMP-ACT-IA§#

3 Quality and efficiency of the implementation

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3.1 Work plan and resources

3.1.1 Overall structure of the SORT4CIRC work plan

The overall structure of the SORT4CIRC work plan is depicted in Figure 11.

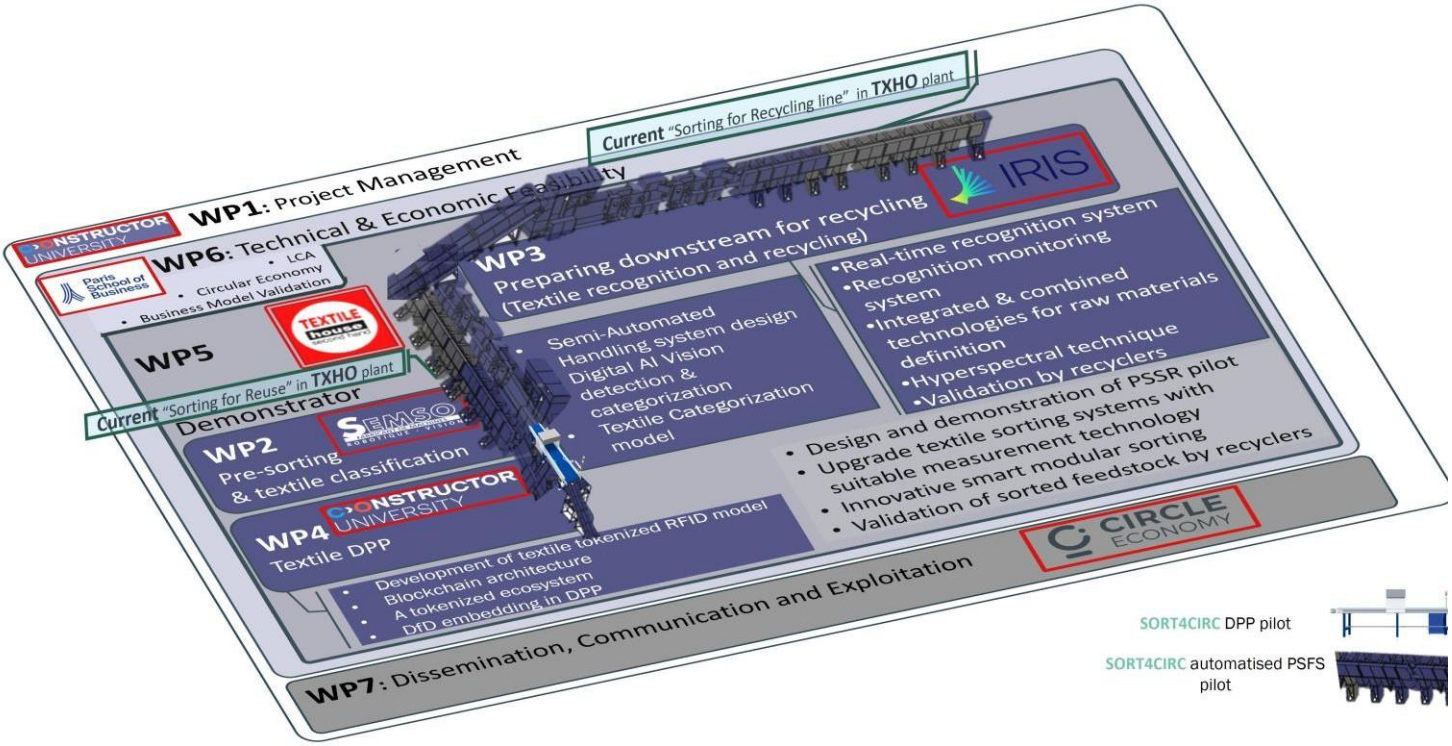

Figure 11 - Work Plan of SORT4CIRC

3.1.2 Planning

| | | |

Environmental impact assessment Cost-benefit analysis of SORT4CIRCBM evaluation, industrial symbiosis, social acceptance and	consumer welfare	ARIS	3	15	**D6.1**		
	Design and operation optimisation of the automated sorting	ARIS	15	34		**D6.2**	
	Technologies benchmarking	ARIS	15	34		**D6.3**	
		PSB	1	36			**D6.4**
		PSB	12	36			**D6.5**
		ARIS	18	36			**D6.6-7**

7.1	Communication, Dissemination and Exploitation activities	CEF	1	36	D7.1-2 D7.3	D7.4	D7.6-7, 10
7.2	Development of knowledge and policy support	CEF	1	36		D7.5	D7.8
7.3	Complementary knowledge transfer	CEF	1	36			D7.9

Table 3.1h. 'Purchase costs' items (travel and subsistence, equipment and other goods, works and services)

The equipment costs reported are depreciation costs only.

1/CU		
	Cost (€)	**Justification**
Travel and subsistence	18 250 €	Consortium meetings (7 meetings) for Project Management (WP1 (7 000 €)), Conference for publications and dissemination events (Year 1,2,3) WP7: (in total 7 Conferences) = 3 * 2 000 € = 6 000 €), Meeting with Consortium partners for 7 Meetings= 3 500 €, Technical Meetings for technology transfer from WP4 to WP5: 1 750 €
Other goods, works and services	29 000 €	Publication and open access support Publication 1 to 4 - WP7 (8 000 €); Chipless RFID Tag alternative 1, 2 and 3 - WP5 (3 * 7 000 € = 21 000 €)
Remaining purchase costs (<15% of pers. Costs)	22 900 €	
Total	70 150 €	
4/RECY		
	Cost (€)	**Justification**
Travel and subsistence	7 000 €	7 Travels = 7 project meetings (1000€ for 2 people per travel) - WP1 (7 000 €)
Equipment	14 728 €	Demonstrator 1000L - WP5 (2 800 000 €*6/60 months*3,52 %=9 868 €); Pilot 100L - WP4 (300 000 €*12/60 months*3,96 %=2 379 €); Pilot 100L - WP5 (300 000 €*6/60 months*5,29 %=1 586 €); Pilot 30L - WP4 (70 000 €*12/60 months*4,62 %=648 €); Pilot 30L - WP5 (70 000 €*6/60 months*3,52 %=247 €)
Other goods, works and services	31 438 €	Analysis on DMT and com-materials (PFAS, heavy metals, formaldehyde, specific / toxic colorants, DSC, ATG, degree of polymerisation, viscosity index…) - IFTH - WP4 (11 598 €); OPEX Cost: Raw material required for pilot 1000L-scale recycling - consumables - WP5 (6 440 €); HPLC analyses - WP4 (5 400 €); GC analyses - WP4 (5 400 €); Analysis of composition of separated co-materials - IFTH - WP3 (2 600 €)
Remaining purchase costs (<15% of pers. Costs)	5 650 €	
Total	58 816 €	
7/CEF		
	Cost (€)	**Justification**
Travel and subsistence	6 000 €	Travel cost (1pax); x7 in-person general consortium meetings - WP7 (3 000 €); Travel cost (2pax); x3 in-person external events (i.e. WCEF, ECESP, CCRI events etc) - WP7 (3 000 €);
Other goods, works and services	10 000 €	External services for Website development, hosting and maintenance – WP7 (10 000 €)
Remaining purchase costs (<15% of pers. Costs)	6 380 €	
Total	22 380 €	
9/SEMSO		
	Cost (€)	**Justification**
Other goods, works and services	€243 080	Supplies for conveyors construction (Motors, belts, accessories, control cabinet, Electric cylinders, Housing, IHM,PalBox, RFID, ect...) - WP2 (199 535 €); Manufacturing (tailor made products) - WP2 (33 545 €); Transport from SEMSO to TXHO - WP5 (10 000 €)
Remaining purchase costs (<15% of pers. Costs)	15 000 €	
Total	€258 080	
11/PSB		
	Cost (€)	**Justification**
Travel and subsistence	10 000 €	Travels to meet partners/stakeholders for data collection and/or output validation (1 person for 3 travels) - WP6 (6 000 €); Travel for conference and dissemination events Year 2 (1 conference for 1 person) - WP6 (2 000 €); Travel for conference and dissemination events Year 1 (1 conference for 1 person) - WP6 (2 000 €)

Remaining purchase costs (<15% of pers. Costs)	47 200 €	
Total	57 200 €	
13/IRIS		
	Cost (€)	**Justification**
15/TXHO		
	Cost (€)	**Justification**
Travel and subsistence	30 000 €	Face to face meeting with each individual Consortium partner, (once a year, 2 persons, TXHO visit the 13 Partners in their location for technical meetings) – WP2-3-5 (21 000 €); Monitoring of the Demonstrator of the automatized Pre-sorting pilot in the sorting plant (only during the pilot construction, 6 month, every 2 week in the sorting plant, 1 person) - WP5 (6 000 €); Event and Marketing travels to the sorting plant (2-3 persons per travel, 2 travels) - WP5 (3 000 €)
Equipment	7 083 €	Blackbox - hard disk- WP5 (10 000 €*17/120 months*100 %=1 417 €); HSI SWIR camera for integration in final equipment- WP5 (35 000 €*17/120 months*100 %=4 958 €); Panel master (control station) - WP5 (5 000 €*17/120 months*100 %=708 €)
Other goods, works and services	119 500 €	Engineering services for implementation of AI Sorting-Software Reverse.Fashion - WP5 (37 500 €); Services for Electric Installation Company - WP5 (30 000 €); Event Agency (Event Planning for Launching of Demonstrator) - WP7 (20 000 €); Market studies by Market agency - WP7 (32 000 €)
Remaining purchase costs (<15% of pers. Costs)	89 200 €	
Total	245 783 €	

Table 3.1i. 'Other costs categories' items (e.g. internally invoiced goods and services)

8/CEA		
	Cost (€)	**Justification**
Internally invoiced goods and services	€32 972	CEA uses specific research and development environments to perform its programs, which calculate hourly rates according to its usual accounting practices. The hourly rates are calculated by dividing the costs of the specific R&D environment used for the project (such as energy and fluids, maintenance costs, small equipment, software, consumables, equipment depreciation costs, etc.) by the total number of hours of work of the researchers using that R&D environment. The amount reported on a project is calculated by multiplying the hourly rate by the number of project hours.
11/PURFI		
	Cost (€)	**Justification**
Internally invoiced goods and services	€30 000	use of existing recycling line + adjustments - WP5 (30 000 €)

#§QUA-LIT-QL§# #§WRK-PLA-WP§#

3.2 Capacity of participants and consortium as a whole

#@CON-SOR-CS@# #@PRJ-MGT-PM@#

3.2.1 Consortium value chain overview

All the 15 partners of SORT4CIRC will have a well-defined role in the project, allowing them to cover the entire value chain from technology providers to the end-users (**Figure 12**)

Advisory Board (AB): SORT4CIRC will set up an AB, and interactions with the AB will occur at least twice a year. SORT4CIRC AB will include members from 4 different groups: *Regulatory*, *overall value chain strategy, technical*

aspects and recyclers. The entities who could be part of the Advisory Board of SORT4CIRC are shown in **Figure 12**. The AB members are chosen to ensure that results from SORT4CIRC will be relevant to the whole TVC. The AB will provide the consortium with advice and recommendations from end-users and other stakeholders/interests.

3.2.2 Consortium description and complementarity

To achieve its specific objectives, SORT4CIRC will gather a multidisciplinary consortium with the specific expertise of each participant to support the development of the project. The consortium comprises 14 beneficiaries and 1 affiliated entity from 8 countries (Germany, Luxembourg, France, Belgium, the Netherlands, the United Kingdom, Slovakia, and Spain). 1 partner is RTOs, 3 partner is UNI, 10 partners are industrial companies (including 7 SMEs).

Figure 12 - Value chain of the SORT4CIRC project

CU: coordinator of the project - Constructor University's three Schools convene scientists across various disciplines. CU research is characterised by multi- and interdisciplinarity, which is also the heart of its educational concept. The research group "EITIE" (Emerging Technologies in Industrial Engineering) has been active around blockchain technology and developing solutions for the sustainability of SC processes for almost 5 years. CU has a solid track record in Horizon Europe project participation and management, including previous involvement in CRM-geothermal project, FAIRiCUBE project, PhotoCaM project, Progeny project, and EuropeanCity2 project. These projects have provided us with both the scientific and administrative experience required to coordinate complex international research collaborations. CU will coordinate the project (WP1) and lead the WP4, and all activities linked to the development of textile DPP.

TXHO: Since beginning its operations over 20 years ago, Textile House has consistently built infrastructure and processes to support the reuse and recycling of post-consumer textiles in line with EU circular economy and waste policies, and it will continue to develop and scale innovative technologies for automated sorting, recognition, monitoring, measuring and classification.

ARIS: AristEng S.à r.l. is an engineering consulting firm that provides tailor-made and fit-for-purpose engineering services to clients from diverse backgrounds. By linking research/experimental results/real production line data with process design and synthesis principles, they perform detailed techno-economic evaluation and environmental impact assessment of novel processes and new products at any project development phase, from embryonic lab-scale concepts to mature industrial-scale facilities. They will lead the tasks related to plantwide process model development, techno-economic evaluation and LCA.

RECY: Recyc'Elit is a French start-up that has developed a patented chemical recycling process dedicated to complex polyester-based textiles. They will assess the quality of the sorting done in the project framework. In 2025, they will be equipped with several pilots / pre-industrial demonstrators. As recyclers, they will also provide the requirements/needs/specifications needed for WP4 (Textile DPP).

RESO – Resortecs - provides DfD solutions that enable high-quality textile recycling on an industrial scale. RESO is the expert on 1) Eco-Design, DfD, and 2) Active-Disassembly in the textile industry, i.e. automatic disassembly of textile products to eliminate recycling disruptors (such as trims) from textile products, with no quality loss. RESO will use its expertise to develop DfD solutions over the DPP model (WP4).

IOM - The Leibniz Institute of Surface Engineering - is a non-profit, non-university research institute in the legal form of a registered association. The Materials Characterization and Analytics Group of IOM has dealt with developing methods for industrial process control based on NIR spectroscopy and NIR hyperspectral imaging. One more recent focus of this work was directed towards textiles, textile finishing, textile identification, etc. Those works were carried out with companies and industry research institutes. IOM will contribute to the development of a cost-effective and reliable inline multirange spectroscopic system for PSSR side of SORT4CIRC project (WP3).
CEF - Circle Economy Foundation - is a non-profit impact organisation based in the Netherlands. CEF builds the evidence base and capacity of stakeholders to accelerate the uptake of circular solutions globally across industry and government. CEF will lead in the extraction of best practices across WPs, development of policy recommendations, as well as the organisation and facilitation of webinars. The affiliate 'Circularity Gap Solutions (**CGS**)' will support the project by providing experts who are able to provide their communications (i.e. social media content creation and management) and design support (i.e. visualisations, infographics, website design) to the project. The support from the experts in this affiliate entity will contribute to the services the project can provide to the beneficiaries.
CEA - Commissariat à l'Energie Atomique et aux énergies alternatives - is a French research centre that develops technological solutions for industry. The CEA teams in Bordeaux and Saclay have strong expertise in developing intelligent imaging solutions exploiting images' spatial and spectral dimensions. The CEA has already developed expertise in analysing the composition of various textiles using hyperspectral imaging and will contribute to the development of PSSR system (WP3)
SEM – SEMSO - is a machine manufacturer with strong experience in several sorting and recycling processes equipment. SEM has all the trades in-house, from the preliminary project to the start-up at the end customer. They have expertise in mechanical studies, electrical studies, automation studies, robotics studies, vision, manufacturing, assembly, wiring, finetune, delivery and after-sales service monitoring. SEM will use its expertise to design and build the metal frame for the automatised PSFS pilot and the integration of technologies into TXHO plants (WP2&5)
PURFI - Purfi is the leader in fibre rejuvenation technology for spun fibres, focusing on the strategic categorisation and classification of textile waste and the subsequent processing of these materials using patented and trade secret-protected machinery and process techniques. Purfi is doing Soft Mechanical Recycling/Rejuvenation. The foundation of their technology is the rejuvenation of waste material back to virgin quality fibre, enabling it to be used in a variety of products and industries. They will assess the quality of the sorted garments and provide the requirements/needs/specifications for WP4 (Textile DPP).
PSB- Paris School of Business - has a dedicated team of researchers with advanced decision-making and problem-solving abilities, supported by extensive expertise in quantitative research methods, including analytical, mathematical, and simulation modelling techniques. PSB will propose and evaluate innovative BMs, emphasising the viability of circular business practices to provide actionable insights and robust solutions to drive sustainable business strategies. PSB will lead technical and economic feasibility, environmental and social assessments (WP6).
TRA - The Textile Recycling Association - is the UK trade association for textile recycling. TRA will engage stakeholders through facilitating/organising relevant workshops/meetings and participating in sustainable textiles events to disseminate findings. TRA will directly engage with policymakers at the national/international level and industry decision-makers, ensuring that lessons learnt from the research will be used to facilitate the delivery of new advanced textile sorting technologies & dramatically help to improve the economics of textile recycling (WP7).
IRIS - Iris Technology Solutions - is the leading European company in manufacturing photonic and AI solutions to control industrial and production processes in real time. The expertise IRIS will bring to the project will be linked to recognising complex textiles through spectroscopy techniques. IRIS will develop a Multirange Spectroscopic Platform, which will be integrated into a semi-automatic sorting process and will lead WP3.
REFA – reverse.fashion - is a company that provides technology for automated textile sorting. The AI-based software uses computer vision and enables the processing of DPP data to sort PCT into the most suitable reuse or recycling category. It increases the sorting efficiency and improves sorting quality according to the waste hierarchy. In this project, REFA develops its system further to retrieve product data by optical identification and similarity detection (WP2). Furthermore, it will provide access to its systems in WP5 for the demonstration of an automated sorting line.

3.2.3 Resources distribution

The SORT4CIRC project is composed of a total project cost of 5,097,767 € (and requested grant of 4,999,999 €). Partners are willing to contribute with in-kind resources as they foresee a huge commercial interest. The complementarity of the consortium is highlighted in **Figure 12**, and each partner will provide adequate resources for fulfilling their roles in the project, as highlighted in Table 3.1.f. The distribution of budget and grant is balanced among the partner types as follows: 39% for industries, 33% for SMEs, 23% for UNI and 5% for RTO. A major part of the budget (61%) is dedicated to direct personal costs, and the other parts are dedicated to indirect costs (20%), purchase costs (18%) and other cost categories (1%).

3.2.4 Consortium critical infrastructures

IRIS: *Spectroscopic laboratory* (equipped with FTIR-ATR, 2 UV-VIS, Extended-SWIR, NIR spectrophotometers, SWIR Hyperspectral imager, Chemometrics modelling suite, and non-commercial chemometric tools), *electronics, mechanical & optical workshops & labs* (Equipped with up-to-date hardware and tools) and *optical photonics-based technologies & PAT systems* (HSI and NIR based cameras suitable for surface and composition analysis), Multirange Spectroscopic Platform (VISUM Fusion). **CEA** possesses a NIR hyperspectral camera (0.9 to 1.7 µm), a SWIR hyperspectral camera (1 to 2.5µm) and a mid-wave infrared (MWIR) camera (1.5 to 5 µm) and RGB/monochrome cameras (useful e.g. in T3.3 and WP2). In addition to an FTIR (Vis->LWIR) for spectral reference measurements. **IOM** has two Raman spectrometers (NIR or VIS excitation), conventional NIR spectrometers (1.1 to 2.2 µm), a NIR hyperspectral camera (1.3 to 1.9 µm), various MIR spectrometers as well as a broad range of analytical instrumentation (helpful, e.g. for reference measurements) available. **REFA:** *A sorting machine that enables automated sorting for reuse (including quality)* will be lent to **TXHO** for the WP5 implementation.

3.2.5 Industrial involvement

SORT4CIRC has a complete value chain for designing and demonstrating an automated pre-sorting system. SORT4CIRC has secured a strong industrial commitment, the consortium including one main sorting actor in the textile field (**TXHO**) and 2 other industrial partners (**SEM** and **PURFI**) with a significant role in the project's achievements to ensure industrial and commercial exploitation of the project results. **CU** is leading the overall project management (WP1). **SEM** will implement all technologies developed in WP3 and 4 into a demonstrator (WP5) in **TXHO** facilities. Recyclers (**PURFI**) will validate the innovative process, representing mechanical recycling (cotton, polycotton, etc.).